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Disrupting the LINC complex for treating laminopathy

Abstract

The present invention relates to use of expression vectors and other compounds in methods to disrupt the Linker of Nucleoskeleton and Cytoskeleton (LINC) complex, uncoupling the nucleus from its linkage to the cytoskeleton, resulting in amelioration of diseases caused by one or more Lmna mutations, so-called laminopathies. More particularly, the invention relates to the expression of dominant negative SUN domain protein and/or dominant negative KASH domain protein to disrupt, for example, the LINC complex in cardiomyocytes for suppressing disease progression in dilated cardiomyopathy (DCM).

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Background/Summary

RELATED APPLICATIONS (1) This application is a national stage filing under 35 U.S.C.

 private use character Ovalhollow 371 of international application number PCT/SG2019/50033, filed Jan. 18, 2019, which claims the benefit of Singapore application number 10201800530Q, filed Jan. 19, 2018, each of which is herein incorporated by reference in its entirety.

FIELD OF THE INVENTION

(1) The present invention relates to use of expression vectors and other compounds in methods to disrupt the Linker of Nucleoskeleton and Cytoskeleton (LINC) complex, uncoupling the nucleus from its linkage to the cytoskeleton, resulting in amelioration of diseases caused by one or more Lmna mutations, so-called laminopathies. More particularly, the invention relates to the expression of dominant negative or mutated SUN domain protein and/or dominant negative or mutated KASH domain protein to disrupt the LINC complex in, for example, cardiomyocytes for suppressing disease progression in dilated cardiomyopathy (DCM).

BACKGROUND

(2) Dilated Cardiomyopathy (DCM) is the most common disease affecting heart muscle, accounting for approximately 60% of all cardiomyopathies. It is characterized by reduced systolic (contractile) function due to enlargement and thinning of the left ventricular wall or in some cases both ventricles. DCM is associated with sudden heart failure and cardiac death, resulting in high rates of hospital admission, the need for heart transplantation and consequently a high cost burden [J. L. Jefferies and J. A. Towbin, *Lancet* 375:752-762 (2010); R. E. Hershberger, et al., *Nat Rev Cardiol* 10: 531-547 (2013)]. The causes of DCM are varied, but include a variety of extrinsic factors, (viral, autoimmune infiltration, alcohol, and drugs). However 30-40% of all cases have a monogenic basis, with mutations in some 40 genes being linked to DCM. The most frequently mutated gene in DCM is TTN, that encodes the giant sarcomeric protein titin, with truncating variants in TTN accounting for almost 15-25% of all congenital forms of DCM [D. S. Herman et al., *N Engl J Med* 366: 619-628 (2012); U. Tayal, S. et al., *Genome Med* 9: 20 (2017)]. The second most frequently mutated gene is Lamin A (LMNA) accounts for as many as 6-8% of congenital DCM patients [U. Tayal, S. et al., *Genome Med* 9: 20 (2017)].

(3) LMNA Induced DCM Is characterized by cardiac conduction disease manifested by electrophysiological (ECG) abnormalities, including atrioventricular block, ventricular arrhythmias and fibrillation. The risk of sudden cardiac death is greater in patients with LMNA-cardiomyopathy than patients with other forms of DCM [J. H. Van Berlo et al., *Hum Mol Genet* 14: 2839-2849 (2005)]. Some 450 different mutations have been Identified in the LMNA gene, most being missense, resulting in the majority of DCM cases being inherited as autosomal dominants, with this diversity of the mutations complicating genetic approaches to treating LMNA induced DCM. To a limited extent LMNA induced DCM can be treated by fitting a pacemaker. Ultimately, however, effective treatment at present is accomplished by heart transplantation (R. E. Hershberger and A. Morales, in *Gene Reviews*((R)), M. P. Adam et al., Eds. (Seattle (Wash.), 1993); G. Captur et al., *Heart* 104: 488-479 (2018)].

(4) Mouse lines carrying Lmna mutations usually die within a few weeks after birth [T. Sullivan et al., *J Cell Biol* 147: 913-920 (1999); A. T. Bertrand et al., *Hum Mol Genet* 21: 1037-1048 (2012); V. Nikolova et al., *J Clin Invest* 113: 357-369 (2004); A. S. Wang, et al., *Differentiation; research in biological diversity*, (2015)]. The cause of early death in mice lacking Lmna is uncertain due to

multiple tissues being affected. Cardiac myopathy is thought to be a major contributing cause, as Lmna mutant mice develop DCM with conduction abnormalities and focal myocyte degeneration [V. Nikolova et al., *J Clin Invest* 113: 357-369 (2004); L. C. Mounkes, et al., *Hum Mol Genet* 14: 2167-2180 (2005)], although effects on other, as yet undefined, skeletal muscles may contribute to early postnatal death.

(5) The lamins are nuclear intermediate filament proteins and are the principal constituents of the nuclear lamina, the proteinaceous matrix underlying the inner nuclear membrane (INM). The lamina consists of the A-type lamins, consisting of 2 predominant forms, lamins A and lamin C, derived by alternate splicing of LMNA, whereas the two B type lamins (LMNB 1 and 2) are each encoded by two genes: LMNB1 and LMNB2 [B. Burke and C. L. Stewart, *Nat Rev Mod Cell Biol* 14: 13-24 (2013)]. The lamina provides structural and mechanical integrity to the nucleus, maintains nuclear shape and position within the cell, as well as being determinants of chromatin organization [T. Sullivan et al., *J Cell Biol* 147: 913-920 (1999); I. Solovei et al., *Cell* 152: 584-598 (2013)]. The lamins interact with numerous INM proteins, inducing Emerin, the Lamina-Associated Polypeptides (LAPs) and the SUN domain proteins [B. van Steensel and A. S. Belmont, *Cell* 169: 780-791 (2017)], many of which when either mutated or present as a variant are linked to heart disease [H. J. Worman, et al., *Cold Spring Harbor perspectives in biology* 2: a000760 (2010); C. L. Stewart, et al., *Exp Cell Res* 313:2144-2156 (2007)]. Together these proteins comprise an integrated protein network, centered on the lamina, where loss or mutation of the lamins can result in either the mislocalization or a change in expression levels of many lamina associated proteins, (emerin, SUN1, LBR and Lap2 α) [T. Sullivan et al., *J Cell Biol* 147: 913-920 (1999); I. Solovei et al., *Cell* 152: 584-598 (2013); C. Y. Chen et al., *Cell* 149: 565-577 (2012); T. V. Cohen et al., *Hum Mol Genet* 22: 2852-2869 (2013); F. Haque et al., *J Biol Chem* 285: 3487-3498 (2010)]. Among these proteins, where expression is affected by the loss of Lmna or mutation, are SUN1 and Lap2, both of whose levels are Increased. In the case of SUN1 the Increased level is due to reduced turnover, rather than increased expression, resulting in high levels accumulating in the Golgi which appeared to be cytotoxic at least in the Lmna.sup.-/- and Lmna Δ 9 mouse disease models [C. Y. Chen et al., *Cell* 149: 565-577 (2012); C. Stewart and B. Burke, WO/2013/158046]. However, when SUN1 levels are genetically ablated in mice with Lmna mutations, this increases the longevity 3-fold and ameliorates much of the pathology [C. Y. Chen et al., *Cell* 149: 585-577 (2012); C. Stewart and B. Burke, WO/2013/158048]. The median survival of wild-type or Sun1.sup.-/- is >210 days in a 7 month follow up; Lmna.sup.-/- mice had median survival of 41 days; Lmna.sup.-/- Sun1.sup.-/- mice had a median survival of 54 days; Lmna.sup.-/- Sun1.sup.-/- mice had a median survival of 104 days (p<0.01 comparing Lmna.sup.-/- and Lmna.sup.-/- Sun1.sup.-/-). Likewise, whereas all Lmna Δ 9 mice expired by 30 days after birth, their Lmna Δ 9Sun1.sup.-/- littermates thrived past this date, and most achieved life spans more than twice this duration [C. Y. Chen et al., *Cell* 149: 565-577 (2012)]. At the cellular level, human fibroblasts harbouring a LMNA mutation resulting in Hutchison-Gilford Progeria Syndrome also exhibited increased Sun1 levels. Depleting Sun1 in these cells alleviated nuclear morphology defects, again suggesting that excess Sun1 resulting from LMNA mutation is cytotoxic [C. Y. Chen et al., *Cell* 149: 565-577 (2012); C. Stewart and B. Burke, WO/2013/158046].

(6) The SUN (Sad1p, UNC-84) domain proteins share a conserved C-terminal SUN domain and localize to the INM [C. J. Malone, et al., *Development* 126:3171-3181 (1999)]. In mammals, SUN1 and SUN2 are the 2 principal SUN proteins that are widely expressed in virtually all tissues. In the perinuclear space, between the INM and outer nuclear membrane (ONM), the C-termini of SUN1 and/or 2 bind to the C-termini (KASH domains) of the different Nesprins/SYNE/KASH proteins that traverse the ONM. Together these 2 families of proteins comprise the LINC complexes that physically couple the interphase nuclei to the cytoskeleton [M. Crisp et al., *J Cell Biol* 172: 41-53 (2006); E. C. Tapley and D. A. Starr, *Curr Opin Cell Biol* 25: 57-62 (2013)]. The N-termini of the SUN domain proteins protrude into the nucleoplasm and with SUN1, this region interacts with pre-

laminA and nucleopore complexes. Whether the N-terminus of SUN2 interacts with any nucleoplasmic/NE protein is unclear. In contrast, the bulk of the Nesprins/KASH domain proteins extend into the cytoplasm adjacent to the ONM. There, depending on the particular Nesprin/KASH protein, they interact directly or indirectly with all 3 cytoskeletal protein networks (microtubules, actin microfilaments and Intermediate filaments) [H. F. Horn, *Current topics in developmental biology* 109: 287-321 (2014)]. Together, the SUN and KASH/Nesprin proteins of the LINC complex establish a direct physical connection between the cytoplasmic cytoskeletal networks (and their connections e.g. cell adhesion complexes at the cell membrane) and the interphase nuclear interior or nucleoplasm. The LINC complex is thought to mediate force transmission between the nucleus and cytoskeleton and consequently regulate changes in gene expression/chromatin organization in response to mechanical/physical stimuli [S. G. Alam et al., *Scientific reports* 6: 38003 (2016)]. Although loss of either SUN1 or SUN2 alone has no overt effect on postnatal growth and viability, SUN1 null mice are infertile and deaf. Simultaneous loss of Sun1 and Sun2 results in perinatal lethality, indicating a degree of redundancy during embryogenesis [K. Lei et al., *Proc Natl Acad Sci USA* 106: 10207-10212 (2009)].

(7) There is a need to develop alternative methods to ameliorate the negative effects over-accumulation of Sun1 has on cells carrying Lmna mutations. The present disclosure aims at providing such a method.

SUMMARY OF INVENTION

(8) Surprisingly, the inventors have found that disruption of the LINC complex rather than removal of accumulated Sun1 protein can ameliorate diseases caused by one or more Lmna mutations. One way of achieving the disruption is via an expression construct/vector comprising an operably linked transgene, the expression of which generates dominant negative SUN domain protein or mutated endogenous SUN domain protein and/or dominant negative KASH domain protein or mutated endogenous KASH domain protein. The exogenous dominant negative SUN domain and KASH domain proteins act as LINC complex binding competitors, thereby uncoupling the nucleus from its linkage to the cytoskeleton. The mutated SUN domain and KASH domain proteins are endogenous Sun and Nesprin proteins that have been mutated in the SUN or KASH domain, respectively, and cannot form a LINC complex because they cannot bind to their cognate LINC complex partner. These strategies may be used to disrupt the LINC complex to treat, for example, laminopathies. The result was achieved without actively reducing the endogenous SUN1 protein levels. Results shown herein support these claims.

(9) According to a first aspect of the invention, there is provided an isolated nucleic acid molecule, wherein the nucleic acid molecule comprises an expression vector and a transgene, whereby the transgene is operably linked to the expression vector, wherein expression of the transgene in a transfected cell results in disruption of a LINC complex in the transfected cell.

(10) In some embodiments, the expression vector is a cardiac- or cardiomyocyte-specific expression vector.

(11) In some embodiments, the expression vector comprises a cardiac- or cardiomyocyte-specific promoter selected from the group comprising a cardiac troponin T promoter (cTnT), a α -myosin heavy chain (α -MHC) promoter and a myosin light chain (MLC2v) promoter. Preferably the promoter is cardiac troponin T promoter (cTnT).

(12) In some embodiments, the cardiomyocyte-specific promoter is chicken cardiac troponin T promoter (cTnT).

(13) In some embodiments, the expression vector has cardiac tropism/is cardiotropic.

(14) In some embodiments, the expression vector is a virus expression vector.

(15) In some embodiments, the virus expression vector is selected from the group comprising Lentivirus, Adenovirus and Adeno-associated virus (AAV). Preferably the virus expression vector is adeno-associated virus (AAV).

(16) In some embodiments, the AAV vector is selected from the group consisting of AAV9

- (serotype 9), AAV1 (serotype 1), AAV6 (serotype 6), AAV8 (serotype 8), AAV218 and AAV9.45.
- (17) In some embodiments, the AAV vector is AAV9 (serotype 9).
- (18) In some embodiments, the transgene comprises nucleic acid sequences for expressing a luminal domain of a SUN domain-containing protein, an N-terminal signal sequence, a signal peptidase cleavage site, and a C-terminal targeting peptide sequence.
- (19) In some embodiments the luminal domain of the SUN domain-containing protein comprises a coiled coil domain and a SUN domain.
- (20) In a preferred embodiment the coiled coil domain is upstream of the SUN domain.
- (21) In some embodiments, the transgene further comprises nucleic acid sequences for expressing an N-terminal signal sequence, a signal peptidase cleavage site, and a C-terminal targeting peptide sequence.
- (22) In some embodiments, the transgene comprises nucleic acid sequences for expressing an N-terminal signal sequence, a signal peptidase cleavage site, and a C-terminal targeting peptide sequence, and either the luminal domain of the SUN domain-containing protein or the SUN domain.
- (23) Preferably, the SUN domain protein is SUN1 or SUN2.
- (24) In some embodiments the luminal domain of Sun1 comprises amino acids 458-913 of full-length mouse Sun1 (Uniprot Q9D666) or its human equivalent comprising the coiled coil domain and the SUN domain and lacking the transmembrane domain. A schematic of the structure of a dominant negative form of Sun is shown in FIG. 7.
- (25) For SUN domain constructs it is expected that the SUN domain alone (crystal structure solved by the Kutay and Schwartz labs [Sosa et al., *Cell* 149(5):1035-47 (2012)]), instead of the entire luminal domain (coiled coil domain and SUN domain) is sufficient to disrupt the SUN-KASH interaction as it is capable of binding to the KASH domain. The human Sun1 SUN domain nucleic acid sequence is set forth in SEQ ID NO: 80. However, the presence of the signal sequence and the KDEL sequence are important for targeting the construct to the perinuclear space.
- (26) In some embodiments, the N-terminal signal sequence is derived from a secretory protein or a Type I transmembrane protein.
- (27) Preferably, the secretory protein or Type I transmembrane protein is selected from the group consisting of human serum albumin, proinsulin, transferrin receptor, EGF receptor, pre-pro-opiomelanocortin, pancreatic digestive enzymes (for example, proteases, amylases and lipases), endoplasmic reticulum luminal proteins, for example protein disulphide isomerases, GRP94 and combinations thereof. More preferably, the N-terminal signal sequence is derived from human serum albumin.
- (28) In some embodiments, the N-terminal signal sequence is not preceded at its N-terminus by any other tags.
- (29) In some embodiments, the signal peptidase cleavage site is a signal peptidase cleavage site derived from or is one of the group consisting of human serum albumin, proinsulin, transferrin receptor, EGF receptor, pre-pro-opiomelanocortin, pancreatic digestive enzymes (for example, proteases, amylases and lipases), endoplasmic reticulum luminal proteins, such as protein disulphide isomerases, GRP94 and combinations thereof. Preferably, the signal peptidase cleavage site is a signal peptidase cleavage site derived from human serum albumin.
- (30) In some embodiments, the C-terminal targeting peptide sequence prevents secretion of a peptide expressed from the transgene according to any aspect of the invention.
- (31) In some embodiments, the C-terminal targeting peptide sequence is a KDEL tetrapeptide Golgi retrieval sequence. Examples of such structures are shown in FIGS. 11 and 12.
- (32) In some embodiments the transgene comprises a humanized Sun1DN nucleic acid sequence or a humanized Sun2DN nucleic acid sequence. In a preferred embodiment, the transgene comprises a signal sequence, a humanized Sun1DN nucleic acid sequence and a KDEL sequence as set forth in SEQ ID NO: 4; or the transgene comprises a signal sequence, a humanized Sun2DN nucleic acid

sequence and a KDEL sequence as set forth in SEQ ID NO: 5.

(33) In some embodiments, the transgene further comprises an epitope tag. Preferably the epitope tag is N-terminal, or located anywhere in the nucleic acid molecule except downstream of (after) the C-terminal targeting peptide sequence [for example KDEL], or located anywhere in the nucleic acid molecule except upstream of (before) the N-terminal signal sequence.

(34) In some embodiments, the epitope tag is selected from the group consisting of cellulose binding domain (CBD), chloramphenicol acetyl transferase (CAT), dihydrofolate reductase (DHFR), one or more FLAG tags, glutathione S-transferase (GST), green fluorescent protein (GFP), haemagglutinin A (HA), histidine (His), Herpes simplex virus (HSV), luciferase, maltose-binding protein (MBP), c-Myc, Protein A, Protein G, streptavidin, T7, thioredoxin, V5, vesicular stomatitis virus glycoprotein (VSV-G), and combinations thereof. Preferably, the epitope tag is haemagglutinin A (HA).

(35) In some embodiments, the nucleic acid molecule of the invention comprises an adeno-associated virus vector (AAV) comprising a chicken cardiac troponin T promoter (cTnT), a transgene according to any aspect of the invention comprising the luminal domain of the SUN domain-containing protein derived from SUN, an N-terminal signal sequence and a signal peptidase cleavage site which are each derived from human serum albumin, a C-terminal targeting peptide sequence which is a KDEL sequence, and wherein the transgene optionally further comprises haemagglutinin (HA) as an N-terminal epitope tag.

(36) According to an embodiment an example of such a vector is shown in FIG. 10 and comprises the nucleic acid sequence set forth in SEQ ID NO: 3.

(37) In some embodiments, the nucleic acid molecule of the invention comprises an adeno-associated virus vector (AAV) comprising a chicken cardiac troponin T promoter (cTnT), a transgene according to any aspect of the invention comprising the luminal domain of the SUN domain-containing protein derived from SUN2, an N-terminal signal sequence and a signal peptidase cleavage site which are each derived from human serum albumin, a C-terminal targeting peptide sequence which is a KDEL sequence, and wherein the transgene optionally further comprises hemagglutinin (HA) as the N-terminal epitope tag.

(38) According to an embodiment an example nucleic acid molecule would comprise the vector structure shown in FIG. 10 and the transgene nucleic acid sequence set forth in SEQ ID NO: 5.

(39) Rather than expressing components of a luminal domain of a SUN domain-containing protein, a KASH domain may be expressed to disrupt a LINC complex by competing with endogenous Nesprins (which comprise a KASH domain) for binding to SUN1 and SUN2 domains.

(40) Accordingly, in some embodiments of the nucleic acid molecule of the Invention, the transgene comprises nucleic acid sequences for expressing a KASH domain, and an N-terminal stabiliser polypeptide sequence.

(41) Preferably, the KASH domain comprises a transmembrane domain and a SUN-interacting peptide.

(42) Preferably the transgene comprises nucleic acid sequences for expressing a KASH domain that traverses the outer nuclear membrane, a SUN-interacting KASH peptide that extends into the perinuclear space at the C-terminus, and an N-terminal stabiliser polypeptide sequence in the cytoplasm.

(43) It would be understood that KASH domain constructs with extensions after the last C-terminal amino acid of the naturally occurring KASH domain are not expected to work. i.e. C-terminal tags, or even an additional carboxy-terminal single amino acid, will disrupt KASH interaction with SUN. In addition, a signal sequence on the N-terminus of SUN domain constructs cannot be preceded by any tags.

(44) In some embodiments, the KASH domain is selected from the group consisting of KASH1 (derived from Nesprin-1 (SYNE1 gene)), KASH2 (derived from Nesprin-2 (SYNE2 gene)), KASH3 (derived from Nesprin-3 (SYNE3 gene)), KASH4 (derived from Nesprin-4 (SYNE4

gene)) and KASH5 (derived from KASH5/CCDC155 (KASH5 gene)).

(45) In preferred embodiments the KASH 1 domain comprises the human amino acid sequence set forth in SEQ ID NO: 7; the KASH 2 domain comprises the human amino acid sequence set forth in SEQ ID NO: 9; the KASH 3 domain comprises the human amino acid sequence set forth in SEQ ID NO: 11; the KASH 4 domain comprises the human amino acid sequence set forth in SEQ ID NO: 13; and the KASH 5 domain comprises the human amino acid sequence set forth in SEQ ID NO: 15. An alignment of the five KASH amino acid sequences is shown in FIG. 14.

(46) In some embodiments the KASH domain nucleic acid sequence has at least 80%, at least 85%, at least 90%, at least 95% sequence identity or 100% sequence identity to the nucleic acid sequence of the KASH1 domain set forth in SEQ ID NO: 6; the nucleic acid sequence of the KASH2 domain set forth in SEQ ID NO: 8; the nucleic acid sequence of the KASH3 domain set forth in SEQ ID NO: 10; the nucleic acid sequence of the KASH4 domain set forth in SEQ ID NO: 12; or the nucleic acid sequence of the KASH5 domain set forth in SEQ ID NO: 14.

(47) More preferably, for the purpose of clinical use, the KASH domain is the human KASH1 domain of SYNE1 having at least 80%, at least 85%, at least 90%, at least 95% sequence identity or 100% sequence identity to the nucleic acid sequence of the human KASH1 domain set forth in SEQ ID NO: 6.

(48) It would be understood that due to the redundancy in the genetic code, a nucleic acid sequence may have less than 100% identity and still encode the same amino acid sequence.

(49) In some embodiments, the KASH domain does not comprise any extensions after the last C-terminal amino acid compared to a naturally occurring KASH domain.

(50) In some embodiments, the N-terminal stabiliser polypeptide sequence is selected from the group consisting of green fluorescent protein (GFP), cellulose binding domain (CBD), chloramphenicol acetyl transferase (CAT), dihydrofolate reductase (DHFR), glutathione S-transferase (GST), luciferase, maltose-binding protein (MBP), Protein A, Protein G, streptavidin, thioredoxin, DHFR, including multiples and combinations thereof.

(51) In some embodiments, the N-terminal stabilizer polypeptide sequence forms a discretely folded domain.

(52) In some embodiments, the vector is the adeno-associated virus vector (AAV) comprising a cardiac troponin T promoter (cTnT), and the transgene comprises nucleic acid sequences for expressing a KASH domain, and an N-terminal stabiliser polypeptide sequence, wherein the KASH domain is selected from the group comprising KASH1, KASH2, KASH3, KASH4 and KASH5.

(53) In some embodiments, the N-terminal stabiliser polypeptide sequence is green fluorescent protein (GFP).

(54) Rather than expressing components of a luminal domain of a SUN domain-containing protein or a KASH domain to disrupt a LINC complex by competing for binding with endogenous Nesprins (which comprise a KASH domain) or Sun1 and Sun2 (which comprise a SUN domain), another approach for disrupting the LINC complex is to modify the endogenous SUN domain or KASH domain so that it fails to bind to, or has reduced binding capacity for, its cognate LINC complex binding partner.

(55) As both the SUN domain and the KASH domain are located at the C-termini of their respective proteins, one way of producing a modified SUN or KASH domain is to use a CRISPR/Cas system to modify the genes encoding SUN or KASH domain proteins to generate a premature stop codon at the 3' end of the respective protein sequences following CRISPR-induced non-homologous end joining. This would result in a truncated protein with its C-terminal SUN or KASH domain mutated. The truncated protein would be expressed and membrane-localized, but unable to interact with its cognate LINC complex partners.

(56) Accordingly, in some embodiments of the nucleic acid molecule of the invention, the transgene comprises nucleic acid sequences for expressing a CRISPR-Cas or other synthetic nuclease system to modify nucleic acid that encodes the SUN domain or KASH domain of

Symbol 17811 c.1771T > A Substitution p.Cys591Ser Substitution Tail — Acrogeria,
 Gottron Type 201200 # 18255 c.418_438dupCTGCTG Duplication p.Leu140_Ala146dup
 Duplication 1B — Arrhythmogenic — — AACTCCAAGGAGGCC cardiomyopathy 17439
 c.1039G > A Substitution p.Glu347Lys Substitution 2B ARVD7 Arrhythmogenic right
 609160 % ventricular cardiomyopathy 18492 c.1930C > T Substitution p.Arg644Cys
 Substitution Tail ARVD7 Arrhythmogenic right 609160 % ventricular cardiomyopathy 9157
 c.1718C > T Substitution p.Ser573Leu Substitution Tail — Arthropathy syndrome, — —
 autosomal recessive 18299 c.1494G > A Substitution p.Trp498X Substitution Tail AF
 Atrial fibrillation — — 18301 c.175C > G Substitution p.Leu59Val Substitution 1A APS
 Atypical progeroid syndrome — — 8885 c.169G > C Substitution p.Ala57Pro Substitution
 1A WRN Atypical Werner syndrome 277700 # 8886 c.398G > T Substitution p.Arg133Leu
 Substitution 1B WRN Atypical Werner syndrome 277700 # 11462 c.398G > T Substitution
 p.Arg133Leu Substitution 1B WRN Atypical Werner syndrome 277700 # 9232 c.398G > T
 Substitution p.Arg133Leu Substitution 1B WRN Atypical Werner syndrome 277700 # 11509
 c.398G > T Substitution p.Arg133Leu Substitution 1B WRN Atypical Werner syndrome
 277700 # 11670 c.398G > T Substitution p.Arg133Leu Substitution 1B WRN Atypical
 Werner syndrome 277700 # 8888 c.419T > G Substitution p.Leu140Arg Substitution 1B
 WRN Atypical Werner syndrome 277700 # 13457 c.506delT Deletion p.Val169GlyfsX7
 Frame shift 1B WRN Atypical Werner syndrome 277700 # 13350 c.898G > A
 Substitution p.Asp300Asn Substitution 2B WRN Atypical Werner syndrome 277700 # 17875
 c.898G > A Substitution p.Asp300Asn Substitution 2B WRN Atypical Werner syndrome
 277700 # 11767 c.1130G > T Substitution p.Arg377Leu Substitution 2B AD-SMA
 Autosomal dominant spinal 182980 # muscular dystrophy 11766 c.1477C > T
 Substitution p.Gln493X Substitution Tail AD-SMA Autosomal dominant spinal 182980 #
 muscular dystrophy 9205 c.? Unknown p.Glu33Asp Substitution 1A — Axonal neuropathy,
 182980 — muscular dystrophy, cardiac disease 8994 c.99G > T Substitution p.Glu33Asp
 Substitution 1A — Axonal neuropathy, — — muscular dystrophy, cardiac disease,
 leuconychia 12416 c.1621C > T Substitution p.Arg541Cys Substitution Tail — Cardiac
 arrhythmia — — 12399 c.673C > T Substitution p.Arg225X Substitution L12 CCD Cardiac
 conduction defect 115080 # 17808 c.695G > T Substitution p.Gly232Val Substitution L12
 CCD Cardiac conduction defect 115080 # 13449 c.799T > C Substitution p.Tyr267His
 Substitution 2B CCD Cardiac conduction defect 115080 # 14260 c.178C > G Substitution
 p.Arg60Gly Substitution 1A — Cardiomyopathy with — — advanced AV block and
 arrhythmia 14256 c.184C > G Substitution p.Arg62Gly Substitution 1A — Cardiomyopathy
 with — — advanced AV block and arrhythmia 14018 c.497G > C Substitution
 p.Arg166Pro Substitution 1B — Cardiomyopathy with — — advanced AV block and
 arrhythmia 14258 c.575A > T Substitution p.Asp192Val Substitution 1B — Cardiomyopathy
 with — — advanced AV block and arrhythmia 14007 c.673C > T Substitution p.Arg225X
 Substitution L12 — Cardiomyopathy with — — advanced AV block and arrhythmia 14015
 c.775T > C Substitution p.Tyr259His Substitution L2 — Cardiomyopathy with — —
 advanced AV block and arrhythmia 14016 c.775T > C Substitution p.Tyr259His
 Substitution L2 — Cardiomyopathy with — — advanced AV block and arrhythmia 14011
 c.815_818delACAAins Indel p.Asp272AlafsX208 Frame shift 2B — Cardiomyopathy with —
 — CCAGAC advanced AV block and arrhythmia 14012 c.815_818delACAAins Indel
 p.Asp272AlafsX208 Frame shift 2B — Cardiomyopathy with CCAGAC advanced AV
 block and — — arrhythmia 11652 c.- Deletion p.Met1_Pro4del Deletion Head CMT2 Charcot-
 Marie-Tooth disease 118210 # 3_12delGCCATGGAGA type 2 CCCC G 13374
 c.1496_1496delC Deletion p.Ala499ValfsX141 Frame shift Tail CMT2 Charcot-Marie-Tooth
 disease 118210 # type 2 17199 c.1910T > C Substitution p.Phe637Ser Substitution Tail
 CMT2 Charcot-Marie-Tooth disease 118210 # type 2 8840 c.8920 > T Substitution

[illegible]

disproportion 18144 c.91G > A Substitution p.Glu31Lys Substitution Head CMD Congenital
muscular — — dystrophy 18148 c.91_93delGAG Deletion p.Glu31X Substitution Head CMD
Congenital muscular — — dystrophy 17813 c.93G > C Substitution p.Glu31Asp Substitution
Head CMD Congenital muscular — — dystrophy 18146 c.94_96delAAG Deletion p.Lys32X
Substitution 1A CMD Congenital muscular — — dystrophy 16402 c.104T > C Substitution
p.Leu35Pro Substitution 1A CMD Congenital muscular — — dystrophy 16287 c.115A > T
Substitution p.Asn39Tyr Substitution 1A CMD Congenital muscular — — dystrophy 18150
c.117T > G Substitution p.Asn39Lys Substitution 1A CMD Congenital muscular — —
dystrophy 18153 c.143G > C Substitution p.Arg48Pro Substitution 1A CMD Congenital
muscular — — dystrophy 18156 c.422T > C Substitution p.Leu141Pro Substitution 1B CMD
Congenital muscular — — dystrophy 13462 c.745C > T Substitution p.Arg249Trp
Substitution 2A CMD Congenital muscular — — dystrophy 16404 c.745C > T Substitution
p.Arg249Trp Substitution 2A CMD Congenital muscular — — dystrophy 17758 c.745C > T
Substitution p.Arg249Trp Substitution 2A CMD Congenital muscular — — dystrophy 18158
c.745C > T Substitution p.Arg249Trp Substitution 2A CMD Congenital muscular — —
dystrophy 11816 c.1072G > A Substitution p.Glu358Lys Substitution 2B CMD Congenital
muscular — — dystrophy 18161 c.1117A > G Substitution p.Ile373Val Substitution 2B CMD
Congenital muscular — — dystrophy 18166 c.1118-T > A Substitution p.Ile373Asn
Substitution 2B CMD Congenital muscular — — dystrophy 13460 c.1139-T > C Substitution
p.Leu380Ser Substitution 2B CMD Congenital muscular — — dystrophy 18169 c.1147G > A
Substitution p.Glu383Lys Substitution 2B CMD Congenital muscular — — dystrophy 18171
c.1147G > A Substitution p.Glu383Lys Substitution 2B CMD Congenital muscular — —
dystrophy 18163 c.1151A > G Substitution p.Glu384Gly Substitution 2B CMD Congenital
muscular — — dystrophy 11817 c.1162C > T Substitution p.Arg388Cys Substitution Tail CMD
Congenital muscular — — dystrophy 17473 c.1330_1338dupGAGG Duplication
p.Glu444_Asp446dup Duplication Tail CMD Congenital muscular — — TGGAT dystrophy
11818 c.1368_1370delCAA Deletion p.Asn456del Deletion Tail CMD Congenital muscular —
— dystrophy 18238 c.1489-14 1489- Deletion p.? Unknown Unknown CMD Congenital
muscular — — 7delTTTCTCCT dystrophy 16951 c.1445G > A Substitution p.Arg482Gln
Substitution Tail T2D DIABETES MELLITUS, 125853 # NONINSULIN-DEPENDENT,
NIDDM 13171 c.? Unknown p.Lys260Asn Substitution L2 CMD1A Dilated cardiomyopathy
1A 115200 # 8832 c.16C > T Substitution p.Gln6X Substitution Head CMD1A Dilated
cardiomyopathy 1A 115200 # 9397 c.28_29insA Insertion p.Thr10AsnfsX31 Frame shift Head
CMD1A Dilated cardiomyopathy 1A 115200 # 11627 c.31delC Deletion p.Arg11AlafsX85
Frame shift Head CMD1A Dilated cardiomyopathy 1A 115200 # 14254 c.73C > G
Substitution p.Arg25Gly Substitution Head CMD1A Dilated cardiomyopathy 1A 115200 #
13155 c.82C > T Substitution p.Arg28Trp Substitution Head CMD1A Dilated
cardiomyopathy 1A 115200 # 13563 c.155T > C Substitution p.Leu52Pro Substitution 1A
CMD1A Dilated cardiomyopathy 1A 115200 # 13379 c.176T > G Substitution p.Leu59Arg
Substitution 1A CMD1A Dilated cardiomyopathy 1A 115200 # 13378 c.176T > G
Substitution p.Leu59Arg Substitution 1A CMD1A Dilated cardiomyopathy 1A 115200 # 8748
c.178C > G Substitution p.Arg60Gly Substitution 1A CMD1A Dilated cardiomyopathy 1A
115200 # 13192 c.203_208delAGGTGG Deletion p.Glu68_Val69del Deletion 1A CMD1A
Dilated cardiomyopathy 1A 115200 # 17742 c.232G > A Substitution p.Lys78Glu
Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 9163 c.244G > A
Substitution p.Glu82Lys Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 16139
c.244G > A Substitution p.Glu82Lys Substitution 1B CMD1A Dilated cardiomyopathy 1A
115200 # 8749 c.254T > G Substitution p.Leu85Arg Substitution 1B CMD1A Dilated
cardiomyopathy 1A 115200 # 8865 c.266G > T Substitution p.Arg89Leu Substitution 1B
CMD1A Dilated cardiomyopathy 1A 115200 # 11618 c.266G > T Substitution p.Arg89Leu

Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 13080 c.266G > T
Substitution p.Arg89Leu Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 13157
c.266G > T Substitution p.Arg89Leu Substitution 1B CMD1A Dilated cardiomyopathy 1A
115200 # 14030 c.266G > T Substitution p.Arg89Leu Substitution 1B CMD1A Dilated
cardiomyopathy 1A 115200 # 16966 c.266G > T Substitution p.Arg89Leu Substitution 1B
CMD1A Dilated cardiomyopathy 1A 115200 # 17777 c.266G > T Substitution p.Arg89Leu
Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 13613 c.274C > T
Substitution p.Leu92Phe Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 8844
c.289A > G Substitution p.Lys97Glu Substitution 1B CMD1A Dilated cardiomyopathy 1A
115200 # 11620 c.289A > G Substitution p.Lys97Glu Substitution 1B CMD1A Dilated
cardiomyopathy 1A 115200 # 13159 c.289A > G Substitution p.Lys97Glu Substitution 1B
CMD1A Dilated cardiomyopathy 1A 115200 # 13082 c.302G > C Substitution p.Arg101Pro
Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 16954 c.302G > C
Substitution p.Arg101Pro Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 8843
c.331G > T Substitution p.Glu111X Substitution 1B CMD1A Dilated cardiomyopathy 1A
115200 # 11619 c.331G > T Substitution p.Glu111X Substitution 1B CMD1A Dilated
cardiomyopathy 1A 115200 # 13199 c.331G > T Substitution p.Glu111X Substitution 1B
CMD1A Dilated cardiomyopathy 1A 115200 # 16935 c.348_349insG Insertion
p.Lys117Glu fsX10 Frame shift 1B CMD1A Dilated cardiomyopathy 1A 115200 # 13089
c.357-1G > T Substitution p.? Unknown Unknown CMD1A Dilated cardiomyopathy 1A
115200 # 9148 c.394G > C Substitution p.Ala132Pro Substitution 1B CMD1A Dilated
cardiomyopathy 1A 115200 # 17029 c.394G > C Substitution p.Ala132Pro Substitution 1B
CMD1A Dilated cardiomyopathy 1A 115200 # 17028 c.394G > C Substitution p.Ala132Pro
Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 9297 c.425_426insGGCACT
Insertion p.Leu141_Asn142ins Insertion 1B CMD1A Dilated cardiomyopathy 1A 115200 #
GGAGGCTCTGCTGAA LysAspLeuAspAlaLeu Leu 9010 c.427T > C Substitution
p.Ser143Pro Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 11473 c.427T
> C Substitution p.Ser143Pro Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 #
11474 c.427T > C Substitution p.Ser143Pro Substitution 1B CMD1A Dilated
cardiomyopathy 1A 115200 # 11475 c.427T > C Substitution p.Ser143Pro Substitution 1B
CMD1A Dilated cardiomyopathy 1A 115200 # 11476 c.427T > C Substitution p.Ser143Pro
Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 11477 c.427T > C
Substitution p.Ser143Pro Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 9150
c.427T > C Substitution p.Ser143Pro Substitution 1B CMD1A Dilated cardiomyopathy 1A
115200 # 16162 c.427T > C Substitution p.Ser143Pro Substitution 1B CMD1A Dilated
cardiomyopathy 1A 115200 # 16163 c.427T > C Substitution p.Ser143Pro Substitution 1B
CMD1A Dilated cardiomyopathy 1A 115200 # 16164 c.427T > C Substitution p.Ser143Pro
Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 16165 c.427T > C
Substitution p.Ser143Pro Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 17038
c.427T > C Substitution p.Ser143Pro Substitution 1B CMD1A Dilated cardiomyopathy 1A
115200 # 17037 c.427T > C Substitution p.Ser143Pro Substitution 1B CMD1A Dilated
cardiomyopathy 1A 115200 # 17036 c.427T > C Substitution p.Ser143Pro Substitution 1B
CMD1A Dilated cardiomyopathy 1A 115200 # 17024 c.427T > C Substitution p.Ser143Pro
Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 17022 c.427T > C
Substitution p.Ser143Pro Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 17021
c.427T > C Substitution p.Ser143Pro Substitution 1B CMD1A Dilated cardiomyopathy 1A
115200 # 17020 c.427T > C Substitution p.Ser143Pro Substitution 1B CMD1A Dilated
cardiomyopathy 1A 115200 # 17019 c.427T > C Substitution p.Ser143Pro Substitution 1B
CMD1A Dilated cardiomyopathy 1A 115200 # 17018 c.427T > C Substitution p.Ser143Pro
Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 17023 c.427T > C

Substitution p.Ser143Pro Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 17661
c.448A > T Substitution p.Thr150Ser Substitution 1B CMD1A Dilated cardiomyopathy 1A
115200 # 8879 c.481G > A Substitution p.Glu161Lys Substitution 1B CMD1A Dilated
cardiomyopathy 1A 115200 # 13161 c.481G > A Substitution p.Glu161Lys Substitution 1B
CMD1A Dilated cardiomyopathy 1A 115200 # 13162 c.481G > A Substitution
p.Glu161Lys Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 13163 c.481G
> A Substitution p.Glu161Lys Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 #
13328 c.481G > A Substitution p.Glu161Lys Substitution 1B CMD1A Dilated
cardiomyopathy 1A 115200 # 13617 c.481G > A Substitution p.Glu161Lys Substitution 1B
CMD1A Dilated cardiomyopathy 1A 115200 # 13091 c.497G > C Substitution p.Arg166Pro
Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 16968 c.497G > C
Substitution p.Arg166Pro Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 #
16969 c.497G > C Substitution p.Arg166Pro Substitution 1B CMD1A Dilated
cardiomyopathy 1A 115200 # 13165 c.548T > C Substitution p.Leu183Pro Substitution 1B
CMD1A Dilated cardiomyopathy 1A 115200 # 18267 c.563T > G Substitution
p.Leu188Arg Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 16160 c.565C
> T Substitution p.Arg189Trp Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 #
8845 c.568C > T Substitution p.Arg190Trp Substitution 1B CMD1A Dilated
cardiomyopathy 1A 115200 # 9011 c.568C > T Substitution p.Arg190Trp Substitution 1B
CMD1A Dilated cardiomyopathy 1A 115200 # 11507 c.568C > T Substitution p.Arg190Trp
Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 9149 c.568C > T
Substitution p.Arg190Trp Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 #
11621 c.568C > T Substitution p.Arg190Trp Substitution 1B CMD1A Dilated
cardiomyopathy 1A 115200 # 11782 c.568C > T Substitution p.Arg190Trp Substitution 1B
CMD1A Dilated cardiomyopathy 1A 115200 # 11790 c.568C > T Substitution p.Arg190Trp
Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 13167 c.568C > T
Substitution p.Arg190Trp Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 #
16167 c.568C > T Substitution p.Arg190Trp Substitution 1B CMD1A Dilated
cardiomyopathy 1A 115200 # 17034 c.568C > T Substitution p.Arg190Trp Substitution 1B
CMD1A Dilated cardiomyopathy 1A 115200 # 17033 c.568C > T Substitution p.Arg190Trp
Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 17032 c.568C > T
Substitution p.Arg190Trp Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 #
17031 c.568C > T Substitution p.Arg190Trp Substitution 1B CMD1A Dilated
cardiomyopathy 1A 115200 # 13093 c.569G > A Substitution p.Arg190Gln Substitution 1B
CMD1A Dilated cardiomyopathy 1A 115200 # 13330 c.569G > A Substitution
p.Arg190Gln Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 16976 c.569G
> A Substitution p.Arg190Gln Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 #
9165 c.575A > G Substitution p.Asp192Gly Substitution 1B CMD1A Dilated
cardiomyopathy 1A 115200 # 14200 c.575A > G Substitution p.Asp192Gly Substitution 1B
CMD1A Dilated cardiomyopathy 1A 115200 # 8750 c.585C > G Substitution p.Asn195Lys
Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 12393 c.585C > A
Substitution p.Asn195Lys Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 #
11668 c.607G > A Substitution p.Glu203Lys Substitution 1B CMD1A Dilated
cardiomyopathy 1A 115200 # 16978 c.607G > A Substitution p.Glu203Lys Substitution 1B
CMD1A Dilated cardiomyopathy 1A 115200 # 8751 c.6084 > G Substitution p.Glu203Gly
Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 13332 c.608A > T
Substitution p.Glu203Val Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 #
13095 c.629T > G Substitution p.Ile210Ser Substitution 1B CMD1A Dilated
cardiomyopathy 1A 115200 # 16971 c.629T > G Substitution p.Ile210Ser Substitution 1B
CMD1A Dilated cardiomyopathy 1A 115200 # 8901 c.640 - 10A > G Substitution p.?

Unknown Unknown CMD1A Dilated cardiomyopathy 1A 115200 # 8864 c.644T > C
 Substitution p.Leu215Pro Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 #
 16980 c.644T > C Substitution p.Leu215Pro Substitution 1B CMD1A Dilated
 cardiomyopathy 1A 115200 # 13169 c.656A > C Substitution p.Lys219Thr Substitution 1B
 CMD1A Dilated cardiomyopathy 1A 115200 # 13335 c.656A > C Substitution p.Lys219Thr
 Substitution 1B CMD1A Dilated cardiomyopathy 1A 115200 # 11669 c.673C > T
 Substitution p.Arg225X Substitution L12 CMD1A Dilated cardiomyopathy 1A 115200 # 17607
 c.673C > T Substitution p.Arg225X Substitution L12 CMD1A Dilated cardiomyopathy 1A
 115200 # 17609 c.673C > T Substitution p.Arg225X Substitution L12 CMD1A Dilated
 cardiomyopathy 1A 115200 # 13097 c.700C > T Substitution p.Gln234X Substitution L12
 CMD1A Dilated cardiomyopathy 1A 115200 # 13201 c.736C > T Substitution p.Gln246X
 Substitution 2A CMD1A Dilated cardiomyopathy 1A 115200 # 17786 c.736C > T
 Substitution p.Gln246X Substitution 2A CMD1A Dilated cardiomyopathy 1A 115200 # 17779
 c.767T > G Substitution p.Val256Gly Substitution 2A CMD1A Dilated cardiomyopathy 1A
 115200 # 11809 c.8004 > G Substitution p.Tyr267Cys Substitution 2B CMD1A Dilated
 cardiomyopathy 1A 115200 # 13173 c.8004 > G Substitution p.Tyr267Cys Substitution 2B
 CMD1A Dilated cardiomyopathy 1A 115200 # 11810 c.855delG Deletion p.Ala287LeufsX191
 Frame shift 2B CMD1A Dilated cardiomyopathy 1A 115200 # 11613 c.908_909delCT
 Deletion p.Ser303CysfsX26 Frame shift 2B CMD1A Dilated cardiomyopathy 1A 115200 #
 12384 c.908_909delCT Deletion p.Ser303CysfsX26 Frame shift 2B CMD1A Dilated
 cardiomyopathy 1A 115200 # 13206 c.936 + 1G > T Substitution p.? Unknown Unknown
 CMD1A Dilated cardiomyopathy 1A 115200 # 16805 c.937 - 11C > G Substitution
 p.Leu313GlylsX31 Frame shift 2B CMD1A Dilated cardiomyopathy 1A 115200 # 8846
 c.949G > A Substitution p.Glu317Lys Substitution 2B CMD1A Dilated cardiomyopathy 1A
 115200 # 11617 c.949G > A Substitution p.Glu317Lys Substitution 2B CMD1A Dilated
 cardiomyopathy 1A 115200 # 13175 c.949G > A Substitution p.Glu317Lys Substitution 2B
 CMD1A Dilated cardiomyopathy 1A 115200 # 13176 c.949G > A Substitution
 p.Glu317Lys Substitution 2B CMD1A Dilated cardiomyopathy 1A 115200 # 13621 c.949G
 > A Substitution p.Glu317Lys Substitution 2B CMD1A Dilated cardiomyopathy 1A 115200 #
 13100 c.952G > A Substitution p.Ala318Thr Substitution 2B CMD1A Dilated
 cardiomyopathy 1A 115200 # 16956 c.952G > A Substitution p.Ala318Thr Substitution 2B
 CMD1A Dilated cardiomyopathy 1A 115200 # 11700 c.959delT Deletion p.Leu320fsX160
 Frame shift 2B CMD1A Dilated cardiomyopathy 1A 115200 # 9409 c.959delT Deletion
 p.Leu320fsX160 Frame shift 2B CMD1A Dilated cardiomyopathy 1A 115200 # 9391
 c.959delT Deletion p.Leu320fsX160 Frame shift 2B CMD1A Dilated cardiomyopathy 1A
 115200 # 14032 c.959delT Deletion p.Leu320fsX160 Frame shift 2B CMD1A Dilated
 cardiomyopathy 1A 115200 # 12389 c.961C > T Substitution p.Arg321X Substitution 2B
 CMD1A Dilated cardiomyopathy 1A 115200 # 14077 c.961C > T Substitution p.Arg321X
 Substitution 2B CMD1A Dilated cardiomyopathy 1A 115200 # 14093 c.961C > T
 Substitution p.Arg321X Substitution 2B CMD1A Dilated cardiomyopathy 1A 115200 # 17737
 c.961C > T Substitution p.Arg321X Substitution 2B CMD1A Dilated cardiomyopathy 1A
 115200 # 17738 c.961C > T Substitution p.Arg321X Substitution 2B CMD1A Dilated
 cardiomyopathy 1A 115200 # 9155 c.976T > A Substitution p.Ser326Thr Substitution 2B
 CMD1A Dilated cardiomyopathy 1A 115200 # 11760 c.[992G > A; =] + Substitution p.
 [Arg331Glu; =] + Substitution 2B CMD1A Dilated cardiomyopathy 1A 115200 # [=;
 1039G > A] [=; Glu347Lys] 11811 c.992G > C Substitution p.Arg331Pro Substitution 2B
 CMD1A Dilated cardiomyopathy 1A 115200 # 14075 c.992G > A Substitution
 p.Arg331Gln Substitution 2B CMD1A Dilated cardiomyopathy 1A 115200 # 17477 c.1003C
 > T Substitution p.Arg335Trp Substitution 2B CMD1A Dilated cardiomyopathy 1A 115200 #
 18486 c.1039G > A Substitution p.Glu347Lys Substitution 2B CMD1A Dilated

cardiomyopathy 1A 115200 # 9012 c.1046G > T Substitution p.Arg349Leu Substitution 2B
 CMD1A Dilated cardiomyopathy 1A 115200 # 13561 c.1048G > C Substitution
 p.Ala350Pro Substitution 2B CMD1A Dilated cardiomyopathy 1A 115200 # 14204 c.1057C
 > A Substitution p.Gln353Lys Substitution 2B CMD1A Dilated cardiomyopathy 1A 115200 #
 14072 c.1063C > T Substitution p.Gln355X Substitution 2B CMD1A Dilated
 cardiomyopathy 1A 115200 # 17475 c.1070A > C Substitution p.Asp357Ala Substitution 2B
 CMD1A Dilated cardiomyopathy 1A 115200 # 13116 c.1072G > T Substitution p.Glu358X
 Substitution 2B CMD1A Dilated cardiomyopathy 1A 115200 # 9152 c.1085_1085delT
 Deletion p.Leu363TrpfsX117 Frame shift 2B CMD1A Dilated cardiomyopathy 1A 115200 #
 17026 c.1085_1085delT Deletion p.Leu362TrpfsX117 Frame shift 2B CMD1A Dilated
 cardiomyopathy 1A 115200 # 16157 c.1102_1130dupGCCC Duplication p.Lys378ProfsX112
 Frame shift 2B CMD1A Dilated cardiomyopathy 1A 115200 # TGGACATGGAGATCC
 ACGCCTACCG 13102 c.1114delG Deletion p.Glu372ArgfsX107 Frame shift 2B CMD1A
 Dilated cardiomyopathy 1A 115200 # 8866 c.1130G > A Substitution p.Arg377His
 Substitution 2B CMD1A Dilated cardiomyopathy 1A 115200 # 8869 c.1130G > A
 Substitution p.Arg377His Substitution 2B CMD1A Dilated cardiomyopathy 1A 115200 # 8880
 c.1130G > A Substitution p.Arg377His Substitution 2B CMD1A Dilated cardiomyopathy
 1A 115200 # 9160 c.1130G > A Substitution p.Arg377His Substitution 2B CMD1A Dilated
 cardiomyopathy 1A 115200 # 9162 c.1130G > A Substitution p.Arg377His Substitution 2B
 CMD1A Dilated cardiomyopathy 1A 115200 # 12330 c.1130G > A Substitution
 p.Arg377His Substitution 2B CMD1A Dilated cardiomyopathy 1A 115200 # 13178 c.1130G
 > T Substitution p.Arg377Leu Substitution 2B CMD1A Dilated cardiomyopathy 1A 115200 #
 13623 c.1130G > A Substitution p.Arg377His Substitution 2B CMD1A Dilated
 cardiomyopathy 1A 115200 # 14034 c.1130G > A Substitution p.Arg377His Substitution 2B
 CMD1A Dilated cardiomyopathy 1A 115200 # 17664 c.1150G > T Substitution p.Glu384X
 Substitution 2B CMD1A Dilated cardiomyopathy 1A 115200 # 17482 C.1157 + 1G > T
 Substitution p.Arg386SerfsX21 Frame shift 2B CMD1A Dilated cardiomyopathy 1A 115200
 # 13104 c.1163G > A Substitution p.Arg388His Substitution Tail CMD1A Dilated
 cardiomyopathy 1A 115200 # 16958 c.1163G > A Substitution p.Arg388His Substitution Tail
 CMD1A Dilated cardiomyopathy 1A 115200 # 13106 c.1195C > T Substitution
 p.Arg399Cys Substitution Tail CMD1A Dilated cardiomyopathy 1A 115200 # 16960
 c.1195C > T Substitution p.Arg399Cys Substitution Tail CMD1A Dilated cardiomyopathy
 1A 115200 # 17782 c.1197_1240delTGGCC Deletion p.Gly400ArgfsX11 Frame shift Tail
 CMD1A Dilated cardiomyopathy 1A 115200 # GTGCTTCCTCTCACTC
 ATCCCAGACACAGGG TGGGGGCA 17789 c.1292C > G Substitution p.Ser431X
 Substitution Tail CMD1A Dilated cardiomyopathy 1A 115200 # 14082 c.1294C > T
 Substitution p.Gln432X Substitution Tail CMD1A Dilated cardiomyopathy 1A 115200 # 14084
 c.1294C > T Substitution p.Gln432X Substitution Tail CMD1A Dilated cardiomyopathy 1A
 115200 # 14085 c.1294C > T Substitution p.Gln432X Substitution Tail CMD1A Dilated
 cardiomyopathy 1A 115200 # 14086 c.1294C > T Substitution p.Gln432X Substitution Tail
 CMD1A Dilated cardiomyopathy 1A 115200 # 13108 c.1307_1308insGCAC Insertion
 p.Ser437HisfsX1 Frame shift Tail CMD1A Dilated cardiomyopathy 1A 115200 # 16962
 c.1307_1308insGCAC Insertion p.Ser437HisfsX1 Frame shift Tail CMD1A Dilated
 cardiomyopathy 1A 115200 # 14088 c.1318G > A Substitution p.Val440Met Substitution Tail
 CMD1A Dilated cardiomyopathy 1A 115200 # 14090 c.1318G > A Substitution
 p.Val440Met Substitution Tail CMD1A Dilated cardiomyopathy 1A 115200 # 14091
 c.1318G > A Substitution p.Val440Met Substitution Tail CMD1A Dilated cardiomyopathy
 1A 115200 # 11812 c.1370delA Deletion p.Lys457SerfsX21 Frame shift Tail CMD1A Dilated
 cardiomyopathy 1A 115200 # 12397 c.1380 + 1G > A Substitution p.? Unknown
 Unknown CMD1A Dilated cardiomyopathy 1A 115200 # 9413 c.1397_1397delA Deletion

p.Asn466IlefsX14 Frame shift Tail CMD1A Dilated cardiomyopathy 1A 115200 # 9161
 c.1397_1397delA Deletion p.Asn466IlefsX14 Frame shift Tail CMD1A Dilated
 cardiomyopathy 1A 115200 # 13110 c.1412G > A Substitution p.Arg471His Substitution Tail
 CMD1A Dilated cardiomyopathy 1A 115200 # 16973 c.1412G > A Substitution
 p.Arg471His Substitution Tail CMD1A Dilated cardiomyopathy 1A 115200 # 13112
 c.1424_1425insAGA Insertion p.Gly474_Asp475ins Insertion Tail CMD1A Dilated
 cardiomyopathy 1A 115200 # Glu 11508 c.1443C > G Substitution p.Tyr481X Substitution
 Tail CMD1A Dilated cardiomyopathy 1A 115200 # 17791 c.1443C > G Substitution
 p.Tyr481X Substitution Tail CMD1A Dilated cardiomyopathy 1A 115200 # 17480 c.1489
 - 1G > T Substitution p.Ile497_Glu536del Deletion Tail CMD1A Dilated
 cardiomyopathy 1A 115200 # 13180 c.1492T > A Substitution p.Trp498Arg Substitution Tail
 CMD1A Dilated cardiomyopathy 1A 115200 # 9153 c.1493_1493delG Deletion
 p.Ala499LeufsX47 Frame shift Tail CMD1A Dilated cardiomyopathy 1A 115200 # 12395
 c.1512_1513insAG Insertion p.Thr505ArgfsX44 Frame shift Tail CMD1A Dilated
 cardiomyopathy 1A 115200 # 17784 c.1526_1527insC Insertion p.Thr510TyrfsX42 Frame shift
 Tail CMD1A Dilated cardiomyopathy 1A 115200 # 17797 c.1526_1527insA Insertion
 p.Thr510TyrfsX42 Frame shift Tail CMD1A Dilated cardiomyopathy 1A 115200 # 17486
 c.1549C > T Substitution p.Gln517X Substitution Tail CMD1A Dilated cardiomyopathy 1A
 115200 # 17484 c.1560G > A Substitution p.Trp520X Substitution Tail CMD1A Dilated
 cardiomyopathy 1A 115200 # 13625 c.1567G > A Substitution p.Gly523Arg Substitution Tail
 CMD1A Dilated cardiomyopathy 1A 115200 # 11624 c.1579_1580insCTGC Insertion
 p.Arg527ProfsX26 Frame shift Tail CMD1A Dilated cardiomyopathy 1A 115200 # 17558
 c.1583C > T Substitution p.Thr528Met Substitution Tail CMD1A Dilated cardiomyopathy
 1A 115200 # 9307 c.1621C > T Substitution p.Arg541Cys Substitution Tail CMD1A Dilated
 cardiomyopathy 1A 115200 # 9283 c.1621C > A Substitution p.Arg541Ser Substitution Tail
 CMD1A Dilated cardiomyopathy 1A 115200 # 9151 c.1621C > A Substitution p.Arg541Ser
 Substitution Tail CMD1A Dilated cardiomyopathy 1A 115200 # 13592 c.1621C > T
 Substitution p.Arg541Cys Substitution Tail CMD1A Dilated cardiomyopathy 1A 115200 #
 14202 c.1621C > A Substitution p.Arg541Ser Substitution Tail CMD1A Dilated
 cardiomyopathy 1A 115200 # 16472 c.1621C > G Substitution p.Arg541Gly Substitution Tail
 CMD1A Dilated cardiomyopathy 1A 115200 # 17774 c.1621C > T Substitution
 p.Arg541Cys Substitution Tail CMD1A Dilated cardiomyopathy 1A 115200 # 17775
 c.1621C > T Substitution p.Arg541Cys Substitution Tail CMD1A Dilated cardiomyopathy
 1A 115200 # 13559 c.1622G > A Substitution p.Arg541His Substitution Tail CMD1A
 Dilated cardiomyopathy 1A 115200 # 9392 c.1718C > T Substitution p.Ser573Leu
 Substitution Tail CMD1A Dilated cardiomyopathy 1A 115200 # 13182 c.1718C > T
 Substitution p.Ser573Leu Substitution Tail CMD1A Dilated cardiomyopathy 1A 115200 #
 13183 c.1718C > T Substitution p.Ser573Leu Substitution Tail CMD1A Dilated
 cardiomyopathy 1A 115200 # 14038 c.1718C > T Substitution p.Ser573Leu Substitution Tail
 CMD1A Dilated cardiomyopathy 1A 115200 # 17667 c.1879C > T Substitution
 p.Arg624Cys Substitution Tail CMD1A Dilated cardiomyopathy 1A 115200 # 13565
 c.1904G > A Substitution p.Gly635Asp Substitution Tail CMD1A Dilated cardiomyopathy
 1A 115200 # 8833 c.1930C > T Substitution p.Arg644Cys Substitution Tail CMD1A Dilated
 cardiomyopathy 1A 115200 # 9018 c.1930C > T Substitution p.Arg644Cys Substitution Tail
 CMD1A Dilated cardiomyopathy 1A 115200 # 11500 c.1930C > T Substitution
 p.Arg644Cys Substitution Tail CMD1A Dilated cardiomyopathy 1A 115200 # 11501
 c.1930C > T Substitution p.Arg644Cys Substitution Tail CMD1A Dilated cardiomyopathy
 1A 115200 # 11502 c.1930C > A Substitution p.Arg644His Substitution Tail CMD1A
 Dilated cardiomyopathy 1A 115200 # 13185 c.1930C > T Substitution p.Arg644Cys
 Substitution Tail CMD1A Dilated cardiomyopathy 1A 115200 # 13337 c.1930C > T

substitution p.Arg644Cys Substitution Tail CMD1A Dilated cardiomyopathy 1A 115200 # 14080 c.1930C > T Substitution p.Arg644Cys Substitution Tail CMD1A Dilated cardiomyopathy 1A 115200 # 13114 c.1960C > T Substitution p.Arg654X Substitution Tail CMD1A Dilated cardiomyopathy 1A 115200 # 16964 c.1960C > T Substitution p.Arg654X Substitution Tail CMD1A Dilated cardiomyopathy 1A 115200 # 11788 c.? Unknown p.Tyr481X Substitution Tail DCM-CD Dilated cardiomyopathy with — — conduction system defects 17886 c.80C > T Substitution p.Thr27Ile Substitution Head DCM-CD Dilated cardiomyopathy with — — conduction system defects 13486 c.106C > T Substitution p.Gln36X Substitution 1A DCM-CD Dilated cardiomyopathy with — — conduction system defects 11792 c.158A > T Substitution p.Glu53Val Substitution 1A DCM-CD Dilated cardiomyopathy with — — conduction system defects 18301 c.175C > G Substitution p.Leu59Val Substitution 1A DCM-CD Dilated cardiomyopathy with — — conduction system defects 11791 c.481G > A Substitution p.Glu161Lys Substitution 1B DCM-CD Dilated cardiomyopathy with — — conduction system defects 13204 c.514 - 1G > A Substitution p.? Unknown Unknown DCM-CD Dilated cardiomyopathy with — — conduction system defects 11793 c.556G > A Substitution p.Glu186Lys Substitution 1B DCM-CD Dilated cardiomyopathy with — — conduction system defects 11789 c.568C > T Substitution p.Arg190Trp Substitution 1B DCM-CD Dilated cardiomyopathy with — — conduction system defects 11787 c.575A > G Substitution p.Asp192Gly Substitution 1B DCM-CD Dilated cardiomyopathy with — — conduction system defects 18470 c.683A > T Substitution p.Glu228Val Substitution L12 DCM-CD Dilated cardiomyopathy with — — conduction system defects 17868 c.871G > A Substitution p.Glu291Lys Substitution 2B DCM-CD Dilated cardiomyopathy with — — conduction system defects 17870 c.949G > A Substitution p.Glu317Lys Substitution 2B DCM-CD Dilated cardiomyopathy with — — conduction system defects 13078 c.1069G > C Substitution p.Asp357His Substitution 2B DCM-CD Dilated cardiomyopathy with — — conduction system defects 13208 C.1157 + 1G > A Substitution p.? Unknown Unknown DCM-CD Dilated cardiomyopathy with — — conduction system defects 18490 c.1412G > A Substitution p.Arg471His Substitution Tail DCM-CD Dilated cardiomyopathy with — — conduction system defects 11814 c.1526_1527insC Insertion p.Thr510TyrfsX42 Frame shift Tail DCM-CD Dilated cardiomyopathy with — — conduction system defects 9008 c.1621C > T Substitution p.Arg541Cys Substitution Tail DCM-CD Dilated cardiomyopathy with — — conduction system defects 17578 c.1711C > A Substitution p.= Silent Not DCM-CD Dilated cardiomyopathy with — — affected conduction system defects 17880 c.1930C > T Substitution p.Arg644Cys Substitution Tail DCM-CD Dilated cardiomyopathy with — — conduction system defects 18305 c.1774G > A Substitution p.Gly592Arg Substitution Tail DAPJ Distal acroosteolysis, — — poikiloderma and joint stiffness 13581 c.1930C > T Substitution p.Arg644Cys Substitution Tail — Distal motor neuropathy — — 17813 c.93G > C Substitution p.Glu31Asp Substitution Head — Dropped head syndrome — — 9020 c.94_96delAAG Deletion p.Lys32del Deletion 1A — Dropped head syndrome — — 13476 c.116A > G Substitution p.Asn39Ser Substitution 1A — Dropped head syndrome — — 13466 c.149G > C Substitution p.Arg50Pro Substitution 1A — Dropped head syndrome — — 13464 c.745C > T Substitution p.Arg249Trp Substitution 2A — Dropped head syndrome — — 13470 c.905T > C Substitution pleu302Pro Substitution 2B — Dropped head syndrome — — 13468 c.1072G > A Substitution p.Glu358Lys Substitution 2B — Dropped head syndrome — — 13482 c.1072G > A Substitution p.Glu358Lys Substitution 2B — Dropped head syndrome — — 13484 c.1072G > A Substitution p.Glu358Lys Substitution 2B — Dropped head syndrome — — 13474 c.1358G > C Substitution p.Arg453Pro Substitution Tail — Dropped head syndrome — — 13472 c.1364G > C Substitution p.Arg455Pro Substitution Tail — Dropped head syndrome — — 13478 c.1366A > G

Substitution p.Asn456Asp Substitution Tail — Dropped head syndrome — 13480 c.1381-2A > G Substitution p.? Unknown Unknown — Dropped head syndrome — 11652 c.-Deletion p.Met1_Pro4del Deletion Head EDMD2 Emery-Dreifuss muscular 181350 # 3_12delGCCATGGAGA dystrophy, autosomal CCCC dominant 8743 c.16C > T Substitution p.Gln6X Substitution Head EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 8796 c.16C > T Substitution p.Gln6X Substitution Head EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 11746 c.31delC Deletion p.Arg11AlafsX85 Frame shift Head EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 9262 c.73C > G Substitution p.Arg25Gly Substitution Head EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 9376 c.74G > C Substitution p.Arg25Pro Substitution Head EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 9177 c.94_96delAAG Deletion p.Lys32del Deletion 1A EDMD2 Emery-Dreifuss muscular 181350 # EDMD2 dystrophy, autosomal dominant 8998 c.94_96delAAG Deletion p.Lys32del Deletion 1A EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 9014 c.94_96delAAG Deletion p.Lys32del Deletion 1A EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 13594 c.98A > G Substitution p.Glu33Gly Substitution 1A EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 11747 c.99G > C Substitution p.Glu33Asp Substitution 1A EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 8999 c.103C > G Substitution p.Leu35Val Substitution 1A EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 11748 c.116A > G Substitution p.Asn39Ser Substitution 1A EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 16329 c.116A > G Substitution p.Asn39Ser Substitution 1A EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 18174 c.116A > G Substitution p.Asn39Ser Substitution 1A EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 9377 c.127G > A Substitution p.Ala43Thr Substitution 1A EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 8797 c.134A > G Substitution p.Tyr45Cys Substitution 1A EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 16331 c.134A > G Substitution p.Tyr45Cys Substitution 1A EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 13605 c.136A > G Substitution p.Ile46Val Substitution 1A EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 13545 c.139G > C Substitution p.Asp47His Substitution 1A EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 9378 c.148C > A Substitution p.Arg50Ser Substitution 1A EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 8798 c.149G > C Substitution p.Arg50Pro Substitution 1A EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 8799 c.188T > G Substitution p.Ile63Ser Substitution 1A EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 9175 c.188T > A Substitution p.Ile63Asn Substitution 1A EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 11633 c.188T > A Substitution p.Ile63Asn Substitution 1A EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 13192 c.203_208delAGGTGG Deletion p.Glu68_Val69del Deletion 1A EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 12477 c.265C > T Substitution p.Arg89Cys Substitution 1B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 13596 c.265C > T Substitution p.Arg89Cys Substitution 1B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 17193 c.266G > T Substitution p.Arg89Leu Substitution 1B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 9410 c.334_336delGAG Deletion p.Glu112del Deletion 1B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 9000 c.334_336delGAG Deletion p.Glu112del Deletion 1B EDMD2 Emery-Dreifuss muscular 181350 # EDMD2 dystrophy, autosomal dominant 16291 c.357C > T Substitution p.= Silent Not EDMD2 Emery-Dreifuss

muscular 181350 # affected dystrophy, autosomal dominant 17182 c.367_369delAAG Deletion
 p.Lys123del Deletion 1B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal
 dominant 9379 c.398G > C Substitution p.Arg133Pro Substitution 1B EDMD2 Emery-
 Dreifuss muscular 181350 # dystrophy, autosomal dominant 11632 c.419T > C Substitution
 p.Leu140Pro Substitution 1B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy,
 autosomal dominant 9015 c.428C > T Substitution p.Ser143Phe Substitution 1B EDMD2
 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 12609 c.428C > T
 Substitution p.Ser143Phe Substitution 1B EDMD2 Emery-Dreifuss muscular 181350 #
 dystrophy, autosomal dominant 8795 c.448A > C Substitution p.Thr150Pro Substitution 1B
 EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 16334 c.448A
 > C Substitution p.Thr150Pro Substitution 1B EDMD2 Emery-Dreifuss muscular 181350 #
 dystrophy, autosomal dominant 13453 c.485T > C Substitution p.Leu162Pro Substitution 1B
 EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 16396
 c.566_567delGGinsCC Indel p.Arg189Pro Substitution 1B EDMD2 Emery-Dreifuss muscular
 181350 # dystrophy, autosomal dominant 16299 c.568_570dupCGG Duplication p.Arg190dup
 Duplication 1B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant
 9380 c.588_596delGCTGCA Deletion p.Arg196_Thr199delin Indel 1B EDMD2 Emery-
 Dreifuss muscular 181350 # GAC sSer dystrophy, autosomal dominant 16301 c.618C > G
 Substitution p.Phe206Leu Substitution 1B EDMD2 Emery-Dreifuss muscular 181350 #
 dystrophy, autosomal dominant 12382 c.625delA Deletion p.Asn209ThrfsX271 Frame shift 1B
 EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 8800 c.665A
 > C Substitution p.His222Pro Substitution L12 EDMD2 Emery-Dreifuss muscular 181350 #
 dystrophy, autosomal dominant 13547 c.694G > C Substitution p.Gly232Arg Substitution
 L12 EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 8801
 c.695G > A Substitution p.Gly232Glu Substitution L12 EDMD2 Emery-Dreifuss muscular
 181350 # dystrophy, autosomal dominant 12485 c.695G > A Substitution p.Gly232Glu
 Substitution L12 EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant
 9001 c.743T > C Substitution p.Leu248Pro Substitution 2A EDMD2 Emery-Dreifuss
 muscular 181350 # dystrophy, autosomal dominant 16336 c.745C > T Substitution
 p.Arg249Trp Substitution 2A EDMD2 Emery-Dreifuss muscular 181350 # dystrophy,
 autosomal dominant 16337 c.745C > T Substitution p.Arg249Trp Substitution 2A EDMD2
 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 16338 c.745C > T
 Substitution p.Arg249Trp Substitution 2A EDMD2 Emery-Dreifuss muscular 181350 #
 dystrophy, autosomal dominant 16342 c.[745C > T; 1930C > T] Substitution p.
 [Arg249Trp; Substitution 2A, Tail EDMD2 Emery-Dreifuss muscular 181350 # Arg644Cys]
 dystrophy, autosomal dominant 8783 c.746G > A Substitution p.Arg249Gln Substitution 2A
 EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 11380 c.746G
 > A Substitution p.Arg249Gln Substitution 2A EDMD2 Emery-Dreifuss muscular 181350 #
 dystrophy, autosomal dominant 11381 c.746G > A Substitution p.Arg249Gln Substitution 2A
 EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 8802 c.746G
 > A Substitution p.Arg249Gln Substitution 2A EDMD2 Emery-Dreifuss muscular 181350 #
 dystrophy, autosomal dominant 9002 c.746G > A Substitution p.Arg249Gln Substitution 2A
 EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 9023 c.746G
 > A Substitution p.Arg249Gln Substitution 2A EDMD2 Emery-Dreifuss muscular 181350 #
 dystrophy, autosomal dominant 11631 c.746G > A Substitution p.Arg249Gln Substitution 2A
 EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 11749 c.746G
 > A Substitution p.Arg249Gln Substitution 2A EDMD2 Emery-Dreifuss muscular 181350 #
 dystrophy, autosomal dominant 11750 c.746G > A Substitution p.Arg249Gln Substitution 2A
 EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 13552 c.746G
 > A Substitution p.Arg249Gln Substitution 2A EDMD2 Emery-Dreifuss muscular 181350 #

dystrophy, autosomal dominant 16344 c.746G > A Substitution p.Arg249Gln Substitution 2A EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 16345 c.746G
 > A Substitution p.Arg249Gln Substitution 2A EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 18176 c.746G > A Substitution p.Arg249Gln Substitution 2A EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 11751 c.775T
 > G Substitution p.Tyr259Asp Substitution L2 EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 9640 c.781_783delAAG Deletion p.Lys261del Deletion L2 EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 9411
 c.781_783delAAG Deletion p.Lys261del Deletion L2 EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 9381 c.781_783delAAG Deletion p.Lys261del Deletion L2 EDMD2 Emery-Dreifuss muscular 181350 # EDMD2 dystrophy, autosomal dominant 13584
 c.781_783delAAGinsG Indel p.Lys261delinsValGlu Indel L2 EDMD2 Emery-Dreifuss muscular 181350 # TGGAGCAGTATAAGA GlnTyrLysLys dystrophy, autosomal AA dominant 16149 c.788T > C Substitution p.Leu263Pro Substitution L2 EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 17403 c.788T > C Substitution p.Leu263Pro Substitution L2 EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 17751 c.788T > C Substitution p.Leu263Pro Substitution L2 EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 17405 c.799T > C Substitution p.Tyr267His Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 9003 c.800A > G Substitution p.Tyr267Cys Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 11809 c.800A
 > G Substitution p.Tyr267Cys Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 13173 c.800A > G Substitution p.Tyr267Cys Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 16303 c.802T
 > C Substitution p.Ser268Pro Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 16305 c.810G > A Substitution p.= Silent Not EDMD2 Emery-Dreifuss muscular 181350 # affected dystrophy, autosomal dominant 16307 c.810G
 > A Substitution p.= Silent Not EDMD2 Emery-Dreifuss muscular 181350 # affected dystrophy, autosomal dominant 16309 c.810 + 1G > A Substitution p.? Unknown Unknown EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 16347 c.812T > C Substitution p.Leu271Pro Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 18186 c.832G > C Substitution p.Ala278Pro Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 8803 c.881A > C Substitution p.Gln294Pro Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 16349 c.881A > C Substitution p.Gln294Pro Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 16311 c.883T > C Substitution p.Ser295Pro Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 8863 c.907T > C Substitution p.Ser303Pro Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 16351 c.907T > C Substitution p.Ser303Pro Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 8786 c.1007G
 > A Substitution p.Arg336Gln Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 16314 c.1064_1066delAGC Deletion p.Gln355del Deletion 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 8804 c.1072G
 > A Substitution p.Glu358Lys Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 11388 c.1072G > A Substitution p.Glu358Lys Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 11816 c.1072G > A Substitution p.Glu358Lys Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 9167 c.1072G > A Substitution p.Glu358Lys Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant

11478 c.1072G > A Substitution p.Glu358Lys Substitution 2B EDMD2 Emery-Dreifuss
 muscular 181350 # dystrophy, autosomal dominant 11479 c.1072G > A Substitution
 p.Glu358Lys Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy,
 autosomal dominant 11480 c.1072G > A Substitution p.Glu358Lys Substitution 2B EDMD2
 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 11752 c.1072G > A
 Substitution p.Glu358Lys Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 #
 dystrophy, autosomal dominant 12479 c.1072G > A Substitution p.Glu358Lys Substitution
 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 12483
 c.1072G > A Substitution p.Glu358Lys Substitution 2B EDMD2 Emery-Dreifuss muscular
 181350 # dystrophy, autosomal dominant 13598 c.1072G > A Substitution p.Glu358Lys
 Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant
 13603 c.1072G > A Substitution p.Glu358Lys Substitution 2B EDMD2 Emery-Dreifuss
 muscular 181350 # dystrophy, autosomal dominant 16353 c.1072G > A Substitution
 p.Glu358Lys Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy,
 autosomal dominant 16354 c.1072G > A Substitution p.Glu358Lys Substitution 2B EDMD2
 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 16355 c.1072G > A
 Substitution p.Glu358Lys Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 #
 dystrophy, autosomal dominant 16316 c.1081G > A Substitution p.Glu361Lys Substitution
 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 8806
 c.1112T > A Substitution p.Met371Lys Substitution 2B EDMD2 Emery-Dreifuss muscular
 181350 # dystrophy, autosomal dominant 18184 c.1124C > G Substitution p.Ala375Gly
 Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant
 11634 c.1130G > T Substitution p.Arg377Leu Substitution 2B EDMD2 Emery-Dreifuss
 muscular 181350 # dystrophy, autosomal dominant 11693 c.1130G > A Substitution
 p.Arg377His Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy,
 autosomal dominant 12405 c.1130G > A Substitution p.Arg377His Substitution 2B EDMD2
 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 18223 c.1130G > A
 Substitution p.Arg377His Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 #
 dystrophy, autosomal dominant 12474 c.1142A > C Substitution p.Glu381Ala Substitution
 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 11629
 c.1157G > A Substitution p.Arg386Lys Substitution 2B EDMD2 Emery-Dreifuss muscular
 181350 # dystrophy, autosomal dominant 13653 c.1157G > T Substitution p.Arg386Met
 Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant
 16357 c.1157G > A Substitution p.Arg386Lys Substitution 2B EDMD2 Emery-Dreifuss
 muscular 181350 # dystrophy, autosomal dominant 16358 c.1157G > A Substitution
 p.Arg386Lys Substitution 2B EDMD2 Emery-Dreifuss muscular 181350 # dystrophy,
 autosomal dominant 16319 c.1158-2A?G Substitution p.? Unknown Unknown EDMD2 Emery-
 Dreifuss muscular 181350 # dystrophy, autosomal dominant 11817 c.1162C > T
 Substitution p.Arg388Cys Substitution Tail EDMD2 Emery-Dreifuss muscular 181350 #
 dystrophy, autosomal dominant 9281 c.1187A > G Substitution p.Gln396Arg Substitution
 Tail EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 8862
 c.1201C > T Substitution p.Arg401Cys Substitution Tail EDMD2 Emery-Dreifuss muscular
 181350 # dystrophy, autosomal dominant 9176 c.1201C > T Substitution p.Arg401Cys
 Substitution Tail EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant
 9302 c.1201C > T Substitution p.Arg401Cys Substitution Tail EDMD2 Emery-Dreifuss
 muscular 181350 # dystrophy, autosomal dominant 9004 c.1337A > T Substitution
 p.Asp446Val Substitution Tail EDMD2 Emery-Dreifuss muscular 181350 # dystrophy,
 autosomal dominant 16321 c.1346G > A Substitution p.Gly449Asp Substitution Tail EDMD2
 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 18192 c.1346G?T
 Substitution p.Gly449Val Substitution Tail EDMD2 Emery-Dreifuss muscular 181350 #

[illegible]

Substitution dystrophy, autosomal dominant 16325 c.1399T > C Substitution p.Trp467Arg
 Substitution Tail EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant
 8791 c.1406T > C Substitution p.Ile469Thr Substitution Tail EDMD2 Emery-Dreifuss
 muscular 181350 # dystrophy, autosomal dominant 18194 c.1466T > C Substitution
 p.Leu489Pro Substitution Tail EDMD2 Emery-Dreifuss muscular 181350 # dystrophy,
 autosomal dominant 16327 c.1488 + 1G > A Substitution p.? Unknown Unknown EDMD2
 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 13180 c.1492T > A
 Substitution p.Trp498Arg Substitution Tail EDMD2 Emery-Dreifuss muscular 181350 #
 dystrophy, autosomal dominant 16375 c.1526dupC Duplication p.Thr510TyrfsX42 Frame shift
 Tail EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 18188
 c.1540T > A Substitution p.Trp514Arg Substitution Tail EDMD2 Emery-Dreifuss muscular
 181350 # dystrophy, autosomal dominant 18190 c.1540T > A Substitution p.Trp514Arg
 Substitution Tail EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant
 12434 c.1558T > G Substitution p.Trp520Gly Substitution Tail EDMD2 Emery-Dreifuss
 muscular 181350 # dystrophy, autosomal dominant 18179 c.1558T > C Substitution
 p.Trp520Arg Substitution Tail EDMD2 Emery-Dreifuss muscular 181350 # dystrophy,
 autosomal dominant 8811 c.1559G > C Substitution p.Trp520Ser Substitution Tail EDMD2
 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 8745 c.1580G > C
 Substitution p.Arg527Pro Substitution Tail EDMD2 Emery-Dreifuss muscular 181350 #
 dystrophy, autosomal dominant 11344 c.1580G > C Substitution p.Arg527Pro Substitution
 Tail EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 8792
 c.1580G > C Substitution p.Arg527Pro Substitution Tail EDMD2 Emery-Dreifuss muscular
 181350 # dystrophy, autosomal dominant 8812 c.1580G > C Substitution p.Arg527Pro
 Substitution Tail EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant
 11391 c.1580G > C Substitution p.Arg527Pro Substitution Tail EDMD2 Emery-Dreifuss
 muscular 181350 # dystrophy, autosomal dominant 8838 c.1580G > C Substitution
 p.Arg527Pro Substitution Tail EDMD2 Emery-Dreifuss muscular 181350 # dystrophy,
 autosomal dominant 8859 c.1580G > C Substitution p.Arg527Pro Substitution Tail EDMD2
 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 11423 c.1580G > C
 Substitution p.Arg527Pro Substitution Tail EDMD2 Emery-Dreifuss muscular 181350 #
 dystrophy, autosomal dominant 11630 c.1580G > C Substitution p.Arg527Pro Substitution
 Tail EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 11754
 c.1580G > C Substitution p.Arg527Pro Substitution Tail EDMD2 Emery-Dreifuss muscular
 181350 # dystrophy, autosomal dominant 16377 c.1580G > C Substitution p.Arg527Pro
 Substitution Tail EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant
 18182 c.1580G > C Substitution p.Arg527Pro Substitution Tail EDMD2 Emery-Dreifuss
 muscular 181350 # dystrophy, autosomal dominant 8793 c.1583C > A Substitution
 p.Thr528Lys Substitution Tail EDMD2 Emery-Dreifuss muscular 181350 # dystrophy,
 autosomal dominant 8814 c.1583C > A Substitution p.Thr528Lys Substitution Tail EDMD2
 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 11392 c.1583C > A
 Substitution p.Thr528Lys Substitution Tail EDMD2 Emery-Dreifuss muscular 181350 #
 dystrophy, autosomal dominant 9006 c.1583C > G Substitution p.Thr528Arg Substitution Tail
 EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 16379 c.1583C
 > A Substitution p.Thr528Lys Substitution Tail EDMD2 Emery-Dreifuss muscular 181350 #
 dystrophy, autosomal dominant 16381 c.1583C > G Substitution p.Thr528Arg Substitution
 Tail EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 16382
 c.1583C > G Substitution p.Thr528Arg Substitution Tail EDMD2 Emery-Dreifuss muscular
 181350 # dystrophy, autosomal dominant 16407 c.1583C > G Substitution p.Thr528Arg
 Substitution Tail EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant
 18196 c.1583C > G Substitution p.Thr528Arg Substitution Tail EDMD2 Emery-Dreifuss

muscular 181350 # dystrophy, autosomal dominant 18297 c.1588C > T Substitution
 p.Leu530Phe Substitution Tail EDMD2 Emery-Dreifuss muscular 181350 # dystrophy,
 autosomal dominant 8747 c.1589T > C Substitution p.Leu530Pro Substitution Tail EDMD2
 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 16384 c.1621C > A
 Substitution p.Arg541Ser Substitution Tail EDMD2 Emery-Dreifuss muscular 181350 #
 dystrophy, autosomal dominant 9007 c.1622G > A Substitution p.Arg541His Substitution Tail
 EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 16387 c.1622G
 > C Substitution p.Arg541Pro Substitution Tail EDMD2 Emery-Dreifuss muscular 181350 #
 dystrophy, autosomal dominant 13588 c.1633C > T Substitution p.Arg545Cys Substitution
 Tail EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 16389
 c.1804G > A Substitution p.Arg541Pro Substitution Tail EDMD2 Emery-Dreifuss muscular
 181350 # dystrophy, autosomal dominant 9383 c.1871G > A Substitution p.Arg624His
 Substitution Tail EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant
 9154 c.1930C > T Substitution p.Arg644Cys Substitution Tail EDMD2 Emery-Dreifuss
 muscular 181350 # dystrophy, autosomal dominant 13570 c.1930C > T Substitution
 p.Arg644Cys Substitution Tail EDMD2 Emery-Dreifuss muscular 181350 # dystrophy,
 autosomal dominant 13571 c.1930C > T Substitution p.Arg644Cys Substitution Tail EDMD2
 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 13572 c.1930C > T
 Substitution p.Arg644Cys Substitution Tail EDMD2 Emery-Dreifuss muscular 181350 #
 dystrophy, autosomal dominant 13573 c.1930C > T Substitution p.Arg644Cys Substitution
 Tail EDMD2 Emery-Dreifuss muscular 181350 # dystrophy, autosomal dominant 16392
 c.1930C > T Substitution p.Arg644Cys Substitution Tail EDMD2 Emery-Dreifuss muscular
 181350 # dystrophy, autosomal dominant 9523 c.664C > T Substitution p.His222Tyr
 Substitution L12 EDMD3 Emery-Dreifuss muscular 604929 # dystrophy, autosomal recessive
 17407 c.674G > A Substitution p.Arg225Gln Substitution L12 EDMD3 Emery-Dreifuss
 muscular 604929 # dystrophy, autosomal recessive 17573 c.1445G > A Substitution
 p.Arg482Gln Substitution Tail EDMD3 Emery-Dreifuss muscular 604929 # dystrophy,
 autosomal recessive 13319 c.1580G > C Substitution p.Arg527Pro Substitution Tail EDMD3
 Emery-Dreifuss muscular 604929 # dystrophy, autosomal recessive 18345 c.? Unknown
 p.Arg482Trp Substitution Tail FPLD2 Familial partial lipodystrophy 151660 # (Dunnigan
 type) 8847 c.82C > T Substitution p.Arg28Trp Substitution Head FPLD2 Familial partial
 lipodystrophy 151660 # (Dunnigan type) 18291 c.139G > A Substitution p.Asp47Asn
 Substitution 1A FPLD2 Familial partial lipodystrophy 151660 # (Dunnigan type) 8861
 c.178C > G Substitution p.Arg60Gly Substitution 1A FPLD2 Familial partial lipodystrophy
 151660 # (Dunnigan type) 14260 c.178C > G Substitution p.Arg60Gly Substitution 1A
 FPLD2 Familial partial lipodystrophy 151660 # (Dunnigan type) 8848 c.184C > G
 Substitution p.Arg62Gly Substitution 1A FPLD2 Familial partial lipodystrophy 151660 #
 (Dunnigan type) 14256 c.184C > G Substitution p.Arg62Gly Substitution 1A FPLD2
 Familial partial lipodystrophy 151660 # (Dunnigan type) 12100 c.398G > T Substitution
 p.Arg133Leu Substitution 1B FPLD2 Familial partial lipodystrophy 151660 # (Dunnigan
 type) 14258 c.575A > T Substitution p.Asp192Val Substitution 1B FPLD2 Familial partial
 lipodystrophy 151660 # (Dunnigan type) 17624 c.667G > A Substitution p.Glu223Lys
 Substitution L12 FPLD2 Familial partial lipodystrophy 151660 # (Dunnigan type) 11775
 c.688G > A Substitution p.Asp230Asn Substitution L12 FPLD2 Familial partial
 lipodystrophy 151660 # (Dunnigan type) 11776 c.1195C > T Substitution p.Arg399Cys
 Substitution Tail FPLD2 Familial partial lipodystrophy 151660 # (Dunnigan type) 17066
 c.1232G > A Substitution p.Gly411Asp Substitution Tail FPLD2 Familial partial
 lipodystrophy 151660 # (Dunnigan type) 12096 c.1315C > T Substitution p.Arg439Cys
 Substitution Tail FPLD2 Familial partial lipodystrophy 151660 # (Dunnigan type) 17208
 c.1315C > T Substitution p.Arg439Cys Substitution Tail FPLD2 Familial partial

[illegible]

[illegible]

c.1445G > A Substitution p.Arg482Gln Substitution Tail FPLD2 Familial partial
lipodystrophy 151660 # (Dunnigan type) 9305 c.1445G > T Substitution p.Arg482Leu
Substitution Tail FPLD2 Familial partial lipodystrophy 151660 # (Dunnigan type) 11663
c.1445G > A Substitution p.Arg482Gln Substitution Tail FPLD2 Familial partial
lipodystrophy 151660 # (Dunnigan type) 11667 c.1445G > A Substitution p.Arg482Gln
Substitution Tail FPLD2 Familial partial lipodystrophy 151660 # (Dunnigan type) 12098
c.1445G > A Substitution p.Arg482Gln Substitution Tail FPLD2 Familial partial
lipodystrophy 151660 # (Dunnigan type) 12387 c.1445G > A Substitution p.Arg482Gln
Substitution Unknown FPLD2 Familial partial lipodystrophy 151660 # (Dunnigan type)
12418 c.1445G > A Substitution p.Arg482Gln Substitution Unknown FPLD2 Familial
partial lipodystrophy 151660 # (Dunnigan type) 12423 c.1445G > A Substitution
p.Arg482Gln Substitution Tail FPLD2 Familial partial lipodystrophy 151660 # (Dunnigan
type) 13150 c.1445G > A Substitution p.Arg482Gln Substitution Tail FPLD2 Familial
partial lipodystrophy 151660 # (Dunnigan type) 13367 c.1445G > A Substitution
p.Arg482Gln Substitution Tail FPLD2 Familial partial lipodystrophy 151660 # (Dunnigan
type) 16285 c.1445G > A Substitution p.Arg482Gln Substitution Tail FPLD2 Familial
partial lipodystrophy 151660 # (Dunnigan type) 16436 c.1445G > A Substitution
p.Arg482Gln Substitution Tail FPLD2 Familial partial lipodystrophy 151660 # (Dunnigan
type) 16437 c.1445G > A Substitution p.Arg482Gln Substitution Tail FPLD2 Familial
partial lipodystrophy 151660 # (Dunnigan type) 16438 c.1445G > A Substitution
p.Arg482Gln Substitution Tail FPLD2 Familial partial lipodystrophy 151660 # (Dunnigan
type) 16439 c.1445G > A Substitution p.Arg482Gln Substitution Tail FPLD2 Familial
partial lipodystrophy 151660 # (Dunnigan type) 16440 c.1445G > A Substitution
p.Arg482Gln Substitution Tail FPLD2 Familial partial lipodystrophy 151660 # (Dunnigan
type) 18138 c.1445G > A Substitution p.Arg482Gln Substitution Tail FPLD2 Familial
partial lipodystrophy 151660 # (Dunnigan type) 8765 c.1458G > C Substitution
p.Lys486Asn Substitution Tail FPLD2 Familial partial lipodystrophy 151660 # (Dunnigan
type) 11366 c.1458G > C Substitution p.Lys486Asn Substitution Tail FPLD2 Familial
partial lipodystrophy 151660 # (Dunnigan type) 8767 c.1458G > T Substitution
p.Lys486Asn Substitution Tail FPLD2 Familial partial lipodystrophy 151660 # (Dunnigan
type) 9319 c.1488 + 5G > C Substitution p.Ile497_Met664delins Indel Tail FPLD2
Familial partial lipodystrophy 151660 # ValThrGlyArgAlaLeu (Dunnigan type)
GlyThrLeuGlyArgPro TrpValAlaMetGlyAla LeuGlyX 17455 c.1683G > C Substitution p.=
Silent Not FPLD2 Familial partial lipodystrophy 151660 # affected (Dunnigan type) 11777
c.1718C > T Substitution p.Ser573Leu Substitution Tail FPLD2 Familial partial
lipodystrophy 151660 # (Dunnigan type) 8780 c.1745G > A Substitution p.Arg582His
Substitution Tail FPLD2 Familial partial lipodystrophy 151660 # (Dunnigan type) 16429
c.1745G > A Substitution p.Arg582His Substitution Tail FPLD2 Familial partial
lipodystrophy 151660 # (Dunnigan type) 16430 c.1745G > A Substitution p.Arg582His
Substitution Tail FPLD2 Familial partial lipodystrophy 151660 # (Dunnigan type) 16431
c.1745G > A Substitution p.Arg582His Substitution Tail FPLD2 Familial partial
lipodystrophy 151660 # (Dunnigan type) 9452 c.1751G > A Substitution p.Arg584His
Substitution Tail FPLD2 Familial partial lipodystrophy 151660 # (Dunnigan type) 11545
c.1751G > A Substitution p.Arg584His Substitution Tail FPLD2 Familial partial
lipodystrophy 151660 # (Dunnigan type) 12429 c.1772G > T Substitution p.Arg156Cys
Substitution 1B FPLD2 Familial partial lipodystrophy 151660 # (Dunnigan type) 17068
c.1892G > A Substitution p.Gly631Asp Substitution Tail FPLD2 Familial partial
lipodystrophy 151660 # (Dunnigan type) 13575 c.1930C > T Substitution p.Arg644Cys
Substitution Tail FPLD2 Familial partial lipodystrophy 151660 # (Dunnigan type) 13576
c.1930C > T Substitution p.Arg644Cys Substitution Tail FPLD2 Familial partial

lipodystrophy 151660 # (Dunnigan type) 13579 c.1930C > T Substitution p.Arg644Cys
 Substitution Tail FPLD2 Familial partial lipodystrophy 151660 # (Dunnigan type) 17461
 c.1930C > T Substitution p.Arg644Cys Substitution Tail FPLD2 Familial partial
 lipodystrophy 151660 # (Dunnigan type) 12601 c.1411C > G Substitution p.Arg471Gly
 Substitution Tail FPLD1 Familial partial lipodystrophy 608600 % (Köbberling) 11482 c.
 [1583C > T; =] + Substitution p.[Thr528Met; =] + Substitution Tail FPLD1 Familial
 partial lipodystrophy 608600 % [=; 1748C > T] [=; Ser583Leu (Köbberling) 9183
 c.1748C > T Substitution p.Ser583Leu Substitution Tail FPLD1 Familial partial
 lipodystrophy 608600 % (Köbberling) 11481 c.1748C > T Substitution p.Ser583Leu
 Substitution Tail FPLD1 Familial partial lipodystrophy 608600 % (Köbberling) 13490 c.29C
 > T Substitution p.Thr10Ile Substitution Head — Generalized lipoatrophy — syndrome
 8867 c.398G > T Substitution p.Arg133Leu Substitution 1B — Generalized lipoatrophy —
 — syndrome 17762 c.1609-12T > G Substitution p.Glu536fsX14 Frame shift Tail HSS
 Hallermann-Streiff syndrome 234100 % 17260 c.1930C > T Substitution p.Arg644Cys
 Substitution Tail HSS Hallermann-Streiff syndrome 234100 % 13153 c.1609-12T > G
 Substitution p.Glu536fsX14 Frame shift Tail — Heart-hand syndrome, 610140 % Slovenian
 Type 14097 c.11C > G Substitution p.Pro4Arg Substitution Head HGPS Hutchinson-Gilford
 progeria 176670 # syndrome 14112 c.11C > G Substitution p.Pro4Arg Substitution Head HGPS
 Hutchinson-Gilford progeria 176670 # syndrome 9172 c.29C > T Substitution p.Thr10Ile
 Substitution Head HGPS Hutchinson-Gilford progeria 176670 # syndrome 14108 c.29C > T
 Substitution p.Thr10Ile Substitution Head HGPS Hutchinson-Gilford progeria 176670 #
 syndrome 14110 c.29C > T Substitution p.Thr10Ile Substitution Head HGPS Hutchinson-
 Gilford progeria 176670 # syndrome 14101 c.331G > T Substitution p.Glu111Lys
 Substitution 1B HGPS Hutchinson-Gilford progeria 176670 # syndrome 14095 c.406G > C
 Substitution p.Asp136His Substitution 1B HGPS Hutchinson-Gilford progeria 176670 #
 syndrome 17070 c.412G > A Substitution p.Glu138Lys Substitution 1B HGPS Hutchinson-
 Gilford progeria 176670 # syndrome 9016 c.428C > T Substitution p.Ser143Phe Substitution
 1B HGPS Hutchinson-Gilford progeria 176670 # syndrome 12609 c.428C > T Substitution
 p.Ser143Phe Substitution 1B HGPS Hutchinson-Gilford progeria 176670 # syndrome 8871
 c.433G > A Substitution p.Glu145Lys Substitution 1B HGPS Hutchinson-Gilford progeria
 176670 # syndrome 14106 c.475G > A Substitution p.Glu159Lys Substitution 1B HGPS
 Hutchinson-Gilford progeria 176670 # syndrome 17626 c.899A > G Substitution
 p.Asp300Gly Substitution 2B HGPS Hutchinson-Gilford progeria 176670 # syndrome 17903
 c.917T > G Substitution pi Substitution 2B HGPS Hutchinson-Gilford progeria 176670 #
 syndrome 14114 c.1303C > T Substitution p.Arg435Cys Substitution Tail HGPS Hutchinson-
 Gilford progeria 176670 # syndrome 9393 c.1411C > T Substitution p.Arg471Cys
 Substitution Tail HGPS Hutchinson-Gilford progeria 176670 # syndrome 12615 c.1411C > T
 Substitution p.Arg471Cys Substitution Tail HGPS Hutchinson-Gilford progeria 176670 #
 syndrome 9394 c.1579C?T Substitution p.Arg527Cys Substitution Tail HGPS Hutchinson-
 Gilford progeria 176670 # syndrome 13119 c.1579C Substitution p.Arg527Cys Substitution Tail
 HGPS Hutchinson-Gilford progeria 176670 # syndrome 13651 c.1579C > T Substitution
 p.Arg527Cys Substitution Tail HGPS Hutchinson-Gilford progeria 176670 # syndrome 11671 c.
 [1583C > T; =] + Substitution p.[Thr528Met; =] + Substitution Tail HGPS Hutchinson-
 Gilford progeria 176670 # [=; 1619T > C] [=; Met540Thr] syndrome 9013 c.1626G >
 C Substitution p.Lys542Asn Substitution Tail HGPS Hutchinson-Gilford progeria 176670 #
 syndrome 9173 c.1733A > T Substitution p.Glu578Val Substitution Tail HGPS Hutchinson-
 Gilford progeria 176670 # syndrome 14099 c.1762-T > C Substitution p.Cys588Arg
 Substitution Tail HGPS Hutchinson-Gilford progeria 176670 # syndrome 14104 c.1762-T >
 C Substitution p.Cys588Arg Substitution Tail HGPS Hutchinson-Gilford progeria 176670 #
 syndrome 11785 c.1821G > A Substitution p.= Silent Not HGPS Hutchinson-Gilford progeria

[illegible]

Tail HGPS Hutchinson-Gilford progeria 176670 # Val607_Gln656del] Deletion syndrome 17393
 c.1824C > T Substitution p.= Silent Not HGPS Hutchinson-Gilford progeria 176670 #
 affected syndrome 17395 c.1824C > T Substitution p.= Silent Not HGPS Hutchinson-Gilford
 progeria 176670 # affected syndrome 17564 c.1824C > T Substitution p.= Silent Not HGPS
 Hutchinson-Gilford progeria 176670 # affected syndrome 17729 c.1824C > T Substitution
 p.= Silent Not HGPS Hutchinson-Gilford progeria 176670 # affected syndrome 17818 c.1824C
 > T Substitution p.= Silent Not HGPS Hutchinson-Gilford progeria 176670 # affected
 syndrome 9009 c.1868C > G Substitution p.[Thr623Ser, Deletion, Tail HGPS Hutchinson-
 Gilford progeria 176670 # Val622_Gln656del] Substitution syndrome 12457 c.1868C > G
 Substitution p.[Thr623Ser, Deletion, Tail HGPS Hutchinson-Gilford progeria 176670 #
 Val622_Gln656del] Substitution syndrome 9174 c.1930C > T Substitution p.Arg644Cys
 Substitution Tail HGPS Hutchinson-Gilford progeria 176670 # syndrome 11615 c.1960C > T
 Substitution p.Arg654X Substitution Tail HGPS Hutchinson-Gilford progeria 176670 %
 syndrome 17159 c.1968G > A Substitution p.= Silent Not HGPS Hutchinson-Gilford progeria
 176670 # affected syndrome 11784 c.1968 + 1G > A Substitution p.Val607_Gln656del
 Deletion Tail HGPS Hutchinson-Gilford progeria 176670 # syndrome 17161 c.1968 + 5G
 > A Substitution p.Val607_Gln656del Deletion Tail HGPS Hutchinson-Gilford progeria 176670
 # syndrome 12615 c.1411C > T Substitution p.Arg471Cys Substitution Tail — Lamin-related
 rigid spine — — muscular dystrophy 17426 c.31delC Deletion p.Arg11AlafsX85 Frame shift
 Head LGMD1B Limb-girdle muscular 159001 # dystrophy type 1B 14254 c.73C > G
 Substitution p.Arg25Gly Substitution Head LGMD1B Limb-girdle muscular 159001 #
 dystrophy type 1B 17886 c.80C > T Substitution p.Thr27Ile Substitution Head LGMD1B
 Limb-girdle muscular 159001 # dystrophy type 1B 18488 c.80C?T Substitution p.Thr27Ile
 Substitution Head LGMD1B Limb-girdle muscular 159001 # dystrophy type 1B 17428
 c.99G > C Substitution p.Glu33Asp Substitution 1A LGMD1B Limb-girdle muscular 159001
 # dystrophy type 1B 17429 c.99G > C Substitution p.Glu33Asp Substitution 1A LGMD1B
 Limb-girdle muscular 159001 # dystrophy type 1B 17430 c.99G > C Substitution
 p.Glu33Asp Substitution 1A LGMD1B Limb-girdle muscular 159001 # dystrophy type 1B
 17434 c.194A > G Substitution p.Glu65Gly Substitution 1A LGMD1B Limb-girdle muscular
 159001 # dystrophy type 1B 17435 c.194A > G Substitution p.Glu65Gly Substitution 1A
 LGMD1B Limb-girdle muscular 159001 # dystrophy type 1B 16283 c.302G > C
 Substitution p.Arg101Pro Substitution 1B LGMD1B Limb-girdle muscular 159001 #
 dystrophy type 1B 17399 c.388G > T Substitution p.Ala130Ser Substitution 1B LGMD1B
 Limb-girdle muscular 159001 # dystrophy type 1B 17755 c.388G > T Substitution
 p.Ala130Ser Substitution 1B LGMD1B Limb-girdle muscular 159001 # dystrophy type 1B
 17432 c.471G > A Substitution p.= Silent Not LGMD1B Limb-girdle muscular 159001 #
 affected dystrophy type 1B 9216 c.513G > A Substitution p.= Silent Not LGMD1B Limb-
 girdle muscular 159001 # affected dystrophy type 1B 17575 c.513 + 1G > A
 Substitution p.? Unknown Unknown LGMD1B Limb-girdle muscular 159001 # dystrophy
 type 1B 17437 c.565C > T Substitution p.Arg189Trp Substitution 1B LGMD1B Limb-
 girdle muscular 159001 # dystrophy type 1B 9639 c.622_624delAAG Deletion p.Lys208del
 Deletion 1B LGMD1B Limb-girdle muscular 159001 # dystrophy type 1B 12403
 c.624_626delGAA Deletion p.Lys208del Deletion 1B LGMD1B Limb-girdle muscular 159001 #
 dystrophy type 1B 16145 c.673C > T Substitution p.Arg225X Substitution L12 LGMD1B
 Limb-girdle muscular 159001 # dystrophy type 1B 17401 c.673C > T Substitution
 p.Arg225X Substitution L12 LGMD1B Limb-girdle muscular 159001 # dystrophy type 1B
 17753 c.673C > T Substitution p.Arg225X Substitution L12 LGMD1B Limb-girdle muscular
 159001 # dystrophy type 1B 13555 c.746G > A Substitution p.Arg249Gln Substitution 2A
 LGMD1B Limb-girdle muscular 159001 # dystrophy type 1B 9164 c.777T > A
 Substitution p.Tyr259X Substitution L2 LGMD1B Limb-girdle muscular 159001 # dystrophy

type 1B 12401 c.777T > A Substitution p.Tyr259X Substitution L2 LGMD1B Limb-girdle
 muscular 159001 # dystrophy type 1B 17760 c.777T > A Substitution p.Tyr259X
 Substitution L2 LGMD1B Limb-girdle muscular 159001 # dystrophy type 1B 11810
 c.855delG Deletion p.Ala287LeufsX191 Frame shift 2B LGMD1B Limb-girdle muscular
 159001 # dystrophy type 1B 9021 c.864_867delCCAC Deletion p.His289ArgfsX190 Frame
 shift 2B LGMD1B Limb-girdle muscular 159001 # dystrophy type 1B 11759
 c.864_867delCCAC Deletion p.His289ArgfsX190 Frame shift 2B LGMD1B Limb-girdle
 muscular 159001 # dystrophy type 1B 12384 c.908_909delCT Deletion p.Ser303CysfsX26
 Frame shift 2B LGMD1B Limb-girdle muscular 159001 # dystrophy type 1B 11811
 c.992G > C Substitution p.Arg331Pro Substitution 2B LGMD1B Limb-girdle muscular
 159001 # dystrophy type 1B 18309 c.1001_1003delGCC Deletion p.Ser334del Deletion 2B
 LGMD1B Limb-girdle muscular 159001 # dystrophy type 1B 8794 c.1130G > A
 Substitution p.Arg377His Substitution 2B LGMD1B Limb-girdle muscular 159001 #
 dystrophy type 1B 8850 c.1130G > T Substitution p.Arg377Leu Substitution 2B LGMD1B
 Limb-girdle muscular 159001 # dystrophy type 1B 9301 c.1130G > A Substitution
 p.Arg377His Substitution 2B LGMD1B Limb-girdle muscular 159001 # dystrophy type 1B
 9024 c.1130G?T Substitution p.Arg377Leu Substitution 2B LGMD1B Limb-girdle muscular
 159001 # dystrophy type 1B 11762 c.1130G > A Substitution p.Arg377His Substitution 2B
 LGMD1B Limb-girdle muscular 159001 # dystrophy type 1B 13557 c.1130G > A
 Substitution p.Arg377His Substitution 2B LGMD1B Limb-girdle muscular 159001 #
 dystrophy type 1B 11755 c.1146C > T Substitution p.= Silent Not LGMD1B Limb-girdle
 muscular 159001 # affected dystrophy type 1B 11756 c.1357C > T Substitution
 p.Arg453Trp Substitution Tail LGMD1B Limb-girdle muscular 159001 # dystrophy type 1B
 17443 c.1357C > T Substitution p.Arg453Trp Substitution Tail LGMD1B Limb-girdle
 muscular 159001 # dystrophy type 1B 11812 c.1370delA Deletion p.Lys457SerfsX21 Frame
 shift Tail LGMD1B Limb-girdle muscular 159001 # dystrophy type 1B 17441 c.1380 +
 1G > A Substitution KO Unknown Unknown LGMD1B Limb-girdle muscular 159001 #
 dystrophy type 1B 8839 c.1441T > C Substitution p.Tyr481His Substitution Tail LGMD1B
 Limb-girdle muscular 159001 # dystrophy type 1B 13376 c.1488 + 5G > A
 Substitution p.? Unknown Unknown LGMD1B Limb-girdle muscular 159001 # dystrophy
 type 1B 9181 c.1494G > T Substitution p.Trp498Cys Substitution Tail LGMD1B Limb-
 girdle muscular 159001 # dystrophy type 1B 9158 c.1494G > T Substitution p.Trp498Cys
 Substitution Tail LGMD1B Limb-girdle muscular 159001 # dystrophy type 1B 16442
 c.1494G > T Substitution p.Trp498Cys Substitution Tail LGMD1B Limb-girdle muscular
 159001 # dystrophy type 1B 9159 c.1535T > C Substitution p.Leu512Pro Substitution Tail
 LGMD1B Limb-girdle muscular 159001 # dystrophy type 1B 11763 c.1535T > C
 Substitution p.Leu512Pro Substitution Tail LGMD1B Limb-girdle muscular 159001 #
 dystrophy type 1B 17449 c.1535-T > C Substitution p.Leu512Pro Substitution Tail
 LGMD1B Limb-girdle muscular 159001 # dystrophy type 1B 17452 c.1535T > C
 Substitution p.Leu512Pro Substitution Tail LGMD1B Limb-girdle muscular 159001 #
 dystrophy type 1B 11758 c.1583C > A Substitution p.Thr528Lys Substitution Tail
 LGMD1B Limb-girdle muscular 159001 # dystrophy type 1B 13446 c.1608 + 1G > A
 Substitution p.? Unknown Unknown LGMD1B Limb-girdle muscular 159001 # dystrophy
 type 1B 9412 c.1608 + 5G > C Substitution p.Glu537ValfsX36 Frame shift Tail
 LGMD1B Limb-girdle muscular 159001 # dystrophy type 1B 13151 c.1609-3C?G
 Substitution p.? Unknown Unknown LGMD1B Limb-girdle muscular 159001 # dystrophy
 type 1B 11764 c.1718C > T Substitution p.Ser573Leu Substitution Tail LGMD1B Limb-
 girdle muscular 159001 # dystrophy type 1B 13505 c.373G > A Substitution p.Gly125Ser
 Substitution 1B LAF Lone atrial fibrillation — — 13509 c.[373G > A; =] + Substitution
 p.[Gly125Ser; =] + Substitution 1B, Tail LAF Lone atrial fibrillation — — [=; 1243G

> A] [=; Val415Ile] 13492 c.1310 + 63C > A Substitution p.? Unknown LAF
 Lone atrial fibrillation — — 13495 c.937 - 46A > G Substitution p.? Unknown
 Unknown LAF Lone atrial fibrillation — — 13501 c.1149G > A Substitution p.= Silent Not
 LAF Lone atrial fibrillation — — affected 13497 c.1158 - 44C > T Substitution p.?
 Unknown Unknown LAF Lone atrial fibrillation — — 13499 c.1158 - 44C > T
 Substitution p.? Unknown Unknown LAF Lone atrial fibrillation — — 13507 c.1243G > A
 Substitution p.Val415Ile Substitution Tail LAF Lone atrial fibrillation — — 13511 c.1462A
 > C Substitution p.Thr488Pro Substitution Tail LAF Lone atrial fibrillation — — 13503
 c.1803C > T Substitution p.= Silent Not LAF Lone atrial fibrillation — — affected 11783
 c.176T > G Substitution p.Leu59Arg Substitution 1A MADA Mandibuloacral dysplasia
 with 248370 # type A lipodystrophy 18470 c.683A > T Substitution p.Glu228Val
 Substitution L12 MADA Mandibuloacral dysplasia with 248370 # type A lipodystrophy
 12380 c.[1318G > A; =] + Substitution p.[Val440Met; =] + Substitution Tail MADA
 Mandibuloacral dysplasia with 248370 # [=; 1580G > A] [=; Arg527His] type A
 lipodystrophy 12615 c.1411C > T Substitution p.Arg471Cys Substitution Tail MADA
 Mandibuloacral dysplasia with 248370 # type A lipodystrophy 13119 c.1579C > T
 Substitution p.Arg527Cys Substitution Tail MADA Mandibuloacral dysplasia with 248370 #
 type A lipodystrophy 17878 c.1579C > T Substitution p.Arg527Cys Substitution Tail
 MADA Mandibuloacral dysplasia with 248370 # type A lipodystrophy 8851 c.1580G >
 A Substitution p.Arg527His Substitution Tail MADA Mandibuloacral dysplasia with 248370 #
 type A lipodystrophy 11419 c.1580G > A Substitution p.Arg527His Substitution Tail
 MADA Mandibuloacral dysplasia with 248370 # type A lipodystrophy 11420 c.1580G
 > A Substitution p.Arg527His Substitution Tail MADA Mandibuloacral dysplasia with
 248370 # type A lipodystrophy 11421 c.1580G > A Substitution p.Arg527His Substitution
 Tail MADA Mandibuloacral dysplasia with 248370 # type A lipodystrophy 11422
 c.1580G > A Substitution p.Arg527His Substitution Tail MADA Mandibuloacral dysplasia
 with 248370 # type A lipodystrophy 8877 c.1580G > A Substitution p.Arg527His
 Substitution Tail MADA Mandibuloacral dysplasia with 248370 # type A lipodystrophy
 11453 c.1580G > A Substitution p.Arg527His Substitution Tail MADA Mandibuloacral
 dysplasia with 248370 # type A lipodystrophy 8995 c.1580G > A Substitution
 p.Arg527His Substitution Tail MADA Mandibuloacral dysplasia with 248370 # type A
 lipodystrophy 9224 c.1580G > A Substitution p.Ala527His Substitution Tail MADA
 Mandibuloacral dysplasia with 248370 # type A lipodystrophy 17602 c.1580G > T
 Substitution p.Arg527Leu Substitution Tail MADA Mandibuloacral dysplasia with 248370 #
 type A lipodystrophy 17603 c.1580G > T Substitution p.Arg527Leu Substitution Tail
 MADA Mandibuloacral dysplasia with 248370 # type A lipodystrophy 17747 c.1580G
 > T Substitution p.Arg527Leu Substitution Tail MADA Mandibuloacral dysplasia with
 248370 # type A lipodystrophy 17748 c.1580G > T Substitution p.Arg527Leu Substitution
 Tail MADA Mandibuloacral dysplasia with 248370 # type A lipodystrophy 12598
 c.1585G > A Substitution p.Ala529Thr Substitution Tail MADA Mandibuloacral dysplasia
 with 248370 # type A lipodystrophy 9317 c.1586C > T Substitution p.Ala529Val
 Substitution Tail MADA Mandibuloacral dysplasia with 248370 # type A lipodystrophy
 18478 c.1620G > A Substitution p Substitution Tail MADA Mandibuloacral dysplasia with
 248370 # type A lipodystrophy 13054 c.82C > T Substitution p.Arg28Trp Substitution
 Head — Metabolic syndrome — — 13056 c.274C > T Substitution p.Leu92Phe Substitution
 1B — Metabolic syndrome — — 13059 c.11590 > G Substitution p.Leu387Val Substitution
 2B — Metabolic syndrome — — 13061 c.1184C > T Substitution p.Ser395Leu Substitution
 Not — Metabolic syndrome — — affected 13063 c.1196G > A Substitution p.Arg399His
 Substitution Tail — Metabolic syndrome — — 13065 c.1262T > C Substitution p.Leu421Pro
 Substitution Tail — Metabolic syndrome — — 13067 c.1315C > T Substitution p.Arg439Cys

Substitution Tail — Metabolic syndrome — — 13069 c.1516C > G Substitution p.His506Asp
 Substitution Tail — Metabolic syndrome — — 13047 c.1698C > T Substitution p.= Silent
 Not — Metabolic syndrome — — affected 13071 c.1961_1962insG Insertion p.Thr655AsnfsX49
 Frame shift Tail — Metabolic syndrome — — 12408 c.73C > T Substitution p.Arg25Cys
 Substitution Head — Muscular dystrophy — — 12410 c.1130G > T Substitution
 p.Arg377Leu Substitution 2B — Muscular dystrophy — — 12412 c.1622G > C Substitution
 p.Arg541Pro Substitution Tail — Muscular dystrophy — — 12414 c.1045C > T Substitution
 p.Arg349Trp Substitution 2B — Muscular dystrophy and — — lipodystrophy 17411
 c.1821G > A Substitution p.= Silent Not WRS Progeroid syndrome, 264090 % affected
 neonatal 18482 c.1940T > G Substitution p.Leu647Arg Substitution Tail WRS Progeroid
 syndrome, 264090 % neonatal 17009 c.1303C > T Substitution p.Arg435Cys Substitution Tail
 RD Restrictive dermatopathy 275210 # 17731 c.1303C > T Substitution p.Arg435Cys
 Substitution Tail RD Restrictive dermatopathy 275210 # 17801 c.1303C > T Substitution
 p.Arg435Cys Substitution Tail RD Restrictive dermatopathy 275210 # 9166 c.1824C > T
 Substitution p.[=, Silent, Deletion Tail RD Restrictive dermatopathy 275210 #
 Val607_Gln656del] 9208 c.1968 + 1G > A Substitution p.Gly567_Gln656del Deletion Tail
 RD Restrictive dermatopathy 275210 # 17980 c.1057C > T Substitution p.Gln353X
 Substitution 2B — Spinal muscular atrophy with — — cardiac involvement 18424 c.868G
 > A Substitution p.Glu290Lys Substitution 2B SCD Sudden cardiac death 115080 # 17189
 c.908_909delCT Deletion p.Ser303CysfsX26 Frame shift 2B SCD Sudden cardiac death
 115080 # 17901 c.1334T > A Substitution p.Val445Glu Substitution Tail SCD Sudden
 cardiac death 115080 # 9022 c.1804G > A Substitution p.Gly602Ser Substitution Tail —
 Type A insulin resistance — — syndrome

(71) In some embodiments, the nucleic acid molecule according to any aspect of the invention is for use in treating cardiovascular disease in a subject.

(72) In some embodiments, the disease or the cardiovascular disease is characterised by the presence of at least one Lmna mutation.

(73) Preferably, the cardiovascular disease is selected from the group consisting of laminopathy, cardiomyopathy, such as dilated cardiomyopathy (DCM), dilated cardiomyopathy 1A, dilated cardiomyopathy with conduction system defects, cardiomyopathy with advanced AV block and arrhythmia, lone atrial fibrillation; muscular dystrophy (often associated with cardiomyopathy), such as cardiomyopathy associated with Emery-Dreifuss muscular dystrophy (autosomal dominant), cardiomyopathy associated with Emery-Dreifuss muscular dystrophy (autosomal recessive), cardiomyopathy associated with Limb-girdle muscular dystrophy type 1B, cardiomyopathy associated with congenital muscular dystrophy; premature aging syndromes (thought to be primarily vascular, but may have cardiac involvement) such as cardiomyopathy associated with Atypical Werner syndrome, cardiomyopathy associated with Hutchinson-Gilford progeria syndrome and the like, as well as diseases presented in bold font in Table 1.

(74) According to a third aspect of the invention there is provided an adeno-associated virus vector (AAV) comprising a cardiac troponin T promoter (cTnT), and the transgene according to any aspect of the invention.

(75) According to a fourth aspect of the invention there is provided a pharmaceutical composition comprising the nucleic acid molecule according to any embodiment of the invention for treating a disease.

(76) In some embodiments the disease is a laminopathy.

(77) In some embodiments the pharmaceutical composition comprising the nucleic acid molecule according to the invention is for use in treating a cardiovascular disease in a subject.

(78) According to a fifth aspect of the invention there is provided a method of treating a disease in a subject, the method comprising administration of a pharmaceutically effective amount of the nucleic acid molecule according to any embodiment of the invention, or the pharmaceutical

composition of the invention.

(79) In some embodiments of the method of treating a disease in a subject, the disease is characterised by the presence of at least one Lmna mutation.

(80) In some embodiments of the method of treating a disease in a subject, the method comprises:

(81) (i) testing a sample obtained from a subject suspected of having a disease for the presence or absence of at least one Lmna mutation;

(82) wherein the presence of at least one Lmna mutation indicates that the subject is to be administered the pharmaceutical composition of the invention or the nucleic acid molecule of the invention.

(83) In some embodiments of the method of treating a disease in a subject, the Lmna mutation(s) affect(s) lamin A isoform, or lamin C isoform of the Lmna gene, or both lamin A/C isoforms.

(84) In some embodiments of the method of treating a disease in a subject, the disease is selected from the group consisting of restrictive dermopathy, familial partial lipodystrophy (for example, Dunnigan type), mandibuloacral dysplasia with type A lipodystrophy, metabolic syndrome, Charcot-Marie-Tooth disease type 2, Charcot-Marie-Tooth disease type 2B1 and diseases presented in normal font in Table 1.

(85) In some embodiments of the method of treating a disease in a subject, the disease is a cardiovascular disease, wherein the cardiovascular disease is selected from the group consisting of laminopathy, cardiomyopathy, such as dilated cardiomyopathy (DCM), dilated cardiomyopathy 1A, dilated cardiomyopathy with conduction system defects, cardiomyopathy with advanced AV block and arrhythmia, lone atrial fibrillation; muscular dystrophy (often associated with cardiomyopathy), such as cardiomyopathy associated with Emery-Dreifuss muscular dystrophy (autosomal dominant), cardiomyopathy associated with Emery-Dreifuss muscular dystrophy (autosomal recessive), cardiomyopathy associated with Limb-girdle muscular dystrophy type 1B, cardiomyopathy associated with congenital muscular dystrophy; premature aging syndromes (thought to be primarily vascular, but may have cardiac involvement) such as cardiomyopathy associated with Atypical Werner syndrome, cardiomyopathy associated with Hutchinson-Gilford progeria syndrome; and diseases presented in bold font in Table 1.

(86) In some embodiments of the method of treating a disease in a subject, the subject is a non-human mammal, such as a mouse, or a human.

(87) In some embodiments of the method, the mouse is an N195K mouse (Lmna N195K/N195K), or a Lmna conditional knockout (Lmna^{flox}/flox).

(88) According to a sixth aspect of the invention there is provided use of the pharmaceutical composition according to the invention or the nucleic acid molecule according to the invention in the manufacture of a medicament for treating a disease caused by one or more Lmna mutations.

(89) In some embodiments the disease is Lmna mutation-related cardiovascular disease.

(90) In some embodiments the disease is selected from the group consisting of restrictive dermopathy, familial partial lipodystrophy (for example, Dunnigan type), mandibuloacral dysplasia with type A lipodystrophy, metabolic syndrome, Charcot-Marie-Tooth disease type 2, Charcot-Marie-Tooth disease type 2B1 and diseases presented in normal font in Table 1.

(91) In some embodiments the cardiovascular disease is selected from the group consisting of laminopathy, cardiomyopathy, such as dilated cardiomyopathy (DCM), dilated cardiomyopathy 1A, dilated cardiomyopathy with conduction system defects, cardiomyopathy with advanced AV block and arrhythmia, lone atrial fibrillation; muscular dystrophy (often associated with cardiomyopathy), such as cardiomyopathy associated with Emery-Dreifuss muscular dystrophy (autosomal dominant), cardiomyopathy associated with Emery-Dreifuss muscular dystrophy (autosomal recessive), cardiomyopathy associated with Limb-girdle muscular dystrophy type 1B, cardiomyopathy associated with congenital muscular dystrophy; premature aging syndromes (thought to be primarily vascular, but may have cardiac involvement) such as cardiomyopathy associated with Atypical Werner syndrome, cardiomyopathy associated with Hutchinson-Gilford

progeria syndrome and the like, as well as diseases presented in bold font in Table 1.

(92) According to a seventh aspect of the invention there is provided a method for screening for drug candidates capable of inhibiting or disrupting the LINC complex in a cell.

(93) Accordingly, in some embodiments there is provided a method for screening for drug candidates capable of inhibiting the interaction of the proteins of a LINC complex in a cell, which comprises: (a) combining the proteins of said LINC complex in the presence of a drug to form a first complex; (b) combining the proteins in the absence of said drug to form a second complex; (c) measuring the amount of said first complex and said second complex; and (d) comparing the amount of said first complex with the amount of said second complex, wherein if the amount of said first complex is less than the amount of said second complex, then the drug is a drug candidate for inhibiting the interaction of the proteins of said LINC complex in a cell.

(94) In some embodiments the drug candidate disrupts the protein-protein interaction between SUN and KASH of the LINC complex. Preferably the drug candidate disrupts the interaction between Sun1 and Nesprin-1 proteins.

(95) In some embodiments said screening is an in vitro screening.

(96) In some embodiments said complex is measured by an ELISA method.

(97) In some embodiments recombinant SUN domain is immobilized on a solid surface and recombinant KASH domain is labelled with an enzyme that can generate a colorimetric or chemiluminescent readout. Compounds that fail to inhibit the SUN-KASH interaction will result in a well in the plate where the recombinant SUN would bind to the enzyme-linked KASH domain. Following wash steps and incubation with colorimetric or chemiluminescent enzyme substrates, the presence of the SUN-KASH interaction can be detected in standard plate readers. If the compound can inhibit SUN-KASH interaction, then following the wash step, the KASH domain would be removed, and there would be reduced or no enzymatic reaction in the well.

(98) In some embodiments if the amount of said first complex is less than the amount of said second complex, then said drug is a drug candidate for inhibiting the interaction of said proteins.

(99) In some embodiments said complex is measured by a fluorescence anisotropy method.

(100) In some embodiments the fluorescence anisotropy method employs recombinant SUN and KASH domains. In some embodiments the KASH domain is fluorescently labelled with a fluorescein moiety and fluorescence anisotropy of the KASH domain interacting with SUN domain may be measured using standard equipment such as a plate reader incorporating a fluorescence spectrometer function.

(101) In some embodiments if the amount of said first complex is less than the amount of said second complex there will be a difference in the fluorescence anisotropy of the fluorescent KASH and said drug is a drug candidate for inhibiting the interaction of said proteins.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) FIG. 1 shows a schematic of the mutations in the lamin A/C gene LMNA and the laminopathies resulting from the mutations.

(2) FIG. 2 shows a schematic of the positioning of components of the nuclear envelope membrane and lamina.

(3) FIG. 3 shows a schematic of the connections between the nucleus and the extracellular matrix via the LINC complex and how mutations in lamin A/C might result in DCM. The plasma membrane, cytoskeleton and nucleus form a mechanically and physically linked entity. In Lmna mutants, the nucleus is structurally weak. It is much more susceptible to mechanical stress from cytoskeletal forces. This leads to severe damage to the myocyte nuclei that in turn leads to a cascade of events such as apoptosis and fibrosis that results finally in DCM.

(4) FIG. 4 shows the effect of microinjection of dextran into the nucleus of *Lmna*.sup.-/- and *Lmna*.sup.-/- mice under low pressure. In the wildtype cells, the dextran stays in the nucleus, while in the *Lmna* mutant cells the dextran leaks out of the nucleus into the cytoplasm.

(5) FIGS. 5A-5B show schematics of a LINC complex (FIG. 5A) and Interaction between KASH and SUN (FIG. 5B).

(6) FIG. 6 shows defects in body weight and longevity in *Lmna*.sup.-/- and *Lmna*Δ9 mice are ameliorated in homozygous *Sun1* knockout *Lmna*.sup.-/- *Sun1*.sup.-/- and *Lmna*Δ9*Sun1*.sup.-/- animals. (A) Body weights are averages from mice with the indicated genotypes. The number (n) of animals used is indicated. (B) Kaplan-Meier graph showing increased life span of *Lmna*.sup.-/- *Sun1*.sup.-/- compared to *Lmna*.sup.-/- mice. Median survival of wild-type or *Sun1*.sup.-/- is >210 days in a 7 month follow up; *Lmna*.sup.-/- mice have median survival of 41 days; *Lmna*.sup.-/- *Sun1*.sup.-/- mice have a median survival of 54 days; *Lmna*.sup.-/- *Sun1*.sup.-/- mice have a median survival of 104 days (p<0.01 comparing *Lmna*.sup.-/- and *Lmna*.sup.-/- *Sun1*.sup.-/-). (C) Body weights of *Lmna*Δ9 mice that are wild-type, heterozygous, or homozygous for *Sun1* deficiency. Wild-type and *Sun1*.sup.-/- cohorts are graphed for comparison. Values are averages ± SEM from animals in each cohort. Number (n) of animals is indicated (p<0.0001 comparing *Lmna*Δ9*Sun1*.sup.+/+ and *Lmna*Δ9*Sun1*.sup.-/-). (D) Kaplan-Meier graph showing increased life span of *Lmna*Δ9*Sun1*.sup.-/- compared to *Lmna*Δ9*Sun1*.sup.+/+ mice. *Lmna*Δ9*Sun1*.sup.-/- mice are also graphed. (p<0.0001 comparing *Lmna*Δ9*Sun1*.sup.+/+ and *Lmna*Δ9*Sun1*.sup.-/-). (E) Cell proliferation of the indicated MEFs. Curves are averages ± SD, representative of >3 independent isolates from embryos of the indicated genotypes. (F) Proliferation curves of MAFs (mouse adult fibroblasts) from WT, *Sun1*.sup.-/-, *Lmna*Δ9*Sun1*.sup.+/+ and *Lmna*Δ9*Sun1*.sup.-/- mice. MAFs were seeded at a density of 1000 cells per well. Growth was measured, and normalized cell indexes (averages ± SD) are presented.

(7) FIG. 7 shows a schematic of the features of the *Sun1* protein and the components used to generate a dominant negative *Sun1* protein, including a signal sequence, oiled-coil sequence, SUN domain sequence and KDEL sequence.

(8) FIG. 8 shows a schematic of a plasmid (SEQ ID NO: 1) used for AAV production.

(9) FIG. 9 shows a schematic of a plasmid (SEQ ID NO: 2) comprising sequences from AAV2 and AAV9 for AAV production.

(10) FIG. 10 shows a schematic of an AAV expression construct (SEQ ID NO: 3) comprising cardiac-specific promoter and *Sun1* dominant negative sequence.

(11) FIG. 11 shows a schematic of the features of the dominant negative *Sun1* protein, including a signal sequence, coiled-coil sequence, SUN domain sequence and KDEL sequence (SEQ ID NO: 4).

(12) FIG. 12 shows a schematic of the features of a dominant negative *Sun2* protein, including a signal sequence, luminal domain sequence and KDEL sequence (SEQ ID NO: 5).

(13) FIG. 13 shows a schematic of the region of *Sun1* protein used in dominant negative constructs.

(14) FIG. 14 shows an alignment of KASH1-KASH5 domain amino acid sequences with conserved residues (SEQ ID Nos: 7, 9, 11, 13 and 15, respectively).

(15) FIG. 15 shows a schematic of the LINC complex in wildtype mice, *Sun1* KO mice, AAV dominant negative SUN mice and mice with altered KASH domain. The schematic for wildtype mice is obtained from Brian Burke, 2012. The schematic for *Sun1* KO mice represents the results from Chen et al., 2012. The AAV dominant negative SUN and the altered KASH domain schematics represent Inventor proposals at the priority date on methods for LINC complex disruption to ameliorate laminopathies, based on data obtained at that time.

(16) FIG. 16 shows a Kaplan Meier curve of *Lmna* KO mice surviving for an average of 28 days, *Sun1* KO mice living beyond 300 days and cardiac *Lmna* KO/*Sun1* KO mice living beyond 300 days.

(17) FIG. 17 shows H&E stained sections of hearts from *Sun1* KO mice, cardiac *Lmna* KO mice

and cardiac Lmna KO/Sun1 KO mice, with LmnaKO/Sun1WT hearts showing enlargement of the left ventricle (DCM) compared to WT and LmnaKO/Sun1KO hearts.

(18) FIG. **18** shows a schematic of disruption of a LINC complex in a Nesprin-1 Δ KASH mouse. LmnaKO Nesprin-WT mice have a lifespan of about 20 days. LmnaKO Nesprin-1- Δ KASH survive about 40 days, which is similar to LmnaKOSun1KO mice.

(19) FIGS. **19A-19B** show a schematic of anticipated AAV-cTNT-DN-SUN expression and competition between exogenous DN-SUN and native Sun1 for binding to the KASH domain (FIG. **19A**) with the DN-SUN shown in 19B (upper panel) and the effect of transfected DN-SUN on native Nesprin2G positioning in cells where the 2 nuclei in the middle panel express the DN-SUN and in the merge panel both show loss of Nesprin2 from the nuclear membranes (FIG. **19B**).

(20) FIG. **20** is a Kaplan Meier curve showing disruption of SUN-KASH interaction in vivo, using AAV9-cTNT-dominant negative Sun1 (DNSun1), extends the longevity of the heart-specific Lmna KO in male and female mice.

(21) FIG. **21** shows C-terminal amino acids of the KASH domain of Nesprin-2 (KASH2). The 14 or 18 amino acid sequence from KASH2 C-terminus are able to physically interact with the SUN domain of SUN2. Loss of the last 4 amino acids from KASH2 or addition of a single alanine amino acid at the C-terminus of KASH2 is sufficient to disrupt interaction of the KASH2 domain with the SUN domain.

(22) FIG. **22** shows a schematic of a screening method for detecting agents that disrupt the LINC complex.

(23) FIG. **23** shows a flowchart showing a more detailed screening method for identifying a small molecule to disrupt the LINC complex.

(24) FIG. **24** is a Kaplan Meier curve showing that wild type (C57/Bl6) mice with or without a Nesprin-1 KASH-disrupting (C'T Δ 8) mutation have a normal lifespan. Mice with a Lmna null/KO mutation (LA-ZP3cre.sup. Δ/Δ) and wildtype (Nesp1.sup.+/-) or heterozygous (Nesp1.sup.+/-C'T Δ 8) for Nesp1-C'T Δ 8 have a median lifespan of 15 or 18 days, which is increased to 38 days in Lmna KO/homozygous Nesp1 mutant (LA-ZP3cre.sup. Δ/Δ ; Nesp1.sup.C'T Δ 8/C'T Δ 8) mice.

(25) FIG. **25** is a Kaplan Meier curve showing that mice with wildtype Lmna (N1.sup.CT Δ 8/CT Δ 8LA.sup.+/-MCre.sup.+/-), or floxed alleles of Lmna but lacking a cardiac-specific Cre driver (N1.sup.CT Δ 8/CT Δ 8LA.sup.f/fMCre.sup.+/- and N1.sup.WT/WTLA.sup.f/fMCre.sup.+/-), live for length of the experiment (~80 days at priority filing, which extended to 120 days unchanged). Mice with a cardiomyocyte-specific deletion of Lmna (N1.sup.WT/WTLA.sup.f/fMCre.sup.+/-) have a lifespan of 22-24 days following induction of the Cre/loxP-mediated deletion by tamoxifen (TMX) delivery, which is increased to the length of the experiment in mice with a cardiomyocyte-specific deletion of Lmna induced by TMX and also homozygous mutant for Nesprin-1 (N1.sup.CT Δ 8/CT Δ 8LA.sup.f/fMCre.sup.+/-).

(26) FIGS. **26A-26D** show Kaplan Meier curves of Sun1 loss extending the longevity of Lmna mutant mice. (FIG. **26A**) Wild type (C57/Bl6) mice with or without Sun1 have a normal lifespan, whereas the average postnatal lifespan of the Lmna.sup.Flx/Flx:Zp3 mice in which LaminA is deleted in all tissues was 17.5 days (***P=<0.0001; Log-rank test). On a Sun1.sup.-/- background longevity is increased to 32.5 days. (FIG. **26B**) When Lmna.sup.Flx/Flx was deleted specifically and constitutively in hearts by crossing the mice with the Cre.sup. α MyHC line, the Lmna.sup.Flx/Flx: α MyHC mice lived on average 26.5 days. On a Sun1.sup.-/- background these mice lived for longer than 6 months. (FIG. **26C**) 3-5 month old Lmna.sup.Flx/Flx were crossed with the Tmx inducible cardiomyocyte specific Cre Tg(Myh6-cre/Esr1), (abbreviated to mcm), after a single injection of Tmx the mice die within 3-4 weeks. On a Sun1.sup.-/- background these mice lived for more than 1 year. (FIG. **26D**) Lmna.sup.N195K/N195K mice lived for an average of 78 days compared to Lmna.sup.N195K/N195K Sun1.sup.-/- mice which had an average lifespan of 111 days. (***P=<0.0001, **P=0.0073 Log-rank test).

(27) FIGS. 27A-27E show the lifespan and phenotype of *Lmna*.sup.Flx/Flx:mcm+Tmx mice. (FIG. 27A) The average lifespan of the *Lmna*.sup.Flx/Flx:mcm mice was 27 days after a single Tmx injection ($***P<0.0001$; Log-rank test). (FIG. 27B) PCR detected the floxed (deleted) *Lmna* gene (arrow head) only in heart tissue after Tmx injection and not in other tissues or when Tmx was not injected (FIG. 27C) *Lmna*.sup.Flx/Flx:mcm+Tmx mice developed kyphosis (arrow head) by 21 days after injection. (FIG. 27D) LaminA/C protein, detected by immunofluorescence, were present in control (i, iii), but reduced/absent in cardiomyocyte (CM) nuclei in both isolated CMs (ii second panel) and heart sections (iv) (white arrowheads) with CM nuclei being detected by PCM-1 staining, 21 days after Tmx. (FIG. 27E) LaminA/C levels were quantified by Western analysis of whole heart lysates 21 days after injection. A significant reduction ($***P<0.0001$; T-test) in A-type Lamin protein was detected, although Lamin C levels were not reduced as much in the *Lmna*.sup.Flx/Flx:mcm+Tmx mice compared to *Lmna*.sup.Flx/Flx:mcm+CTL. (FIG. 27F) Quantitative analysis was performed at 21 days post Tmx. The presence of the LoxP sites in the WT-*Lmna* gene (*Lmna*.sup.Flx/Flx) results in a reduction in *Lmna* transcript levels compared to *Lmna*.sup.Wt/Wt levels, although this had no overt effect on longevity or growth/viability.

(28) FIGS. 28A-28D show echocardiograms, heart function and histology of *Lmna*.sup.Flx/Flx:mcm+Tmx mice. (FIG. 28A) *Lmna*.sup.Flx/Flx:mcm+Tmx mice show reduced cardiac contractile function. (FIG. 28B) *Lmna*.sup.Flx/Flx:mcm hearts show reduced EF % and FS %, and increased LVID ($***P<0.0001$, $**P=0.0010$ Two way ANOVA). (FIG. 28C) Histological analysis of the hearts revealed increased infiltration of nucleated cells and intercellular spaces in *Lmna*.sup.Flx/Flx:mcm hearts (i and ii). Significantly fewer viable (brick-like) CMs were isolated from *Lmna*.sup.Flx/Flx:mcm hearts compared to *Lmna*.sup.Flx/Flx:mcm controls (iii). With higher magnification, the isolated cardiomyocytes from *Lmna*.sup.Flx/Flx:mcm hearts contained large intracellular vacuoles (arrow head, iv). (FIG. 28D) The left ventricular lumen in *Lmna*.sup.Flx/Flx:mcm hearts was enlarged (i) together with increased fibrosis (ii) ($*P=0.0007$ visible as lighter grey areas in the 28D ii, middle panels and iv left panel) and apoptotic nuclei revealed by TUNEL staining ($*P=0.0220$; One way ANOVA) (iii and iv right panel). All samples and analyses were performed on hearts 21 days post Tmx injection.

(29) FIGS. 29A-29D shows changes in nuclear morphologies and heart structure with and without Sun1 in the *Lmna*.sup.Flx/Flx:mcm after Tmx injection. (FIG. 29A) CM nuclei with reduced or absent Lamin A/C expression are indicated by white arrow heads (1, 3). CM nuclei (2, 4) with normal Lamin A/C levels are indicated by grey arrowheads. LMNA protein levels, measured by both fluorescence intensity (5) and Western blot (6), were significantly reduced in *Lmna*.sup.Flx/Flx:mcm Sun1.sup.+/-+Tmx ($***P=0.0009$; T-test) and *Lmna*.sup.Flx/Flx:mcm Sun1.sup.-/-+Tmx ($*P=0.0359$; T-test) compared to *Lmna*.sup.Flx/Flx:mcm Sun1.sup.+/- controls (lower graph, 6) (FIG. 29B). Left ventricular (LV) enlargement was apparent in the *Lmna*.sup.Flx/Flx:mcm Sun1.sup.+/-+Tmx hearts (panel 1) but not in the LV of the *Lmna*.sup.Flx/Flx:mcm Sun1.sup.-/-+Tmx hearts (panel 2). The *Lmna*.sup.Flx/Flx:mcm Sun1.sup.+/-+Tmx mice had significantly increased fibrosis (panel 3, fibrosis in grey) compared to controls, but there was no significant increase in fibrosis in the *Lmna*.sup.Flx/Flx:mcm Sun1.sup.-/-+Tmx hearts (panel 3) compared to controls (panel 4, quantified in panel 5, $**P=0.0001$; One way ANOVA). Cardiac papillary muscle active force measurements were significantly reduced from the *Lmna*.sup.Flx/Flx:mcm Sun1.sup.+/-+Tmx mice compared to *Lmna*.sup.Flx/Flx:mcm Sun1.sup.+/- controls ($**P=0.0047$; T-test) and *Lmna*.sup.Flx/Flx:mcm Sun1.sup.+/-+Tmx ($*P=0.0113$; T-test) (panel 6). (FIG. 29C) CM nuclear morphologies were significantly altered in *Lmna*.sup.Flx/Flx:mcm Sun1.sup.-/-+Tmx mice (Panel 1, solid arrow heads). In the absence of TMX, control heart sections (CTL, panel 2) display few nuclear abnormalities. In the absence of Sun1, *Lmna*.sup.Flx/Flx:mcm Sun1.sup.-/-+Tmx cardiomyocytes showed no nuclear abnormalities (Panels 3 and 4). In summary FIG. 29C panel 5 reveals that, 70% of CM in *Lmna*.sup.Flx/Flx:mcm Sun1.sup.+/-+Tmx mice had NE ruptures/distortions or misshapen nuclei

compared to less than 1% of CM nuclei in *Lmna*.sup.Flx/Flx:mcm Sun1.sup.-/-+Tmx mice. (FIG. 29D) Echo analyses on TMX-treated and control mice were performed following Tmx induction. Echocardiograms (ECGs) performed at 28 days after Tmx injection on 3-5 month old mice (panel 1). ECGs performed before and after Cre induction revealed a progressive worsening of cardiac contractility in *Lmna*.sup.Flx/Flx:mcm Sun1.sup.+/++Tmx mice (solid black line) compared to *Lmna*.sup.Flx/Flx:mcm Sun1.sup.-/-+Tmx mice (panels 2-4). The loss of SUN1 preserved EF (panel 2), FS (panel 3) and Global Longitudinal Strain (GLS, panel 4) in *Lmna*.sup.Flx/Flx:mcm Sun1.sup.-/-+Tmx mice compared to *Lmna*.sup.Flx/Flx:mcm Sun1.sup.+/++Tmx mice.

(30) FIGS. 30A-30B show Kaplan Meier graph and heart function effects of deletion of SUN1 on cardiac pathology induced by a missense mutation in the *Lmna* gene (N195K). (FIG. 30A) The absence of Sun1 significantly increases the lifespan of *Lmna*.sup.N195K/Flx:mcm Sun1.sup.-/-+Tmx mice compared to *Lmna*.sup.N195K/Flx:mcm Sun1.sup.+/++Tmx mice (*P=0.0101; Log-rank test). Mice with only one copy of the N195K mutation (*Lmna*.sup.N195K/-:mcm Sun1.sup.+/++Tmx) had an average lifespan of 47 days, approximately half the lifespan of mice homozygous i.e. with two copies of the N195K allele. (FIG. 30B) Echocardiograms (ECGs) performed before and after Cre induction revealed progressive worsening of cardiac contractility in *Lmna*.sup.N195K/-:mcm Sun1.sup.+/++Tmx mice compared to *Lmna*.sup.N195K/-:mcm Sun1.sup.-/-+Tmx mice over time. ECGs images were recorded at 28 days after Tmx injection (left-hand side panels). The loss of SUN1 preserved EF, FS and GLS in *Lmna*.sup.N195K/Flx:mcm Sun1.sup.-/-+Tmx mice compared to *Lmna*.sup.N195K/Flx:mcm Sun1.sup.+/++Tmx mice (right-hand side bottom 3 panels).

(31) FIGS. 31A-31G show *Lmna*.sup.Flx/Flx:mcm+Tmx mice expressing an AAV transduced DNSun1 exhibit improved cardiac function and increased longevity. (FIG. 31A) Protocol for AAV-mediated transduction of the DN-Sun1 miniprotein into *Lmna*.sup.Flx/Flx:mcm+Tmx mice. A single Tmx (IP) injection is given at D14 postnatally to induce *Lmna* deletion. AAV9-DNSun1 or AAV9-GFP viral particles are then injected into the chest cavity on D15 postnatally. The experimental endpoint was set at 100 days after Tmx. (FIG. 31B) The DNSun1 miniprotein competes with endogenous Sun1 for binding to the KASH domain of the Nesprins (in CMs this is Nesprin1). The miniprotein competes with endogenous SUN1 in binding to the KASH domain of the Nesprins. As the DNSun1 miniprotein is not anchored in the INM this effectively disconnects the endogenous SUN proteins from binding to the KASH domains so breaking the LINC. (FIG. 31C) The presence of the recombined *Lmna* gene following Tmx injection was confirmed by PCR of the heart tissues (upper panel). Robust expression of both AAV9-DNSun1 and AAV9-GFP protein (Dosage: 5×10^{10} vg/g of mouse) was detected in extracts from whole hearts 99 days post AAV injection (lower panel). (FIG. 31D) CMs derived from human iPS stem cells were transduced with the DNSun1 using AW-DJ as the vector. In CMs expressing high levels of DNSun1, indicated by grey arrows, Nesprin1 localization to the NE is reduced or absent. Nesprin localization to the NE is maintained in CMs either not expressing the AW-DJ-DNSun1 or when expressed at lower levels (white arrow heads). (FIG. 31E) The *Lmna*.sup.Flx/Flx:mcm+Tmx+AAV9-GFP mice lived for an average of 34.5 d after Tmx induction, whereas *Lmna*.sup.Flx/Flx:mcm+Tmx mice injected with AA9-DNSun1 (5×10^{10} vg/g/mouse) lived significantly longer (**P=0.0038; Log-rank test) to at least 100 D post Tmx, after which the mice were sacrificed for analysis. This set of data was derived from that shown in FIG. 20, adjusted by removing mice that were female and those with a different dose of virus. FIG. E(i) represents male mice and FIG. E(ii) represents female mice. (FIG. 31F) At 35 d after Tmx, extensive fibrosis (blue in original image, grey here) and ventricular enlargement was detected in *Lmna*.sup.Flx/Flx:mcm+Tmx+AAV9-GFP hearts compared to *Lmna*.sup.Flx/Flx:mcm+Tmx+AAV9-DNSun1 hearts. (FIG. 31G) ECG analysis confirmed *Lmna*.sup.Flx/Flx:mcm+Tmx+AAV9-DNSun1 hearts had better cardiac function compared to the *Lmna*.sup.Flx/Flx:mcm+Tmx+AAV9-GFP hearts at 35 d days after Tmx injection.

(32) FIGS. 32A-32D shows models of how breaking the LINC by disrupting Sun1 protects

cardiomyocytes from contraction induced stress (FIG. 32A) Cardiomyocyte nuclei expressing LmnaA/C, are able to withstand mechanical stress and tension forces transmitted via the LINC complex from the cytoplasm to the NE. (FIG. 32B) The loss of or introduction of a mutation within the Lmna gene results in loss/or incorrect assembly of the nuclear lamina, which weakens the Lamina/NE. The weakened nuclei are damaged due to the tension/stress forces exerted via the LINC complex from the contractile sarcomeres of the cardiomyocytes. (FIGS. 32C, D). In the absence of SUN1 or by disrupting its binding to the KASH domains by expression of DNSun1, the untethered LINC complexes exert less tensional force on the cardiomyocyte nuclei, enabling survival of the Lmna mutant cardiomyocytes.

(33) FIG. 33: shows the structure of the Lmna.sup.Flx/Flx conditional allele. Primer locations for genotyping the Lmna gene both before and after Cre recombination are indicated for the Lmna.sup.Flx allele (Flox), the Lmna deleted allele (A) and the wildtype allele [A. S. Wang, et al., *Differentiation* 89: 11-21 (2015)].

(34) FIG. 34 shows a diagram of the recombinant AAV9-DNSun1 and AAV9-GFP miniproteins. The DN-Sun1 includes the Sun domain, an HA tag, a Signal Sequence (SS, for targeting the protein to the ER), and the KDEL (ER retention signal) [M. Crisp et al., *J Cell Biol.* 172: 41-53 (2006)]. The AAV9-GFP includes the SS and KDEL sequences. GFP was used as a control in place of the Sun1L-KDEL.

(35) FIG. 35 shows photomicrographs of cardiomyocyte specific expression of Cre recombinase after Tmx injection. The Lmna.sup.Flx/Flx:mcm mice were crossed with the mT/mG (JAX: Gt(ROSA)26Sortm4(ACTB-tdTomato,-EGFP)Luo/J) reporter mice. In the absence of Cre, RFP is expressed. When Cre is induced, GFP is expressed. Only CMs of Lmna.sup.Flx/Flx:mcm mice express GFP upon TMX injection. Heart tissues were analyzed 7 days after Tmx injection.

(36) FIG. 36 shows that loss of Sun2 does not rescue loss of Lmna. Loss of Sun2 does not extend the lifespan of Lmna.sup. Δ/Δ Sun2.sup.-/- mice.

(37) FIGS. 37A-37E show the phenotypes of Lmna.sup.Flx/Flxmcm Sun1.sup.+/+ and Lmna.sup.Flx/Flxmcm Sun1.sup.-/- hearts at 12-14 months after Tmx injection. (FIG. 37A) Histological analysis of the aged Lmna.sup.Flx/Flx:mcm Sun1.sup.+/+ hearts, 12-14 months after the Tmx injection, revealing no significant morphological changes e.g. LV enlargement or (FIG. 37B) in fibrosis compared to the controls. (FIG. 37C) PCR analysis confirmed the sustained deletion of Lmna gene. (FIG. 37D) Protein quantification revealed a significant reduction of LMNA levels in Lmna.sup.Flx/Flx:mcm Sun1.sup.-/-+Tmx hearts at 14 months after TMX. (FIG. 37E) Echocardiograms (left-hand side panel) from the aged mice showed reduced EF and FS (right-hand side panels) in both Lmna.sup.Flx/Flx:mcm Sun1.sup.+/+CTL and Lmna.sup.Flx/Flx:mcm Sun1.sup.-/-+Tmx aged mice.

(38) FIG. 38 shows the rescue by AAV9-DNSun1 depends on the dosage of viral particles injected. Lifespan of Lmna.sup.Flx/Flx:mcm+TMX mice depends of the dosage of AAV9-DNSun1 with, with a lower concentrations resulting in shorter lifespans. Each dot represents a mouse, horizontal lines indicate mean.

(39) FIGS. 39A-39C show levels of LamiA/C following Tmx induction and expression of AAV-expressed proteins. (FIG. 39A) LaminA/C levels were significantly reduced following Tmx induction, and the presence of either AAV9-DNSun1 or AAV9-GFP protein did not alter LMNA protein levels (Quantification of LaminA/C immunofluorescence intensity). The amount of LaminA/C, DNSun1 and GFP protein in whole hearts were also quantified by Western analysis (lower 3 graphs). (Analysis performed 35 days after Tmx). (FIG. 39B) The expression of both DNSun1 and GFP proteins were dependent on the concentration of viral particles injected. (FIG. 39C) Immunofluorescence revealed the majority of CMs were successfully infected and expressed GFP with 5×10^{10} vg/g of AAV9-GFP (left image) compared to infection with a 10-fold lower (5×10^9 AAV9-GFP, right image) concentration of viral particles.

(40) FIGS. 40A-40C show CRISPR targeting of Sun1 SUN domain results in loss of Sun1 protein.

(A, B) Clustal alignment of Sun1 DNA (FIG. 40A) and amino acid (FIG. 40B) sequence (SEQ ID Nos: 69 and 72, respectively) adjacent to CRISPR-induced mutation in wildtype Sun1, Sun1 with 4 bp insertion (Sun1_plus4; SEQ ID NOs: 70 and 73, respectively) and Sun1 with 7 bp deletion (Sun1_del7; SEQ ID NOs: 71 and 74, respectively). Numbering is of Sun1 coding sequence (A) and protein sequence (B). Bold letters in (B) indicate SUN domain. (FIG. 40C)

Immunofluorescence staining of mouse adult fibroblasts derived from wildtype and Sun1 mutant mice. Sun1 expression is lost in mutant mice, but Sun2 and Nesprin-1 expression is similar in all 3 genotypes. Scale bar=10 μ m.

(41) FIG. 41A-D shows CRISPR targeting of Syne1 C-terminus results in expression of a mutant Nesprin-1 protein. (A, B) Clustal alignment of wildtype Nesprin-1 DNA (SEQ ID NO; 75) and Nesprin-1C'T Δ 8 (Nesprn1_CTdel8) (SEQ ID NO: 76) (A) and amino acid sequence adjacent to CRISPR-induced mutation in wildtype Nesprin-1 (SEQ ID NO: 77) and Nesprin-1C'T Δ 8 (Nesprn1_CTdel8) (SEQ ID NO: 78) (B). TGA in bold indicates stop codon of Syne1/Nesprin-1 gene. (C, D) Immunoblots of Nesprin-1 from Syne1/Nesprin-1 wildtype and Syne1/Nesprin-1C'T Δ 8 mutant heart and muscle tissue.

(42) FIG. 42A-B are photomicrographs showing CRISPR-Induced Syne1 mutation results in mislocalized, "KASH-less" Nesprin-1 protein. Immunofluorescence staining of mouse adult fibroblasts (A) and primary myotubes (B) derived from wildtype (WT) and Syne1C'T Δ 8 mutant mice. Nesprin-1 is mislocalized from the nuclear envelope in the mutant samples. Merged images shows Nesprin-1 and DNA staining. Scale bar=10 μ m.

(43) FIG. 43A-C are photomicrographs showing Syne1 mutation does not disrupt localization of certain nuclear envelope proteins. (A-C) Immunofluorescence staining of mouse primary myotubes derived from wildtype (WT) and Syne1C'T Δ 8 mutant mice. Sun1 (A), Sun2 and emerin (B) and lamin A/C (C) localize normally to the nuclear envelope. Merged images show protein and DNA staining. Arrows indicate examples of normally localized nuclear envelope proteins. Scale bar=10 μ m.

(44) FIG. 44A-C are photomicrographs showing Syne1 mutation disrupts localization of nuclear-envelope-localized centrosomal proteins. (A-C) Immunofluorescence staining of mouse primary myotubes derived from wildtype (WT) and Syne1C'T Δ 8 mutant mice. Pcm1, Pericentrin (Pcnt), and Akap450, which normally localize to the nuclear envelope in myotubes, are displaced from the nuclear envelope in Syne1C'T Δ 8 mutant myotubes. MF20 is an antibody for myosin heavy chain, a myotube marker. Merged images show protein and DNA staining. Arrows indicate typical nuclear envelope staining for these centrosomal proteins. Scale bar=10 μ m.

(45) FIG. 45A-C shows Syne1 mutation does not affect mouse phenotype. (A-B) Representative images of 12-week-old male (A) and female (B) mice. (C) Bodyweight of male and female, wildtype (WT) and Syne1C'T Δ 8 mutant, mice over 6 weeks.

(46) FIGS. 46A-46C shows Syne2 constructs and Syne1/Syne2 double mutant mice experience perinatal lethality. (FIG. 46A) Design of IRES- β gal PGK-Neo targeting construct for generating Syne2 mutation. (FIG. 46B) Immunofluorescence staining of mouse adult fibroblasts derived from wildtype (WT) and Syne2 mutant mice showing loss of Nesprin-2. (FIG. 46C) Images of newborn pups. Top row are of mice with at least 1 wildtype Syne1 or Syne2 allele that appear a healthy pink. Bottom row shows cyanotic double mutant Syne1.sup.C'T Δ 8/C'T Δ 8:Syne2.sup.-/- pups which appear blue and die at birth.

(47) FIG. 47 is a Kaplan Meier graph showing a Syne2 mutation does not ameliorate Lmna pathology. Kaplan-Meier survival curve showing that regardless of their Syne2 mutation status (wildtype, heterozygous or mutant), Lmna.sup. Δ / Δ mice die within 3 weeks of birth.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

Definitions

(48) Certain terms employed in the specification, examples and appended claims are collected here for convenience.

(49) The terms “amino acid” or “amino acid sequence,” as used herein, refer to an oligopeptide, peptide, polypeptide, or protein sequence, or a fragment of any of these, and to naturally occurring or synthetic molecules. Where “amino acid sequence” is recited herein to refer to an amino acid sequence of a naturally occurring protein molecule, “amino acid sequence” and like terms are not meant to limit the amino acid sequence to the complete native amino acid sequence associated with the recited protein molecule.

(50) As used herein, the term “comprising” or “including” is to be interpreted as specifying the presence of the stated features, integers, steps or components as referred to, but does not preclude the presence or addition of one or more features, integers, steps or components, or groups thereof. However, in context with the present disclosure, the term “comprising” or “including” also includes “consisting of”. The variations of the word “comprising”, such as “comprise” and “comprises”, and “including”, such as “include” and “includes”, have correspondingly varied meanings.

(51) As used herein, the terms “CRISPR-Cas” and “CRISPR” system are used somewhat interchangeably to refer to a microbial adaptive immune system that uses RNA-guided nucleases to cleave foreign genetic elements. It comprises clustered regularly interspaced short palindromic repeats (CRISPRs), a CRISPR-associated (Cas) endonuclease and a synthetic guide RNA that can be programmed to identify and introduce a double strand break at a specific site within a targeted gene sequence. The palindromic repeats are interspaced by short variable sequences derived from exogenous DNA targets known as protospacers, and together they constitute the CRISPR RNA (crRNA) array. Within the DNA target, each protospacer is always associated with a protospacer adjacent motif (PAM), which can vary depending on the specific CRISPR system. CRISPR-Cas9 is a specific version of the system referring to use of RNA-guided Cas9 nuclease, originally derived from *Streptococcus pyogenes*, whereby the target DNA must immediately precede a 5'-NGG PAM. Variations of the CRISPR-Cas9 system are known [Ran F A, et al., *Nat. Protoc* 8, 2281-2308 (2013); Ran F A, et al., *Cell* 154, 1380-1389 (2013)], including CRISPR-Cpf1, and although CRISPR-Cas9 has been used herein in the Examples, it is not intended that the present invention be limited to a particular CRISPR-Cas system.

(52) As used herein, the term “dominant negative” refers to a mutation whose gene product adversely affects the normal, wild-type gene product within the same cell. This usually occurs if the product can still interact with the same elements as the wild-type product, but block some aspect of its function. In one example, the transgene is expressed as a protein, and said protein that is functional as a dimer. A mutation that removes the functional domain, but retains the dimerization domain would cause a dominant negative phenotype, because some fraction of protein dimers would be missing one of the functional domains.

(53) As used herein, the term “normal font” in reference to diseases listed in Table 1 refers to those diseases that are in plain text and not in bold text. The term “bold text” has its ordinary meaning.

(54) As used herein, the term “stabiliser polypeptide” or “stabiliser protein” refers to an inert polypeptide which folds into a discrete domain, thereby ensuring that the remainder of the peptide maintains, for example, the proper topology. In one example, the stabiliser protein ensures that the KASH protein maintains proper topology on the endoplasmic reticulum membrane and the outer nuclear membrane. In another example, the stabiliser polypeptide prevents an attached polypeptide from translocating into, for example, the perinuclear space.

(55) As used herein, the term “operably linked” means that the components to which the term is applied are in a relationship that allows them to carry out their inherent functions under suitable conditions. For example, a control sequence which is “operably linked” to a protein coding sequence is ligated thereto, so that expression of the protein coding sequence is achieved under conditions compatible with the transcriptional activity of the control sequences. By way of an example, a first nucleic acid sequence is operably linked with a second nucleic acid sequence when the first nucleic acid sequence is placed in a functional relationship with the second nucleic acid sequence. For instance, a promoter is operably linked to a coding sequence if the promoter affects

the transcription or expression of the coding sequence. Generally, operably linked DNA sequences are contiguous and, where necessary to join two protein-coding regions, in the same reading frame. (56) As used herein, the term “extension” refers to one or more amino acids that can be found attached to the N- or the C-terminus of a desired peptide.

(57) As used herein, the terms “polypeptide”, “peptide” or “protein” refer to one or more chains of amino acids, wherein each chain comprises amino acids covalently linked by peptide bonds, and wherein said polypeptide or peptide can comprise a plurality of chains non-covalently and/or covalently linked together by peptide bonds, having the sequence of native proteins, that is, proteins produced by naturally-occurring and specifically non-recombinant cells, or genetically-engineered or recombinant cells, and comprise molecules having the amino acid sequence of the native protein, or molecules having deletions from, additions to, and/or substitutions of one or more amino acids of the native sequence. A “polypeptide”, “peptide” or “protein” can comprise one (termed “a monomer”) or a plurality (termed “a multimer”) of amino acid chains.

(58) The term “subject” is herein defined as vertebrate, particularly mammal, more particularly human. For purposes of research, the subject may particularly be at least one animal model, e.g., a mouse, rat and the like. In particular, for treatment or prophylaxis of a laminopathy, such as DCM, the subject may be a human.

(59) The term “treatment”, as used in the context of the invention refers to ameliorating, therapeutic or curative treatment.

(60) Without being bound by theory, the inventors submit that the whole basis of the therapy is that disrupting LINC complex function suppresses Lmna mutation. It is further noted that the target of the claimed methods is the SUN-KASH interaction in the LINC complex. The endogenous protein levels should not be affected.

(61) It is also further noted that the transgene (for example, the dominant negative transgene) will not work if the full length SUN domain protein is inserted between the signal sequence and the KDEL, as it will invert the membrane topology of the protein such that the SUN domain is no longer in the perinuclear space/ER lumen. Only the regions following the transmembrane domain can be used, i.e. the luminal domain.

(62) A person skilled in the art will appreciate that the present invention may be practiced without undue experimentation according to the methods given herein. The methods, techniques and chemicals are as described in the references given or from protocols in standard biotechnology and molecular biology text books.

EXAMPLES

Example 1

(63) Materials and Methods

(64) Mice were maintained at the A*STAR Biological Resource Centre facility and the NUS Animal Facility in accordance with the guidelines of the Institutional Animal Care and Use Committee for each facility. The Lmna.sup.Flx/Flx mice were generated and characterized as previously described [A. S. Wang, et al., *Differentiation; research in biological diversity*, (2015); I. Solovei et al., *Cell* 152: 584-598 (2013)] (FIG. 33). To derive mice with a global deletion Lmna (Lmna.sup.Δ/Δ), we crossed the floxed allele (Lmna.sup.Flx/Flx) to mice in which Cre recombinase is driven by the regulatory sequences of the mouse zona pellucida 3 gene (Zp3;Tg(Zp3-cre)93Kw, JAX stock 003651) [W. N. de Vries et al., *Genesis* 28:110-112 (2000)]. To obtain cardiomyocyte-specific deletion of Lmna (Lmna.sup.Flx/Flx/NlMhc), we first crossed the Lmna.sup.Flx/Flx mice to mice in which Cre expression was driven by the cardiac-specific murine alpha myosin-heavy chain (Myh6, myosin, heavy polypeptide 6, cardiac muscle, alpha) promoter (MyHC;Tg(Myhca-cre)2182Mds, JAX stock 011038). To obtain a tamoxifen inducible cardiomyocyte-specific deletion of Lmna (LmnFlx/Flx:mcm), we crossed the Lmna.sup.Flx/Flx with mice in which Cre expression was driven by the mouse cardiac-specific alpha-myosin heavy chain promoter (αMHC or alpha-MHC;Myh6) that expressed a tamoxifen-inducible Cre

recombinase (MerCreMer) specifically in juvenile and adult cardiac myocytes (mcm;Tg(Myh6-cre/Esr1*)1Jmk, JAX stock 005657). The specificity of mcm Cre expression to cardiomyocytes was confirmed by crossing Cre lines to the mT/mG reporter mice [M. D. Muzumdar, et al., *Genesis* 45: 593-605 (2007)] (FIG. 35). Generation of the Sun1.sup.-/-mice was previously described [Y. H. Chi et al., *Development* 136: 965-973 (2009)] as were the Lmna.sup.N195K/N195K mice [L. C. Mounkes, et al., *Hum Mol Genet* 14: 2167-2180 (2005)]. The Lmna.sup.Δ/Δ:Sun1.sup.-/- and Lmna.sup.Flx/Flx:mcm:Sun1.sup.-/- mice were obtained by crossing the respective Lamin-Cre mice strains with Sun1.sup.+/- mice as Sun1.sup.-/- mice are infertile.

(65) To test for the insertion of loxP sites and conditional deleted allele, genotyping was performed with a duplex PCR protocol with the following primers were used:

(66) TABLE-US-00002 FLX/FLX-F1: SEQ ID NO: 16 5'-CCAGCTTACAGAGCACCGAGCT-3', FLX/FLX-F2: SEQ ID NO: 17 5'-TCCTTGCAGTCCCTCTTGCATC-3', FLX/FLX-R1: SEQ ID NO: 18 5'-AGGCACCATTGTCACAGGGTC-3'.

To test for Sun1 deletion, the following primers were used:

(67) TABLE-US-00003 Sun1-F: SEQ ID NO: 19 5'-GGC AAG TGG ATC TCT TGT GAA TTC TTG AC-3' Sun1-R: SEQ ID NO: 20 5'-GTA GCA CCC ACC TTG GTG AGC TGG TAC-3' Sun1-E8: SEQ ID NO: 21 5'-AGC CAC ATA ACC ACC TGG AG-3'

To test for the MyHC transgene, the following primers were used:

(68) TABLE-US-00004 MyHC-tF: SEQ ID NO: 22 5'-ATG ACA GAC AGA TCC CTC CTA TCT CC-3' MyHC-tR: SEQ ID NO: 23 5'-CTC ATC ACT CGT TGC ATC ATC GAC-3' MyHC-F: SEQ ID NO: 24 5'-CAA ATG TTG CTT GTC TGG TG-3' MyHC-R: SEQ ID NO: 25 5'-GTC AGT CGA GTG CAC AGT TT-3'

To test for the presence of mcm transgene, the following primers were used:

(69) TABLE-US-00005 mcm-3798t: SEQ ID NO: 26 5'-AGG TGG ACC TGA TCA TGG AG-3' mcm-8346t: SEQ ID NO: 27 5'-ATA CCG GAG ATC ATG CAA GC-3' mcm-7338: SEQ ID NO: 28 5'-CTA GGC CAC AGA ATT GAA AGA TCT-3' mcm-7339: SEQ ID NO: 29 5'-GTA GGT GGA AAT TCT AGC ATC ATC C-3'

Tamoxifen Injection and Tissue Collection

(70) Young mice (14 days old) and adult mice (3-5 months old) were injected once with 40 mg/kg of Tamoxifen (Sigma) dissolved in Corn Oil (Sigma). Mice were sacrificed by CO.sub.2 euthanasia or anesthetised with a gaseous mixture of 1.5% Isoflurane (BioMac) and 1.5LO.sub.2 at various time points after tamoxifen injection. Cardiac arrest was induced by injection of 15% KCl, followed by flushing with PBS to remove blood. Hearts for paraffin embedding were additionally flushed with 4% paraformaldehyde (PFA), left in 4% paraformaldehyde (PFA) overnight, dehydrated in 70% ethanol for at least 24 hr and embedded in paraffin. Hearts for cryosection were embedded in tragacanth gum (Sigma), frozen in isopentane (BDH-AnalaR) cooled in liquid N.sub.2, cut 9 μm sections by cryostat (Leica CM3050), collected onto charged slides and stored at -20° C. for histological and immunofluorescence staining. Hearts for protein and RNA extraction were snap frozen in liquid N2 and stored for further processing.

(71) Cardiomyocyte Isolation

(72) Cardiomyocyte isolation was carried out as per standard protocol [M. Ackers-Johnson et al., *Circulation Research* 119: 909 (2016)]. Briefly, mice were anaesthetised with isoflurane (100% O.sub.2 at 0.5 L/min, isoflurane atomiser dial at 4%). Mice hearts were stopped with 15% KCl, descending aorta was cut and hearts were flushed with 7 mL of EDTA buffer into the right ventricle. Ascending aorta was clamped using Reynolds forceps, the entire heart removed and placed in a 60 mm dish containing fresh EDTA buffer. Hearts were digested by sequential

injections of 10 mL EDTA buffer, 3 mL Perfusion buffer and 30-50 mL Collagenase buffer into the left ventricle. Forceps were used to gently pull the digested heart into smaller pieces ~1 mm and gentle trituration. Enzymatic activity was inhibited by addition of 5 ml of Stop buffer. Cell suspension was passed through a 100 µm filter, and four sequential rounds of gravity settling to enrich for myocytes, ultimately obtaining a highly pure myocyte fraction. The myocyte pellet was snap frozen in liquid N.sub.2 and stored at -80° C. until further processing.

(73) Histological and Immunofluorescence Microscopy

(74) For histological studies, sections (9 µm) were stained with standard Hematoxylin and Eosin for cell morphology, Masson's trichrome stain to detect collagen and TUNEL assay to detect apoptotic nuclei. Images were obtained on a Zeiss Axio Imager Microscope. For immunofluorescence on frozen heart sections, sections were warmed to room temperature, rehydrated with PBS, blocked with M.O.M block (Vector Shields) and donkey serum (Sigma-Aldrich), incubated with primary antibodies overnight at 4° C. The slides were then washed in PBS and incubated with secondary antibodies and Hoechst dye (Sigma-Aldrich) for 60 mins, washed with PBS and mounted in Prolong-Gold Anti-fade reagent (Invitrogen). Primary antibodies: LMNA/C N-18 (goat, 1:50, Santa Cruz), Sun1 monoclonal (mouse, neat, from B. Burke), PCM-1 (rabbit, 1:200, Sigma) and sarcomere- α -actinin (mouse, 1:100, abcam); Secondary antibodies were: Alexa Fluor 488, 568 and 847 (1:250, Invitrogen). For isolated cardiomyocyte immunofluorescence, myocytes were stained in suspension and spun down gently for each solution change then plated on glass slides for imaging with a Zeiss LSM510 inverted confocal microscope.

(75) Western Analysis for LMNA, SUN1, Ha-Tag and GFP.

(76) Whole Hearts and Quadriceps muscles were homogenized in RIPA lysis buffer and spun at 13,200 g, 10 min, 4° C. Total cell lysates were electrophoresed and transferred to PVDF membrane and blocked with Odyssey Blocking Buffer (Li-Cor Biosciences). The membrane was incubated with primary antibodies for 2 h at room temperature. After which, membrane was washed in TBST washing solution and incubated in Odyssey IRDye secondary antibodies for 1 h before visualization on the Odyssey Infrared Imaging System (Li-Cor Biosciences). The primary antibodies used for detection of LMNA/C (Rabbit, Cell Signalling) that is specific to an epitope in the first 50 amino acids in LMNA, Sun1 monoclonal (mouse, 1:500, Burke) and control beta-tubulin (rabbit, 1:1000, Abcam).

(77) Active Force Measurement of Cardiac Papillary Muscle.

(78) Mouse papillary muscle from mouse left ventricle was prepared according to the methods described before [C. N. Toepfer, et al., *J Physiol* 594: 5237-5254 (2016)]. Briefly, explanted mouse heart was immediately rinsed with oxygenated ice-cold Krebs-Henseleit solution with 12 unit/mL heparin sodium (EDQM) and 30 mM 2,3-Butanedione monoxime (BDM, Sigma) and excess blood was removed. After that, the heart was transferred to ice-cold Krebs-Henseleit solution in a glass petri-dish under a dissection microscope with a cooling stage. Cylindrical papillary (200-300 µm in diameter and 1.5-2 mm in length) were excised from the left ventricle. T-shaped aluminium clips with a hole were crimped onto the ends of a papillary preparation and the prepared papillary chunks were fixed using pins onto a glass petri-dish with a layer of PDMS sylgard 184 (Dow Corning). Papillary preparations were immersed in a 2% Triton X-100 solution at 4° C. overnight.

(79) Force measurement was performed as previously described [C. Toepfer et al., *J Biol Chem* 288: 13448-13454 (2013)]. The T-shaped aluminium clips at the ends of the papillary preparations were attached to the hooks of a force transducer (AE801, HJK Sensoren+Systeme) and servomotor in the experimental rig and were glued with shellac in ethanol (Sigma) to minimize the movement during the experiment. Papillary contraction force was measured at 20° C. The max contraction force was measured in activating solution (100 mM TES, 6.5 mM MgCl₂, 25 mM Ca-EGTA, 5.7 mM Na₂ATP, 20 mM Glutathione, 21.5 mM sodium creatine phosphate, pH=7.1, Ionic strength is 150 mmol/L) with 32 µmol/L free Ca²⁺. The data were collected and processed from the force transducer and DAQ data acquisition device (National Instrument) using a customized

software programmed by LabVIEW 2013 (National instrument). At least 5 fibres were tested in each mouse, and at least 3 mice were tested for each experimental group.

(80) AAV9-N-Sun1 and AAV9-GFP Virus

(81) The DN-Sun1 (SS-HA-Sun1L-KDEL) and GFP (SS-GFP-KDEL) vectors were as described [M. Crisp et al., *J Cell Biol.* 172: 41-53 (200)]. Briefly, almost the entire luminal domain of Sun1 was tagged at its NH.sub.2 terminus with HA (HA-Sun1L). To introduce the HA-Sun1L as a soluble form into the lumen of the ER and PNS, signal sequence and signal peptidase cleavage site of human serum albumin was fused onto the NH.sub.2 terminus of HA-Sun1L to yield SS-HA-Sun1L. To prevent its secretion, a KDEL tetrapeptide was fused to the COOH terminus of SS-HA-Sun1L to form the final SS-HA-Sun1L-KDEL. The HA-Sun1L region was replaced with GFP sequence to generate the SS-GFP-KDEL.

(82) The DN-Sun1 and GFP fragments were amplified with the primers listed below (same forward primer was used for both fragments) and ligated into pENN-AAV-cTnT-Pl-eGFP plasmid (kind gift from Dr J. Jian), digested with NcoI and KpnI, to produce Penn-AAV-cTnT-Sun1DN (FIG. 10; SEQ ID NO: 3).

(83) TABLE-US-00006 aav Sun1 F SEQ ID NO: 30 5'-

CgagaattcacgcgggccgccATGAAGTGGGTAACCTTTATTTC-3' aav Sun1 R SEQ ID

NO: 31 5'-CgggtcgactctagaggtaccttaCTACAACCTCATCTTTCTGGATG-3' aav GFP Sun R
SEQ ID NO: 32 5'-CgggtcgactctagaggtacttaCTACAACCTCATCTTIGGATCC-3'

(84) All restriction enzymes were purchased from NEB. PCR reactions were conducted using Q54 Hot Start High-Fidelity 2× Master Mix (NEB, M0494L). Ligations were conducted using isothermal assembly with NEBuilder® HiFi DNA Assembly Master Mix (NEB, E2621L). Primers used for constructing the plasmids were ordered from IDT.

(85) AAV Viruses were produced as per standard protocol [H. Wakimoto, et al., in *Current Protocols in Molecular Biology*. (John Wiley & Sons, Inc., 2001)]. Materials supplied by R. Foo: pAAV2/9- the trans-plasmid encoding AAV replicase and capsid gene (SEQ ID NO: 2; available from University of Pennsylvania Penn Vector Core); pAdDeltaF6—the adenoviral helper plasmid (SEQ ID NO: 1) (available from University of Pennsylvania Penn Vector Core); QIAGEN Plasmid Maxi Kit; HEK293T cells (ATCC); Transfection reagent (polyethylenimine e.g., Polysciences). AAV-DJ capsid was obtained from Cell Biolabs, Inc. The pAAV2/9, AAV-DJ, pAdDeltaF6, DN-Sun1 and GFP plasmids were purified using a QIAGEN Plasmid Maxi Kit HEK293T cells were transfected with the virus combination of pAAV2/9, pAdDeltaF6 and either DN-Sun1 or GFP plasmids. Cells were collected and virus purified by iodixnol gradient ultracentrifugation.

(86) The following timeline was used for infection of the mouse hearts. Mice were genotyped at 10 days postnatally. They were then subjected to 1 IP injection of Tmx (40 mg/kg of mouse weight) at 14 days postnatally, followed by a concentration of 5×10^{10} vg/g AAV9-DN-Sun1 or AAV9-GFP virus injected into the thoracic cavity at 15 days postnatally. Adult mice (3-5 months old) were injected IP with a single dose of Tmx (40 mg/kg of mouse weight), followed by injection of AAV at a concentration of 5×10^{10} vg/g AAV9-DN-Sun1 or AAV9-GFP virus into the thoracic cavity. Young and adult mice were anesthetised with a gaseous mixture of 1.5% Isoflurane (BioMac) and 1.5 L O.sub.2 before virus injections.

(87) Plasmid Construction and Generation of Cas9 mRNA and sgRNAs

(88) pX330 was obtained from Addgene (#42230, Cambridge, MA, USA). The 20 nt Sun1 and Syne1 single guide RNA (sgRNA) sequences were designed with the help of CRISPR Design Tool (crispr.genome-engineering.org). A region of the gene of interest was submitted to the tool to identify suitable target sites. Since off-target mutations are possible in CRISPR/Cas9-mediated targeted mutagenesis in the mouse, the CRISPR Design Tool is able to experimentally assess off-target genomic modifications for each gRNA target sites and provide computationally predicted off-target sites for each intended target, ranking the target sequence according to quantitative specificity analysis on the effects of base-pairing mismatch identity, position and distribution.

Complimentary oligonucleotides containing the gRNA target sequences were annealed and cloned into the BbsI site of pX330. Guide RNA sequences were as follows:

(89) TABLE-US-00007 (SEQ ID NO: 33) 5'-GCACAATAGCCTCGGATGTCG-3' for Sun1 Δ SUN, (SEQ ID NO: 34) 5'-CCGTTGGTATATCTGAGCAT-3' for Syne1-stop, (SEQ ID NO: 35) 5'-GGTTATGGCCGATAGGTGCAT-3' for Tyrosinase4a

(90) These plasmids (pSun1 Δ SUN, pSyne1-stop and pTyrosinase4a) were then sequenced to verify correct insertion of the target sequences. For in vitro transcription, PCR was performed to generate the appropriate transcription templates using a common reverse primer

(AAAAGCACCGACTCGGTGCC-3'; SEQ ID NO: 36) and gRNA-specific forward primers that encoded the T7 promoter sequence as follows:

(91) TABLE-US-00008 Sun1 Δ SUN: (SEQ ID NO: 37) 5'-

TTAATACGACTCACTATAGCACAATAGCCTCGGATGTCG-3'; Syne1-stop: (SEQ ID

NO: 38) 5'-TTAATACGACTCACTATAGCCGTTGGTATATCTGAGCAT-3'; Tyrosinase4a:

(SEQ ID NO: 39) 5'-TTAATACGACTCACTATAGGTTATGGCCGATAGGTGCAT-3'

(92) The gRNA PCR products were then subjected to agarose gel electrophoresis (1.5% agarose) to confirm successful PCR, gel purified and used as templates for in vitro transcription using the MEGAshortscript T7 kit (Life Technologies). The gRNAs were purified using MEGAclean kit (Life Technologies) and eluted in RNase-free water. A sample of purified gRNAs were then subjected to agarose gel electrophoresis for quality checks before injecting into zygotes.

(93) Generation of Mutant Mice Using CRISPR/Cas9

(94) 3 to 4 weeks old C57BL/6N females were superovulated with Pregnant Mare Serum gonadotropin (Calbiochem, 38722, 5 IU/ml). 48 hours later, the females were injected with human chorionic gonadotropin (Sigma, CG10, 5 IU/ml) and were mated with C57BL6 males. The following day, fertilized 0.5 dpc embryos were collected from the oviducts. Cas9 mRNA (Sigma, CAS9MRNA, 100 ng/ul), Tyrosinase4a gRNA (50 ng/ul) and gene-specific gRNA (50 ng/ul) were co-injected into the cytoplasm of the embryos in M2 medium (EmbryoMax® Sigma) using a microinjection system (Nikon). Syne1-stop sgRNA were used to derive Syne1 C'T mutant mice and Sun1 Δ SUN sgRNA were used to derive Sun1 Δ SUN mutant mice. The injected zygotes were cultured in KSOM with amino acids (EmbryoMax® Sigma) in an incubator maintained at 37° C. with 5% CO₂ and 5% O₂ for 2 hours before implanting into 0.5 dpc pseudopregnant C3H-ICR females.

(95) DNA Extraction for Genotyping of CRISPR/Cas9 Mice

(96) Mouse tails were clipped and each placed in a 1.5 ml Eppendorf tube. 80 μ l of lysis buffer (25 mM NaOH, 0.2 mM EDTA, pH 12) was dispensed into the tube and heated at 95° C. for 60 minutes. After heating, the buffer was neutralized with an equal volume of 40 mM Tris-HCl, pH 5. For certain applications, DNA was extracted and purified from mouse tails using DNeasy Blood and Tissue Kit (QIAGEN).

(97) Genotyping of CRISPR/Cas9 Mice

(98) CRISPR modified mutant mice were genotyped by PCR followed by gel electrophoresis using a high resolution agarose (2% MetaPhor agarose, Lonza).

(99) Primers for Syne1CT' Δ 8 Mice were:

(100) TABLE-US-00009 Forward: SEQ ID NO: 40 5'-TGCTCCTGCTGCTGCTTATT-3' and Reverse: SEQ ID NO: 41 5'- ACATGGTGGAGCATTTGTCTCC -3'

Primers for Sun1 CRISPR Mice were:

(101) TABLE-US-00010 Forward: SEQ ID NO: 42 5'-TGACCTTGAGCTGAAACTGC-3' and Reverse: SEQ ID NO: 43 5'-TCAGAACACTGGCACACACA-3'

(102) Lmna mutant mice were genotyped as described in Example 1. To determine sequence of CRISPR-Induced mutations, PCR products from mouse tail DNA were subjected to TOPO cloning (Zero Blunt™ TOPO™ PCR Cloning Kit, 450245, Thermo Fisher Scientific). Plasmid DNA from at least 10 bacterial colonies were isolated using a mini-prep kit (QIAGEN, QIAprepSpin,

Miniprep Kit) and sent for Sanger sequencing.

(103) Derivation of Myoblasts, Fibroblasts and Cell Culture for CRISPR/Cas9 Study

(104) To isolate myoblasts, limbs were obtained from euthanized mice and muscles were dissected from bone. Tissue digestion was performed by incubating the muscle tissues in enzyme solution consisting of equal volumes of dispase II (Roche, cat. 04942078001) at a concentration of 2.4 U/ml and 1% collagenase II (GIBCO® Invitrogen, cat 17101-015) in a 37° C. water bath for 30 minutes, with occasional mixing at 10 minutes interval. After 30 minutes, enzyme solution was neutralized in D10 media (Dulbecco's Modified Eagle Medium (DMEM) with 10% fetal bovine serum).

Mixture is then filtered through 70 µm sterile filter (BD Falcon™, cat 352350) and 40 µm sterile filter (BD Falcon™, cat 352340). The suspension was then centrifuged, supernatant removed and subsequently resuspended in F10 media (GIBCO® Invitrogen, cat. 11550043) supplemented with 10 µg/ml bFGF (GIBCO®, cat PHG0264) and plated in 100 mm plates. Mouse adult fibroblasts were allowed to settle for 2 to 3 hours before collecting the supernatant (with floating myoblasts) and replated into 60 mm plates coated with 0.15% Gelatin (Sigma, cat G1393). D10 media was added to the 100 mm plates with MAFs. To terminally differentiate myoblasts to myotubes, the media was changed to DMEM supplemented with 2% horse serum (Thermo Fisher Scientific GIBCO®, cat 16050122).

(105) Immunoblotting for CRISPR/Cas9 Study

(106) Whole cell lysates were generated using the Lysis-M kit solution (cOmplete; Roche). Cells were washed in ice-cold PBS and lysed with Roche Lysis M buffer, and centrifuged at 14,000 g for 10 minutes to remove cell debris. To extract protein from tissue sample, small slices of tissue were rapidly placed into Lysing Matrix D tubes (MP Biomedicals), and snap frozen in liquid nitrogen. After snap freezing, the tubes were either stored at -80° C. or used directly for protein analysis. Protein extraction buffer (50 mM Tris (pH7.4), 500 mM NaCl, 0.4% SDS, 5 mM EDTA (pH7.4), 1× Protease inhibitor (cOmplete™ EDTA-free Protease Inhibitor cocktail, Cat no. 04693159001, Roche), 2% Triton, 1 mM Dithiothreitol, in distilled water) was added to tissues, which were then homogenized using the FastPrep™-24 Instrument (MP Biomedicals). Samples were then centrifuged at 14,000 g for 10 minutes to remove cell debris. Protein concentration was quantified using bicinchoninic acid (BCA) protein kit (Bio-Rad) before loading protein samples onto a polyacrylamide gel to ensure equal amounts were being analyzed. All protein samples were resolved by SDS-PAGE gel analysis and transferred onto polyvinylidene fluoride (PVDF) membrane (Millipore) by wet transfer for 48 hours at 20V at 4° C. Membranes were blocked in TBS containing 0.1% Tween 20 (TBST) supplemented 5% milk powder (Anlene) for 1 hour at room temperature. Western Blot analysis was performed using primary antibodies diluted in 5% milk powder (diluted in TBST). Membranes were incubated for 2 hours at room temperature or overnight at 4° C. For secondary antibodies, horseradish-peroxidase (HRP) (Invitrogen) conjugated antibodies were used for chemiluminescent imaging. The membranes were incubated for 1 hour at room temperature with the secondary antibodies. For immunoblots visualized by chemiluminescence, membranes were incubated in ECL substrate (Pierce) for 1 minute before being exposed to a chemiluminescence sensitive film (Thermo Scientific) and subsequently processed.

(107) Immunofluorescence for CRISPR/Cas9 Study

(108) Cells were grown in 8-well slides (Ibidi) and fixed in ice-cold methanol for 15 minutes at -20° C. They were then rinsed in PBS twice and permeabilized and blocked with 0.1% Triton X, 3% BSA in PBS for 15 minutes at room temperature. The fixed and permeabilized cells were then rinsed in PBS three times. Samples were then incubated with primary antibodies (Table 2) for 2 hours at room temperature or overnight at 4° C. Samples were then washed with PBS three times and subsequently incubated with secondary antibodies (Life Technologies) and DAPI (Life Technologies) for 1 hour at room temperature. After three washes in PBS, cells were mounted in Anti-fade (1% DABCO, 90% Glycerol, 10% PBS) and inspected using a Zeiss 510 Meta Confocal

microscope or Axiovert 200 inverted epifluorescence microscope (Zeiss). Images were recorded and analysed using Zeiss ZEN, Metamorph or Image J (NIH) software.

(109) TABLE-US-00011 TABLE 2 Antibodies used for immunofluorescence study. Antibody Type and Source Concentration Akap450, HPA-026109 Polyclonal, Sigma 1:500 MF20 Monoclonal, DSHB 1:25 Nesp1 (MANNES1A) Monoclonal, Glenn Morris 1:1000 (Western) 1:50 (IF) Nesp1-C'T Monoclonal, Brian Burke Undiluted supernatant LaminA, ab8984 Monoclonal, Abcam 1:200 LaminA, SSD Monoclonal, Brian Burke 1:200 Pcm-1, HPA-023374 Polyclonal, Sigma 1:100 Pcnt, ab4448 Polyclonal, Abcam 1:100 Sun1-9F10 Monoclonal, Brian Burke 1:200 Sun2-3.1E Monoclonal, Brian Burke 1:500 Nesprin-2 Polyclonal, MyBiosource.com 1:500

Mouse Genetics

(110) Lmna mice and tamoxifen injection were described in Example 1. To obtain Lmna.sup.Δ/Δ:Syne1.sup.C'TΔ8/C'TΔ8 and Lmna.sup.Flx/Flxmcm:Syne1.sup.C'TΔ8/C'TΔ8 double mutant mice, Lmna.sup.Δ/+ or Lmna.sup.Flx/Flx:mcm mice were intercrossed with Syne1.sup.C'TΔ8/C'TΔ8 mice. In the Syne2 mouse model, a IRES-β-gal neomycin selectable cassette (PgkNeo) flanked by IoxP sites was inserted into the Syne2 gene, resulting in deletion of part of axon 102 and all of exons 103-104. The neomycin cassette was subsequently removed by crossing with Cre recombinase mice. Syne1.sup.C'TΔ8/+ or Syne1.sup.C'TΔ8/C'TΔ8 mice were crossed with Syne2.sup.+/- or Syne2.sup.-/- mice to obtain mice with mutant Syne1 and Syne2 alleles, which were intercrossed to obtain double mutant mice. Kaplan-Meier method was used to draw the survival curves.

(111) Human Guide RNA Sequences

(112) Potential guide RNA sequences to disrupt human SYNE1 KASH domain or SUN1 SUN domain were determined using CRISPR tool in Benchling software (Benchling Inc. USA) and are shown in Table 3.

(113) TABLE-US-00012 TABLE 3 Potential guide RNA sequences to target final exons in human SYNE1 (Nesprin-1) or SUN1 genes

SEQ ID	Gene Name	ENSEMBL gene ID	Chromosome	Position	Strand	Sequence	PAM	CRISPR enzyme	NO:
SYNE1	ENSG00000131018	6	515330	-	TCGTGTATCTGAGCATGGGG	TGGAAT	saCas9	44	
SYNE1	ENSG00000131018	6	515335	-	GCCATTCGTGTATCTGAGCA	TGGGGT	saCas9	45	
SYNE1	ENSG00000131018	6	515340	+	TCCACCCCATGCTCAGATAC	ACGAAT	saCas9	46	
SYNE1	ENSG00000131018	6	515320	-	GAGCATGGGGTGGGAATGACC	GGG	spCas9	47	
SYNE1	ENSG00000131018	6	515321	-	TGAGCATGGGGTGGGAATGAC	CGG	spCas9	48	
SYNE1	ENSG00000131018	6	515330	-	TCGTGTATCTGAGCATGGGG	TGG	spCas9	49	
SYNE1	ENSG00000131018	6	515333	-	CATTCGTGTATCTGAGCATG	GGG	spCas9	50	
SYNE1	ENSG00000131018	6	515334	-	CCATTCGTGTATCTGAGCAT	GGG	spCas9	51	
SYNE1	ENSG00000131018	6	515335	-	GCCATTCGTGTATCTGAGCA	TGG	spCas9	52	
SYNE1	ENSG00000131018	6	515345	+	CCCATGCTCAGATACACGAA	TGG	spCas9	53	
SUN1	ENSG00000131018	6	515333	+	CCCGGTCATTCCACCCCATG	TTTG	Cpf1	54	
SUN1	ENSG00000164828	7	873276	+	TTTTTCTAACTGGGGCCATC	CTGAGT	saCas9	55	
SUN1	ENSG00000164828	7	873285	-	CCGATACAGACAGGTATACT	CAGGAT	saCas9	56	
SUN1	ENSG00000164828	7	873266	+	AACTTCGGATTTTTTCTAAC	TGG	spCas9	57	
SUN1	ENSG00000164828	7	873267	+	ACTTCGGATTTTTTCTAACT	GGG	spCas9	58	
SUN1	ENSG00000164828	7	873268	+	CTTCGGATTTTTTCTAACTG	GGG	spCas9	59	
SUN1	ENSG00000164828	7	873280	-	ACAGACAGGTATACTCAGGA	TGG	spCas9	60	
SUN1	ENSG00000164828	7	873296	+	CTGAGTATACCTGTCTGTAT	CGG	spCas9	61	
SUN1	ENSG00000164828	7	873281	+	TTCTAACTGGGGCCATCCTG	TTTT	Cpf1	62	
SUN1	ENSG00000164828	7	873282	+	TCTAACTGGGGCCATCCTGA	TTTT	Cpf1	63	
SUN1	ENSG00000164828	7	873283	+	CTAACTGGGGCCATCCTGAG	TTTT	Cpf1	64	
SUN1	ENSG00000164828	7	873284	+	TAACTGGGGCCATCCTGAGT	TTTC	Cpf1	65	

PAM, protospacer adjacent motif; saCas9, *Staphylococcus aureus* Cas9; spCas9, *Streptococcus pyogenes*

Cas9; Cpf1, CRISPR from *Prevotella* and *Francisella* 1.

Statistical Analysis

(114) All statistical analysis was performed using Graphpad Prism software.

Example 2

(115) Cardiomyocyte Specific Loss of Lmna Results in the Rapid Onset of Heart Failure

(116) To further define the interaction between Sun1 and Lmna in postnatal pathology in mice, we specifically ablated the Lmna gene in different tissues by using a conditional Lmna.sup.Flx/Flx line of mice (FIG. 33), that when recombined by Cre activation, results in the complete loss of LaminA/C protein [A. S. Wang, et al., *Differentiation; research in biological diversity*, (2015); I. Solovei et al., *Cell* 152: 584-598 (2013)]. When Lmna.sup.Flx/Flx was constitutively deleted in all tissues by crossing the Lmna.sup.Flx/Flx mice with Zp3-Cre mice [W. N. de Vries et al., *Genesis* 26: 110-112 (2000)], the mean postnatal lifespan was 17.5 days (FIG. 26A). When the same deletion was induced in the absence of Sun1, the Lmna.sup.Δ/Δ:Sun1.sup.-/- mice lived to a mean of 32.5 days, almost a doubling in longevity (FIG. 28A). Performing the same Lmna deletion on a Sun2 null background did not extend the longevity of Lmna.sup.Δ/Δ mice, revealing the longevity extension is specific to the loss of Sun1 (FIG. 36). Since the A-type lamins are widely expressed in almost all adult tissues, we then determined to what extent, Lmna deletion, specifically in cardiomyocytes, contributes to the early postnatal death of Lmn.sup.Δ/Δ mice. Furthermore, the inventors wished to ascertain whether loss of Sun1 would increase longevity in these mice harbouring Lmna deficient cardiomyocytes. We first crossed the Lmna.sup.Flx/Flx with a constitutive myh6 Cre [R. Agah et al., *J Clin Invest* 100: 169-179 (1997)], in which Cre expression, though constitutive, is restricted to cardiomyocytes but commences during embryogenesis. These mice survived slightly longer than the Lmna.sup.Δ/Δ to an average of 26.5 days postnatally (FIG. 26C). When the same cardiomyocyte specific deletion was performed on a Sun1.sup.-/- background, this resulted in a significant increase in longevity to at least 6 months and beyond after birth (FIG. 26C). To further define the loss of Lmna and its effect in postnatal/adult cardiomyocytes we derived mice homozygous for the Lmn.sup.Flx/Flx allele carrying the inducible cardiomyocyte specific Cre Tg(Myh6-cre/Esr1), (here abbreviated to mcm) in which Cre is induced by a single injection of tamoxifen (Tmx) [D. S. Sohal et al., *Circ Res* 89: 20-25 (2001)]. From this cross, the average lifespan of 3-5 month old Lmna.sup.Flx/Flx:mcm mice, following Cre induction was 27 days (FIG. 27A). Controls were unaffected by Tmx injection. PCR and immunofluorescence analysis confirmed the Lmna deletion was specific to the Lmna.sup.Flx/Flx:mcm cardiomyocytes, with no detectable recombination occurring in the brain, diaphragm, lung, liver and skeletal muscle, or in wild-type control animals (FIG. 27B). By 21 days post injection, Lmna.sup.Flx/Flx:mcm mice showed laboured breathing, a dishevelled, ungroomed appearance, increased lethargy and kyphosis (FIG. 27C). Immunofluorescence analysis of isolated cardiomyocytes (CM) and sections of Lmna.sup.Flx/Flx:mcm hearts showed reduced levels of LaminA protein and cardiomyocyte nuclei without any LaminA expression (FIG. 27D). LaminA protein levels were decreased 3.5 fold in Lmna.sup.Flx/Flx:mcm hearts after Cre induction compared to uninduced Lmna.sup.Flx/Flx:mcm and Lmna.sup.+/+/mcm hearts (FIG. 27E). By sampling Lmna.sup.Flx/Flx:mcm mice at specific time points after Tmx injection, it was estimated that it takes 7-14 days after Cre induction for LMNA protein levels to fall by 50% (data not shown), a rate consistent with a study using siRNA LMNA knockdown in human fibroblasts by 1.3-fold after 48 hrs and a further 4-fold reduction after 10.5 days [A. Buchwalter and M. W. Hetzer, *Nature communications* 8: 328 (2017); T. Sieprath et al., *Nucleus* 6: 236-248 (2015)]. Echocardiograms (ECGs) performed at 21 days after Cre induction revealed poor cardiac contractility in Lmna.sup.Flx/Flx:mcm mice compared to Lmna.sup.Flx/Flx:mcm controls (FIG. 28A). There was a significant reduction in the Ejection Fraction (EF %) and Fractional shortening (FS %) (P<0.0001) (FIG. 288). The left systolic and diastolic ventricular internal diameters (LVID) were enlarged, compared to Lmna.sup.Flx/Flx:mcm controls (FIG. 28B). Significantly fewer viable

(brick-like) cardiomyocytes were isolated from *Lmna.sup.Flx/Flx:mcm+Tmx* hearts compared to *Lmna.sup.Flx/Flx:mcm* controls (FIG. 28C). Visual analysis revealed the isolated cardiomyocytes from *Lmna.sup.Flx/Flx:mcm+Tmx* hearts contained large intracellular vacuoles (FIG. 28C). Histological analysis of *Lmna.sup.Flx/Flx:mcm+Tmx* hearts, revealed infiltration of nucleated cells and increased intercellular spaces between cardiomyocytes compared to *Lmna.sup.Flx/Flx:mcm* control hearts (FIG. 28D). The left ventricular lumen in *Lmna.sup.Flx/Flx:mcm+Tmx* hearts was significantly enlarged, together with significantly increased levels ($P=0.0098$) of fibrosis were noted in *Lmna.sup.Flx/Flx:mcm+Tmx* hearts compared to in *Lmna.sup.Flx/Flx:mcm* controls (FIG. 28D). Increased numbers of apoptotic cells were also identified in *Lmna.sup.Flx/Flx:mcm+Tmx* hearts compared to control hearts (FIG. 28D). However there was no evidence of extensive DNA damage detectable in the cardiomyocytes, as assessed by Rad51, MRE11, H2AX phosphor-Ser and 53BP1 immunostaining (Data not shown).

Example 3

(117) Deletion of Sun1 Ameliorates Cardiac Pathology Induced by *Lmna* Loss

(118) Mice with *Lmna* mutations show a significant increase in longevity and health in the absence of Sun1 [C. Y. Chen et al., *Cell* 149: 565-577 (2012)]. As described, induced deletion of *Lmna* in cardiomyocytes (*Lmna.sup.Flx/Flx:mcm+Tmx*) results in death within 1 month post Cre induction (FIG. 26C). Strikingly, when the same deletion was induced on a Sun1 null background the mice survived for more than 1 year after Cre induction (FIG. 26C). Hearts from *Lmna.sup.Flx/Flx:mcmSun1.sup.-/-+Tmx* mice, 3 weeks after induction, were compared to those from *Lmna.sup.Flx/Flx:mcm Sun1.sup.+/-+Tmx* to determine the extent to which SUN1 loss ameliorated the pathological changes induced by *Lmna* loss in cardiomyocytes. Immunofluorescent imaging for Lamin A/C identified many elongated and distorted nuclei. In some of these, residual Lamin A/C was displaced to one pole of the nucleus (FIG. 29A panel 1 and insert) in the *Lmna.sup.Flx/Flx:mcm Sun1.sup.+/-+Tmx* hearts. In contrast in the *Lmna.sup.Flx/Flx:mcm Sun1.sup.-/-+Tmx* hearts, while there were many elongated nuclei, these showed few if any distortions, even when there was no Lamin A/C staining (FIG. 29A panel 3 yellow arrow heads). Western analysis of whole hearts revealed a significant reduction in Lamin A/C in the *Lmna.sup.Flx/Flx:mcmSun1.sup.-/-+Tmx* hearts ($P=0.0359$) lysates compared to *Lmna.sup.Flx/Flx:mcmSun1.sup.+/-* controls (FIG. 29A lower panels). The *Lmna.sup.Flx/Flx:mcmSun1.sup.+/-+Tmx* cardiomyocyte nuclei exhibited increased longitudinal length, together with a segmented appearance, with the segments connected by narrow bridges (FIGS. 29A and C, panel 1 arrows). However, in the absence of Sun1, *Lmna.sup.Flx/Flx:mcm Sun1.sup.-/-* cardiomyocyte nuclei exhibited no abnormalities or segmentation (FIG. 29C panels 3 and 4). In total, 70% of cardiomyocytes in *Lmna.sup.Flx/Flx:mcmSun1.sup.+/-* mice had ruptured or misshapen nuclei compared to fewer than 1% of the cardiomyocytes from the *Lmna.sup.Flx/Flx:mcmSun1.sup.-/-* (FIG. 29C panel 5).

(119) Clear enlargement of the left ventricle (LV) was evident in the *Lmna.sup.Flx/Flx:mcmSun1.sup.+/-* mice but not in the LVs of the *Lmna.sup.Flx/Flx:mcmSun1.sup.-/-+Tmx* hearts (FIG. 29B, panels 1 and 2). The *Lmna.sup.Flx/Flx:mcmSun1.sup.+/-* hearts exhibited significantly increased levels of fibrosis ($P<0.0001$) compared to controls, whereas there was no significant fibrosis in the *Lmna.sup.Flx/Flx:mcmSun1.sup.-/-* hearts (FIG. 29B, panels 3-5).

(120) As a model for ventricular muscle mechanics we measured the active force in cardiac papillary muscle. The active force was significantly reduced by 66% in *Lmna.sup.Flx/Flx:mcm:Sun1.sup.+/-+Tmx* papillary muscle ($P=0.0028$) compared to *Lmna.sup.Flx/Flx:mcm Sun1.sup.+/-CTL*. In the absence of SUN1, *Lmna.sup.Flx/Flx:mcmSun1.sup.-/-+Tmx* cardiac papillary active force was maintained at levels not significantly different from those of controls (FIG. 29B panel 6).

(121) Echocardiograms performed before and after Cre induction revealed progressive worsening

of cardiac contractility in the *Lmna.sup.Flx/Flx:mcm Sun1.sup.+/++Tmx* mice compared to *Lmna.sup.Flx/Flx:mcmSun1.sup.-/+Tmx* mice (FIG. 29D). Loss of SUN1 preserved both EF, FS and Global Longitudinal Strain (GLS) (GLS is a separate parameter used to assess myocardial contractility, and is a better predictor of heart failure) in *Lmna.sup.Flx/Flx:mcmSun1.sup.-/+Tmx* mice compared to *Lmna.sup.Flx/Flx:mcm Sun1.sup.+/++Tmx* mice.

(122) PCR analysis of the aged *Lmna.sup.Flx/Flx:mcmSun1.sup.-/-* Tmx hearts 12-14 months after Tmx injection confirmed the sustained deletion of *Lmna* gene (FIG. 37C), while protein quantification revealed a significant reduction of LMNA levels in *Lmna.sup.Flx/Flx:mcm Sun1.sup.-/+Tmx* hearts 12-14 months after Tmx (FIG. 37D). Histological analysis of the 12-14 month *Lmna.sup.Flx/Flx:mcm Sun1.sup.-/+Tmx* hearts revealed no significant increase in fibrosis compared to controls (FIGS. 37A and B). However, echocardiograms on these aged mice showed reduced EF and FS in both *Lmna.sup.Flx/Flx:mcm Sun1.sup.+/++CTL* and *Lmna.sup.Flx/FlxmcmSun1.sup.-/+Tmx* mice (FIG. 37E), although the average lifespan of *Lmna.sup.Flx/Flx* mice is 13-14 months (FIG. 26C) and so the reduced contractile function may have been due to ageing. Together these findings demonstrate that loss of *Lmna*, in adult (2-3 month old) cardiomyocytes is sufficient to result in cardiac failure within 3-4 weeks after Cre activation, but the pathology is strikingly reduced by deleting *Sun1*, with this reduction being sustained for a year.

Example 4

(123) Loss of SUN1 Extends Longevity of *Lmna* Missense Mutants

(124) As most cases of LMNA induced DCM result from missense mutations, we determined what effect loss of SUN1 had on the longevity and cardiac function of a previously described *Lmna* mutant mouse line carrying the N195K missense mutation that dies from DCM [L. C. Mounkes, et al., *Hum Mol Genet* 14: 2167-2180 (2005)], with this mutation having been identified in 2 unrelated patients diagnosed with AD-EDMD [D. Fatkin et al., *N Engl J Med* 341: 1715-1724 (1999); J. P. van Tintelen et al., *Am Heart J* 154: 1130-1139 (2007)]. Here too, we found that the absence of SUN significantly extended the lifespan of this mutant mouse line with improved cardiac function (FIG. 26D). We extended these findings by deriving mice heterozygous for the N195K mutation, with the WT-*Lmna* allele being floxed i.e. *Lmna.sup.N195K/Flx* × *Sun1.sup.+/+*. Inducing the Tmx inducible cardiomyocyte Cre allele in these mice

(*Lmna.sup.N195K/Flx:mcm+Tmx*) resulted in the deletion of the WT floxed *Lmna* allele making the cardiomyocytes hemizygous for the *Lmna.sup.N195K/-* mutation. These mice had a mean lifespan of less than 50 days, a longevity half that of the original *Lmna.sup.N195K/N195K* homozygotes (FIG. 30A). When the *Lmna.sup.N195K/flx:mcm+Tmx* mutation was induced on a *Sun1* null background longevity was significantly extended from <50 days to >200 days (FIG. 30A), revealing that loss of *Sun1* is also effective at preventing DCM caused by *Lmna* missense mutations specifically in cardiomyocytes.

(125) Echocardiograms performed before and after Cre induction revealed progressive worsening of cardiac contractility in the *Lmna.sup.N195K/Flx:mcm Sun1.sup.+/+* mice compared to *Lmna.sup.N195K/Flx:mcm Sun1.sup.-/-* mice (FIG. 30B). Loss of SUN1 preserved both EF, FS and Global Longitudinal Strain (GLS) in *Lmna.sup.N195K/-:mcm Sun1.sup.-/-* mice compared to *Lmna.sup.N195K/-:mcm Sun1.sup.+/+* mice (FIG. 308).

Example 5

(126) AAV9 Mediated Transduction and Expression of a *DNSun1* Prolongs the Lifespan of the *Lmna.sup.Flx/Flx:mcm+Tmx* Mice.

(127) The above results demonstrated that genetically ablating SUN1's functions or reducing SUN levels could be of therapeutic value in treating DCM. We then tested whether this was due to the complete ablation of SUN1's functions to overcome its toxic over-abundance versus leaving its levels untouched and specifically disrupting its LINC complex-associated role in tethering KASH-domain proteins in the ONM, thereby tethering the nucleus to components of the cytoskeleton. To

distinguish between these 2 possibilities, Adenovirus Associated Virus (AAV) was utilized to transduce and express, specifically in cardiomyocytes, a dominant negative SUN minigene whose protein product would compete with both SUN1- and SUN2-KASH binding in the cardiomyocyte perinuclear space [M. Crisp et al., *J Cell Biol* 172: 41-53 (2006)]. A region corresponding to the entire luminal domain of the Sun1 gene was tagged at its N terminus with an HA (HA-Sun1L) epitope. To localize the resulting protein product to the endoplasmic reticulum (ER) and perinuclear space (between the INM and ONM-PNS), the signal sequence and signal peptidase cleavage site of human serum albumin was fused to the N terminus of HA-Sun1L to yield SS-HA-Sun1L. To prevent the miniprotein's secretion, a KDEL tetrapeptide was linked to the C-terminus of SS-HA-Sun1L, forming SS-HA-Sun1L-KDEL (FIG. 34). The signal sequence would ensure the HA-Sun1KDEL accumulates intracellularly within the contiguous peripheral ER and PNS lumen. The cDNA sequence encoding the minigene was fused to the chicken cardiomyonin promoter (cTnT) to ensure the minigene is only transcribed in cardiomyocytes [K. M. Prasad, et al., *Gene Ther* 18: 43-52 (2011)]. A diagram of how SS-HA-Sun1L (DN-Sun1) displaces the KASH domain proteins from the LINC complex in the PNS to the ER is presented in FIG. 15 (third panel) and FIG. 31B.

(128) To verify that the DN-Sun1 functioned in cardiomyocytes (CM) we initially transduced human CMs derived from PS stem cells using the AAV-DJ system [D. Grimm, et al., *J Virol* 82(12):5887-911 (2008)] that provides for a higher infectivity rate in cultured cells than the AAV9 serotype used to transduce the DN-Sun1, under transcriptional control of the cTnT promoter, in the mouse hearts. The DN-Sun1 was effective at displacing Nesprin-1 from the nuclear envelopes in the CMs that were expressing the DN-Sun1 as shown in FIG. 31D. Cells expressing high levels and low levels of DN-Sun1 are indicated by grey and white arrowheads respectively. High levels of DN-Sun1 expression resulted in the displacement of Nesprin-1 from the nuclear envelope. This confirmed that the DN-Sun1 was effective at disrupting the LINC complex in CMs.

(129) We used AAV (serotype 9) to transduce and express the DN-Sun1 minigene in the hearts of postnatal mice by intrathoracic injection. The procedure is summarized in FIG. 31A and all mice were sacrificed at 100 days after Tmx injection for analysis. Detection by PCR of the Lmna deletion in the hearts confirmed Cre induction by Tmx injection (FIG. 31C). To determine the localization and expression levels of the DN-Sun1 minigene, total protein was extracted from half the heart. Western analysis revealed robust expression of both AAV9-DNSun1 and AAV9-GFP control protein (Dose injected: 5×10^{10} vg/g of mouse) 99 days after AAV injection (FIG. 31C) with the expression levels of both proteins being dependent on the dose of viral particles injected (FIG. 38). The expression of either AAV9-DNSun1 or AAV9-GFP proteins did not affect LMNA protein levels (FIG. 39A). Immunofluorescence analysis revealed that a larger percentage of cardiomyocytes were expressing GFP with 5×10^{10} vg/g of AAV9-GFP compared to the levels resulting from a 10-fold lower dose of viral particles (5×10^9 AAV9-GFP) (FIG. 39B and FIG. 39C).

(130) The Lmna.sup.Flx/Flx:mcm+Tmx mice, injected with AAV9-GFP control, lived an average of 34.5 days after Tmx, whereas Lmna.sup.Flx/Flx:mcm+Tmx mice injected with AAV9-DNSun1 (5×10^{10} vg/g of mouse) lived significantly longer with the majority surviving at least 100 days after Tmx, before their termination for analysis ($P=0.0002$) (FIG. 20 shows early time period results with male and female mice; FIG. 31E shows results at 100 days with separate graphs for male and female mice and mice with different virus injection titre removed). Echo analysis confirmed Lmna.sup.Flx/Flx:mcm+Tmx+AAV9-DNSun1 hearts were functioning better than Lmna.sup.Flx/Flx:mcm+Tmx+AAV9-GFP hearts at 35 days post Tmx (FIG. 31G). Although the Lmna.sup.Flx/Flx:mcm+Tmx+AAV9-DNSun1 mice were alive at 100 days after induction both EF % and FS % were significantly lower compared to control Lmna.sup.Flx/Flx:mcm+Tmx mice (FIG. 31G). At 35 days post Tmx, increased fibrosis was detected in both the Lmna.sup.Flx/Flx:mcm+Tmx+AAV9-DNSun1 and Lmna.sup.Flx/Flx:mcm+Tmx+AAV9-GFP

hearts (FIG. 31F), although fibrosis in the Lmna.sup.Flx/Flx:mcm+Tmx+AAV9-DNSun1 hearts was significantly lower than in Lmna.sup.Flxx/Flx:mcm+Tmx+AAV9-GFP hearts (FIG. 31F lower panels).

Example 6

(131) Disruption of the LINC Complex in Mice Using CRISPR/Cas9

(132) Mice harboring a variety of Lmna mutations, both global and cardiac-specific, show a significant increase in longevity and health in the absence of Sun1 [(Chen et al., *Cell* 149: 565-577 (2012) and Examples 2-4]. Prior to the findings described in Examples 2-5, the mechanism of this rescue was unclear, but was speculated to be due to the toxic effects of excess Sun1 in Lmna mutants [Chen et al., *Cell* 149: 55-577 (2012)]. AAV-mediated expression of a dominant negative LINC-complex-disrupting transgene ameliorates the pathology associated with Lmna mutation [Example 5]. The findings in Examples 2-5 are consistent with the idea that LINC complex function, rather than excess Sun1, is the molecular driver of Lmna pathology. This was surprising, as genetic disruption of the LINC complex via loss of Sun1 and Sun2 [K. Lei et al., *Proc Nat Acad Sci USA* 106: 10207-10212 (2009)], or cardiac-specific disruption of Nesprin-1 and Nesprin-2 [Banerjee et al., *PLOS Genet* 10(2): e1004114 (2014)], in mice, resulted in various pathologies. (133) To develop alternative means of disrupting the LINC complex in vivo, the possibility of using CRISPR/Cas9 genome editing to disable the SUN and KASH domains of the proteins constituting the LINC complex was examined. As both the SUN domain and the KASH domain are located at the C-termini of their respective proteins, we hypothesized that a CRISPR guide RNA targeted to the 3' end of the genes encoding SUN or KASH domain proteins would result in a premature stop codon following CRISPR-induced non-homologous end joining. This would result in a truncated protein with its C-terminal SUN or KASH domain mutated. The truncated protein would be expressed and membrane-localized, but unable to interact with its cognate LINC complex partners. In Example 2, we found that loss of Sun2 did not ameliorate Lmna-associated pathologies. Thus we chose to target the Sun1 SUN domain using CRISPR as Sun1 appears to be the dominant SUN domain protein mediating Lmna pathology. Of the KASH domain proteins, only Nesprin-1, Nesprin-2 and Nesprin-3 are broadly expressed [H. F. Horn, *Current topics in developmental biology* 109: 287-321 (2014)]. Nesprin-1 and Nesprin-2 are close paralogues that are functionally redundant. They interact with the actin and microtubule cytoskeleton, whereas Nesprin-3 appears to interact specifically with intermediate filaments [Kim et al., *Biol. Chem.* 396: 295-310 (2015)]. As we already had Nesprin-2 and Nesprin-3 mutant mouse strains derived by conventional gene targeting available in the laboratory, we chose to target the KASH domain of Nesprin-1 using CRISPR to test the possibility of using CRISPR/Cas9 in vivo for treatment of laminopathies.

(134) The Sun1 gene and the Syne1 gene encoding Nesprin-1 protein were directly targeted in vivo by microinjecting C57/B16 mouse zygotes with Cas9 mRNA and gRNA targeting the SUN1 (5'-GCACAATAGCCTCGGATGTCG-3'; SEQ ID NO: 66) or KASH1 (5'-CCGTTGGTATATCTGAGCAT-3' SEQ ID NO: 67) domains, followed by implantation into surrogate mothers. Note the SUN gRNA targeted Sun1 upstream of the SUN domain so as to ablate the SUN domain. A gRNA (5'-GGTTATGGCCGATAGGTGCAT-3'; SEQ ID NO: 68) targeting the tyrosinase gene was co-injected—progeny that had undergone CRISPR genome editing would have white or mosaic coat color resulting from tyrosinase disruption. These pups were genotyped to confirm successful gene disruption and used as founder animals to establish Sun1 or Nesprin-1 mutant colonies.

(135) Characterization of Mutant Mice

(136) Following Sanger sequencing of founder animals and F1 progeny, we focused on characterizing Sun1 mutant alleles (FIG. 40A) with a 7 bp deletion (Sun1_del7, or Sun1 Δ 7; SEQ ID NO: 71) and 4 bp insertion (Sun1_plus4; SEQ ID NO: 70), and a Syne1 (Nesprin-1) mutant allele (FIG. 41A) with a 8 bp deletion (Syne1_CTdel8, or Syne1 C'T Δ 8; SEQ ID NO: 78). The

Sun1 mutant alleles were predicted to produce mRNA with premature stop codons resulting in a truncated Sun1 protein lacking a SUN domain (FIG. 40B). Tail tip fibroblasts were isolated from Sun1 homozygous mutant animals. Immunofluorescence staining revealed loss of Sun1 protein (FIG. 40C), suggesting that the indels generated by CRISPR caused nonsense-mediated decay of Sun1 mRNA. It is unclear whether the site of mutation, being outside the SUN domain rather than inside the SUN domain, had an effect on the expression of the mutated gene. As we were unable to obtain Sun1 mutant alleles that produced Sun1 protein lacking the SUN domain, instead obtaining essentially Sun1 null animals, we did not further characterize these mutant lines.

(137) The Syne1 C'TΔ8 allele is predicted to produce a protein where the final 11 amino acids in the wildtype sequence (SEQ ID NO: 77) are mutated and are followed by an additional 50 amino acids encoded by an alternate reading frame (FIG. 41B; SEQ ID NO: 78). Immunoblotting performed on Syne1WT and Syne1C'TΔ8 heart and muscle tissues revealed a ~120 kDa band in WT corresponding to the striated muscle-enriched Nesprin-1α isoform of the Syne1 gene (FIG. 41C, D). In the C'TΔ8 heart and muscle tissues, the presumptive Nesprin-1a polypeptide appeared to be less abundant and of lower electrophoretic mobility than in the wildtype (FIG. 41C, D). This is consistent with the 8 bp deletion in the Syne1C'TΔ8 allele introducing a novel stop codon downstream, resulting in a protein of higher molecular weight. In addition, a ~1 MDa band likely corresponding to Nesprin-1Giant was observed in heart tissue from both Syne1WT and Syne1C'TΔ8 mice.

(138) Immunofluorescence analysis of mouse adult fibroblasts (MAFs) derived from 12 week old mice revealed that Nesprin-1 was mis-localized from the nuclear envelope to the cytoplasm in the Syne1C'TΔ8 MAFs (FIG. 42A). Similarly, in myotubes, Nespr-1 redistributes to the cytoplasm in the Syne1C'TΔ8 myotubes as compared to Syne1WT (FIG. 42B). Other LINC complex and NE proteins such as SUN, SUN2, Emerin and LaminA remained localized to the NE (FIG. 43A-C). Consistent with previous reports [Gimpel et al., *Curr. Biol.* 27: 2999-3009.e9. (2017)], disruption of Nesprin-1 in myotubes led to mislocalization of centrosomal proteins PCM1, Pcnt and Akap450 from the myotube nuclear envelope (FIG. 44A-C). Mislocalization of Nesprin-1 from the nuclear envelope is consistent with disruption of the Nesprin-1 KASH domain, preventing Nesprin-1C'TΔ8 mutant protein from interacting with the SUN domains of Sun1 and Sun2, which would normally restrict Nesprin-1 to the nuclear envelope. As the transmembrane region is not disrupted, it is likely that Nesprin-1 is mislocalized to the endoplasmic reticulum (ER) in the C'TΔ8 mutant, as the ER and the perinuclear space form a contiguous membrane system.

(139) Similar to one previously reported Nesprin-1 mouse model [Zhang et al., *Development* 134(5): 901-8 (2007)], and in contrast to two other models [Puckelwartz et al., *Hum Mol Genet* 18: 607-620 (2009); Zhang et al., *Hum Mol Genet* 19: 329-341 (2010)], the disrupted KASH domain of Nesprin-1 results in no overt phenotypic differences between the Syne1 wildtype (WT) and Syne1C'TΔ8 mutant (FIG. 45A-B). Both male and female homozygous mutants were fertile with no significant differences in body weight between the Syne1WT and Syne1C'TΔ8 mice (FIG. 45C). Syne1C'TΔ8 mice also did not exhibit any growth retardation or obvious muscle dystrophy, nor did they display any difficulty in movement or grooming, which can be indications of muscle deterioration.

(140) In order to probe the role of other KASH domain proteins in Lnma pathology, mice mutant for Syne2, encoding Nesprin-2, were generated by conventional gene targeting (FIG. 46A). To characterize the mutation, immunofluorescence microscopy of tail tip fibroblasts was carried out. Syne2+homozygous mutant fibroblasts expressed little to no Nesprin-2 (FIG. 46B). Consistent with previous findings [Zhang et al., *Development* 134(5): 901-8 (2007)], while Syne2.sup.-/- mice were overtly normal, with no growth retardation or infertility, Nesprin-1/2 double mutant mice (Syne1.sup.C'TΔ8/C'TΔ8;Syne2.sup.-/-) were perinatal lethals (FIG. 46C).

(141) Disruption of Nesprin-1 KASH Domain Ameliorates Lnma Pathologies

(142) Even though Nesprin-1 was still expressed, Nesprin-1-containing LINC complexes would

not be formed in Syne1.sup.C'TΔ8/C'TΔ8 cells and animals. Since AAV-mediated disruption of the LINC complex using dominant negative Sun1 in vivo rescues Lmna pathologies (Example 5), we reasoned that the “KASH-less” Nesprin-1 mutant allele we generated might also rescue Lmna pathology. To test this hypothesis, mice heterozygous for a Lmna null (Lmna.sup.Δ/Δ) allele (Example 1) were intercrossed with Syne1C'TΔ8 mice to obtain Lmna.sup.Δ/Δ:Syne1.sup.C'TΔ8/C'TΔ8 double mutant mice. While Lmna.sup.Δ/Δ mice lived for 15-17 days, Lmna.sup.Δ/Δ:Syne1.sup.C'TΔ8/C'TΔ8 double mutant mice lived for up to 42 days (FIG. 24). Lmna null mice heterozygous for the Syne1.sup.C'TΔ8 allele did not experience any lifespan extension. Lmna.sup.Δ/Δ mice on a Syne2.sup.-/- homozygous mutant background also did not experience lifespan extension (FIG. 47), indicating that Lmna pathology is mediated primarily by Nesprin-1/Sun1 LINC complexes.

(143) To examine the effect of the Syne1.sup.C'TΔ8/C'TΔ8 allele in mice with cardiac-specific loss of Lmna, mice homozygous for a conditional Lmna.sup.Flx/Flx allele carrying the inducible cardiomyocyte specific Cre Tg(Myh6-cre/Esr1) (here abbreviated to mcm), in which Cre is induced by a single injection of tamoxifen (Tmx), were used as described in Examples 1-2. Cardiac-specific deletion of Lmna results in death within a month, but mice with the same deletion induced on a homozygous Syne1.sup.C'TΔ8/C'TΔ8 background lived for at least 120 days after Tmx induction (FIG. 25; no change from day 80-120).

Example 7

(144) Method for Screening Small Molecules that Block SUN-KASH Interactions

(145) Crystallographic studies of human SUN2 reveal that the SUN domain is assembled as a clover-like trimeric structure [Sosa et al., *Cell* 149(5): 1035-47 (2012)]. Trimerization is mediated by a triple-helical coiled-coil, with an estimated length of 40-45 nm. This is sufficient to bridge the perinuclear space (PNS), allowing SUN and KASH domains to directly interact [Sosa et al., *Cell* 149(5): 1035-47 (2012)]. The KASH binding site is formed primarily within a groove formed at the interface between adjacent SUN domains (FIG. 5B; FIG. 21 left panel). This groove accommodates part of the KASH domain, about 18 residues, in an extended conformation. However, it is the C-terminal tetrapeptide of the KASH domain, featuring three proline residues followed by a terminal aliphatic residue, Leu or Thr (for Nesp1 and Nesp2 respectively), that is crucial for the SUN-KASH interaction (FIG. 21 right panel, adapted from FIG. 1 of Sosa et al., *Cell* 149(5): 1035-47 (2012)). The significance of this tetrapeptide is that it is situated in a well-defined pocket formed within a single SUN monomer. Modification of this peptide in any way, including the addition of a single residue (an Ala) at the C-terminus, completely eliminates the SUN-KASH association over the entire SUN-KASH contact region (Sosa et al., *Cell* 149(5): 1035-47 (2012) and FIG. 22 left panel). The conclusion is that while stable binding of the KASH domain requires 18-20 residues, it is the C-terminal tetrapeptide that actually initiates binding. Thus, blocking the tetrapeptide binding-pocket within the SUN monomer will abolish SUN-KASH association. We have described in this disclosure an AAV-based gene therapy strategy to break endogenous SUN-KASH interactions as a treatment for laminopathies, including dilated cardiomyopathy. Alternatively, a small molecule that blocks the SUN-KASH interaction at the SUN binding pocket would disrupt LINC complexes and similarly treat laminopathies. A variety of standard methods exist to screen for small molecule drugs in vitro.

(146) An in vitro screen can be set up employing recombinant SUN and KASH domains or KASH peptide, for which methods of production have been previously published [Sosa et al., *Cell* 149(5): 1035-47 (2012)]. One such screen involves an assay technique analogous to an enzyme-linked immunosorbent assay (FIG. 22 right panel, similar to Lepourcelet et al., *Cancer Cell*. 5(1):91-102 (2004)). Recombinant SUN domain is immobilized on a solid surface, typically in 96-well plates, and then complexed with recombinant KASH domain linked to an enzyme that can generate a colorimetric or chemiluminescent readout. One method for enabling this linkage is to synthesize a biotinylated KASH peptide, which can then be linked with commercially available streptavidin-

horseradish peroxidase (HRP) conjugate. Candidate compounds are obtained from appropriate suppliers and screened for their ability to inhibit KASH-SUN associations in vitro. Compounds that fail to inhibit the SUN-KASH interaction will result in a well in the plate where the recombinant SUN binds to the enzyme-linked KASH domain. Following wash steps and incubation with colorimetric or chemiluminescent HRP substrates, the presence of the SUN-KASH interaction is detected in standard plate readers. If the compound can inhibit SUN-KASH interaction then, following the wash step, the KASH domain is removed and there would be reduced or no enzymatic reaction in the well.

(147) Alternatively, fluorescence anisotropy or polarization can be used to screen for small molecule inhibitors of SUN-KASH interactions in vitro [Lea, W. A., and Simeonov, A. *Expert Opin Drug Discov* 6: 17-32 (2011)]. This assay also employs recombinant SUN and KASH domains. The KASH domain is fluorescently labeled; for example a chemically synthesized KASH peptide could be readily functionalized with a fluorescein moiety. Fluorescence anisotropy of the interacting KASH domain interacting with SUN domain can be measured using standard equipment such as a plate reader. A small molecule inhibitor that disrupts the SUN-KASH interaction can be readily detected as the fluorescence anisotropy of the fluorescent KASH will change if it is not bound to SUN.

(148) As is typical in drug screening campaigns, the compounds which successfully pass the in vitro primary screen will then be subjected to cell-based secondary screens (FIG. 23). In this case, immunofluorescence microscopy will be employed to identify those compounds that can dissociate LINC complexes. This is manifest as dispersal of the KASH component to the peripheral endoplasmic reticulum while the cognate SUN protein is retained in the inner nuclear membrane. This microscopy-based assay can be performed first on HeLa cells. Active compounds are then evaluated on cultured cells from disease-relevant tissue, such as cardiac cells. An additional secondary screen may include the ability of the identified compound to rescue proliferation defects in Lmna knockout cells. Following hit-to-lead optimization of the identified compound using standard methods, the compound can be tested in mouse models of laminopathies such as those described herein for Lmna dilated cardiomyopathy. Efficacy of the leads can be evaluated using lifespan of the mutant mice and echocardiograms, as described herein, to assess heart function.

DISCUSSION

(149) DCM caused by LMNA is regarded as being aggressive, and often leads to premature death or cardiac transplantation [M. Pasotti et al., *J Am Coll Cardiol* 52: 1250-1260 (2008); M. R. Taylor et al., *J Am Coll Cardiol* 41: 771-780 (2003)]. By 60 years, 55% of LMNA mutation carriers die of cardiovascular failure or receive a heart transplant, compared with 11% of patients with idiopathic cardiomyopathy. Attempts to ameliorate DCM by fitting a pacemaker have been at best of transient benefit. Consequently it is necessary develop new therapeutic avenues to treat DCM caused by LMNA mutations.

(150) The majority of LMNA mutations causing DCM are dominant negative missense. Treatment by conventional gene therapy to repair each mutation would be daunting and removal of the mutated allele, leaving the patient hemizygous for the remaining normal WT allele may also result in heart failure [G. Bonne et al., *Nature genetics* 21: 285-288 (1999)]. Various other routes downstream of the Lamin gene have been explored for potential therapeutic intervention, and have included mTOR inhibition with rapamycin/rapalogues [J. C. Choi et al., *Science translational medicine* 4: 144ra102 (2012); F. J. Ramos et al., *Science translational medicine* 4: 144ra103 (2012)] and inhibition of the MEK1/2 kinase pathway [W. Wu, et al., *Circulation* 123: 53-61 (2011)]. Both avenues, resulted in improved ventricular function and increased longevity (10-40%) but the extent and long-term efficacy was significantly less than that we observed with the loss of Sun1.

(151) The molecular mechanisms underlying the varied phenotypes of the laminopathies are still not well understood, though two alternative hypotheses have been proposed to explain the tissue-

specific pathologies. The first “gene regulation hypothesis” proposes that LMNA mutations/loss disrupt the equilibrium of various molecular pathways due to the mutations altering interactions with NE proteins and chromatin, which in turn alter gene expression. Evidence in support of this hypothesis comes from studies reporting changes in signalling pathways including the AKT-MTOR pathway [J. C. Choi et al., *Science translational medicine* 4: 144ra102 (2012)], WNT/ β -catenin pathway [L. Hernandez et al., *Dev Cell* 19: 413-425 (2010); C. Le Dour et al., *Hum Mol Genet* 26: 333-343 (2017)], TGF- β /Smad [J. H. Van Berlo et al., *Hum Mol Genet* 14: 2839-2849 (2005); T. V. Cohen et al., *Hum Mol Genet* 22: 2852-2869 (2013)], MAP Kinase pathway [A. Brull, et al., *Front Physiol* 9: 1533 (2018)] and the ERK1/2-CTGF/CCN2 pathway [M. Chatzifrangkeskou et al., *Hum Mol Genet* 25: 2220-2233 (2016)]. While these changes have been documented, none has clearly established whether these changes are not a secondary compensatory effect of a diseased tissue. Sun1 also fits into this rubric of disrupted expression levels as Sun1 protein, but not mRNA, is upregulated in laminopathies, leading to the proposal that laminopathy phenotypes are caused by toxicity from excess Sun1 [C. Y. Chen et al., *Cell* 149: 565-577 (2012)].

(152) The second hypothesis suggested Lmna loss or mutation leads to Increased nuclear fragility. As a result mechanical stress and tension forces transmitted via the LINC complex from the cytoplasm to the NE causes damage to the NE [J. Lammerding et al., *J Clin Invest* 113: 370-378 (2004)]. This hypothesis is similar to that proposed for Duchenne muscular dystrophy (DMD), where loss of dystrophin increases the fragility of the muscle cell membrane and when tension-stress forces are applied during muscle contraction this results muscle cell rupture and death [D. J. Blake, et al., *Physiol Rev* 82: 291-329 (2002)]. Lmna mutant fibroblasts show nuclear deformation, defective mechanotransduction, and reduced viability when subjected to mechanical strain, together with increased nuclear rupture at low and moderate pressures when compared to WT nuclei [J. Lammerding et al., *J Clin Invest* 113: 370-378 (2004); J. Lammerding et al., *J Cell Biol* 170: 781-791 (2005); J. Lammerding et al., *J Biol Chem* 281: 25768-25780 (2006)]. In contracting mouse cardiomyocytes, mechanical stress and tension forces caused by 500-600 contractions per minute are transmitted to the NE via the LINC complex, resulting in nuclear distortion, damage and eventual death/loss as described in FIGS. 28 and 29. Presumably, such forces would cause significant damage to the fragile NE of Lmna null cardiomyocytes, resulting in CM death. If the tension-stress hypothesis is damaging to the NE, then unlinking the LINC complex, by disrupting SUN1, would reduce the tension-stress on the CM nuclei, and prevent CM cell death in the mutant CMs (FIG. 32A-C). One caveat here is that complete disruption of the LINC complex, as would be the case following overexpression of DN-Sun1, could potentially be deleterious rather than therapeutic. At the cellular level, multiple mechanical phenomena including intracellular force transmission, cell polarization and migration, were impacted following LINC complex disruption by dominant negative SUN and KASH constructs [Lombardi et al., *J Biol Chem* 286(30):26743-53 (2011)]. In animal models, Sun1/Sun2 [Lai et al., *Proc Natl Acad Sci* 106(25):10207-12 (2009)] and Nesprin-1/Nesprin-2 [Zhang et al., *Development* 134(5):901-8 (2007)] double mutant mice experience perinatal lethality and cardiac-specific disruption of the KASH domains of Nesprin-1 and Nesprin-2 using an embryonic cardiac Cre driver (Nkx2.5-Cre) results in early onset cardiomyopathy [Banerjee et al., *PLOS Genet* 10(2):e1004114 (2014)].

(153) We attempted to distinguish the tension-stress hypothesis from the expression level hypothesis in cardiomyocytes, using a DN-Sun1 construct to compete with endogenous Sun1 and Sun2 proteins for KASH-domain-binding and so unlink the LINC complex without directly altering Sun1 levels (FIGS. 32D & 34). The AAV9 vector, which has a high affinity for CM, was used to deliver DN-Sun1 under the cTnT promoter to CMs [C. Zincarelli, et al., *Mol Ther* 16: 1073-1080 (2008)]. Our results showed the successful delivery of GFP to cardiomyocytes (FIG. 31C), and robust expression of both the control GFP and DN-Sun1 proteins (FIG. 31C) with the latter resulting in the dispersal of the KASH domain proteins from the cardiomyocyte nuclei (FIG. 31D). Surprisingly, not only did AAV-DN-Sun1 ameliorate the pathology in mice with depleted

cardiac Lmna levels, it also had no discernible effect on the cardiac health of wildtype mice, which would be expected to also experience complete LINC complex disruption in their hearts (FIGS. 31E & G). This suggests that an intact LINC complex may be required in embryonic development, but not postnatally.

(154) In addition, using CRISPR/Cas9 in mice, we generated a Syne1 mutant allele (C'TΔ8) that gave rise to a truncated Nesprin-1 protein with a disrupted, non-functional, KASH domain. Mice lacking Lmna globally or in the heart have a shortened lifespan, but the presence of a homozygous Syne1.sup.C'TΔ8/C'TΔ8 mutation resulted in significant lifespan extension. Loss of Sun1 or AAV-mediated disruption of the LINC complex by dominant negative transgenes in vivo resulted in similar rescue of Lmna pathology (Example 2-5), while Sun2 and Nesprin-2 mutations did not. Taken together, these data suggest that LINC complexes comprised of Sun1 and Nesprin-1 drive the pathology in Lmna mutant cells and animals.

(155) There have been a number of reports on the use of AAV to deliver CRISPR/Cas components in vivo for treating diseases. Our results predict that AAV-mediated CRISPR/Cas, such as CRISPR/Cas9, delivery to target the Nesprin-1 KASH domain in disease-affected tissue can be used to treat laminopathies, including dilated cardiomyopathy. For instance, cardiotropic AAVs (e.g. AAV9) can be used to deliver transgene cassette(s) containing a cardiac-specific promoter (e.g. cTnT) driving Cas endonuclease enzyme expression and an appropriate promoter (e.g. U6) driving gRNA expression to treat LMNA DCM. Since the packaging capacity of AAV is limited to 4.7 kb, a smaller Cas9 derived from *Staphylococcus aureus* (saCas9) rather than the larger, more commonly used, *Streptococcus pyogenes* Cas9 may be preferred. Alternatively, other CRISPR enzymes such as Cpf1, which is small enough for AAV packaging and has a more commonly found protospacer adjacent motif (PAM) than saCas9, could be used [Zetsche, B., et al., Cpf1 Is a Single RNA-Guided Endonuclease of a Class 2 CRISPR-Cas System. *Cell* 163, 759-771 (2015)].

(156) Guide RNAs would target the 3' region of the Nesprin-1 gene encoding the KASH domain (Table 3). While we have targeted the region adjacent to the stop codon, in principle any gene region encoding the KASH domain could be targeted as indels generated by CRISPR would likely result in frameshift mutations that disable the KASH domain. However, as the final 4 amino acids in the KASH domain are known to be absolutely required for SUN domain interaction and hence LINC complex formation [Sosa et al., *Cell* 149(5):1035-47 (2012)], it is prudent to select gRNA in the vicinity of the stop codon as even indels that do not result in a frameshift could still mutate the relevant KASH amino acids required for the SUN-KASH interaction. Furthermore, it should be noted that because the Syne1 gene encoding Nesprin-1 is very large and has multiple splice isoforms and alternative start sites, guide RNAs targeted outside the KASH domain, while giving rise to some mutant Nesprin-1 isoforms, may not perturb expression of other isoforms of Nesprin-1 protein, including KASH-containing isoforms. This would result in formation of functional or partially functional Nesprin-1/Sun1 LINC complexes that would still be able to drive pathology in Lmna mutants. This CRISPR/Cas9 strategy likely cannot be extended to the KASH domain of Nesprin-2, since Lmna.sup.Δ/Δ:Syne2.sup.-/- mice are phenotypically indistinguishable from Lmn.sup.Δ/Δ mice.

(157) We did not further investigate the Sun1 mutant mice generated in this study as instead of mice with Sun1 lacking the SUN domain, we essentially obtained Sun1 null mice, which have already been well characterized. We suspect that inducing CRISPR mutation in Sun1 resulted in nonsense-mediated decay (NMD) of Sun1 transcript. Occurrence of a premature termination codon (PTC) 50-55 nucleotides upstream of an exon-exon junction is a trigger for NMD [Popp, M. W., and Maquat, L. E. *Cell* 165: 1319-1322 (2016)]. Also, PTCs occurring in the middle of a transcript are more likely to result in NMD [Eberle et al., *PLOS Biology* 6: e92 (2008); Reber et al., *MBoC* 29: 75-83 (2018)]. In the Sun1_{plus4} mutant, the PTC is more than 55 nucleotides upstream of the exon-exon junction and is likely to trigger NMD. For the Sun1Δ7 mutant, the PTC is less than 50 nucleotides from the exon-exon junction. However for both mutants, since we targeted upstream of

the sizeable SUN domain, the PTCs are roughly $\frac{2}{3}$ of the way along the length of the transcript, and hence also likely to trigger NMD. In order to specifically disrupt the SUN domain in Sun1 without inducing a null mutation, we can adopt a similar strategy as for Nesprin-1—directing the guide RNA at the very 3' end of the coding region of the transcript (Table 3). Earlier work demonstrated that mutation of a tyrosine residue to phenylalanine at the C-terminus of SUN2 (Y707F) abolished KASH binding [Sosa et al., *Cell* 149(5):1035-47 (2012)]. This critical tyrosine residue is conserved in SUN1 (Y812 in Uniprot E9PHI4) and present in the final coding exon of the SUN transcript. Selection of a gRNA 5' proximal to the codon for Y812 would produce Indel mutations that cause a frameshift mutation that would mutate Y812 and disrupt KASH binding. As the gRNA would be in the final coding exon, the likelihood of triggering NMD would be low. One can thus envision a CRISPR/Cas9-based strategy to treat laminopathies by targeting a critical residue required for KASH-binding in the SUN1 SUN domain. AAVs could be used to deliver CRISPR enzyme and gRNA targeting SUN in appropriate disease tissue, such as the heart. Incapacitation of SUN KASH binding would then ameliorate the deleterious effects of Lmna mutations.

(158) From these results we propose that the loss of or mutations within Lmna causes instability in the CM nuclei due to loss or incorrect assembly of the nuclear lamina. This makes the nuclei susceptible to the tension/stress forces exerted via the LINC complex from the contractile sarcomeres of the CMs. In the absence of SUN1, or following mutation of Nesprin-1 KASH domain, the untethered LINC complexes exert less tensional force on the CM nuclei, enabling survival of the lamin deficient cardiomyocyte.

(159) These results provide an opportunity to use the AAV-mediated delivery of DN-Sun, DN-KASH, or direct mutation of endogenous SUN or KASH proteins as potential therapeutics for laminopathy-related DCM in patients. The AAV system, as a therapeutic delivery route in patients is established and has been approved by the FDA for treating some diseases. It is becoming more widely used with multiple on-going clinical trials, including the introduction into patients with heart disease. However, even though tension-stress may be the primary cause for Lmna deficient CM death, disrupting SUN1 may not be effective in preventing LMNA mutation induced cell death in skeletal muscle, as *Lmna*.sup. Δ/Δ :*Sun*.sup.-/- die at an earlier age than those mice where *Lmna* was specifically deleted in the CMs. Which muscle groups (or even other tissues lacking *Lmna*) result in the early lethality remain to be identified. However in most of the LMNA DCM patients it is heart failure that is the cause of death, and our results show that disrupting the LINC complex in CMs could be effective at preventing heart failure for an extended period.

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Claims

1. A method of treating a laminopathy in a subject in need thereof, wherein the laminopathy is a disease caused by a mutation in the Lamin A (LMNA) gene, the method comprising administering to the subject an effective amount of an adeno-associated virus (AAV) expression vector operably linked to a transgene; wherein the transgene comprises a nucleic acid sequence encoding a dominant negative Sun domain containing protein; wherein the AAV expression vector comprises a cardiac-or cardiomyocyte-specific promoter; wherein the AAV expression vector is administered intravascularly, wherein expression of the dominant negative Sun domain containing protein disrupts the protein-protein interactions of the Linker of Nucleoskeleton and Cytoskeleton (LINC) complex, which uncouples the nucleus from its linkage to the cytoskeleton, resulting in treatment of the laminopathy.
2. The method according to claim 1, wherein the protein-protein interaction is between a Sad Ip UNC-84 (SUN) domain and a Klarsicht, Anc-1 and a Syne Homology (KASH) domain of the LINC complex.
3. The method according to claim 1, wherein the nucleic acid further comprises a luminal domain of a Sun domain containing protein, an N-terminal signal sequence, a signal peptidase cleavage site, and a C-terminal targeting peptide sequence preventing secretion of the SUN domain-containing protein, wherein the luminal domain comprises a coiled coil domain and a Sun domain.
4. The method according to claim 3, wherein the SUN domain-containing protein is SUN1 or SUN2.
5. The method according to claim 3, wherein the C-terminal targeting peptide sequence is a KDEL sequence.
6. The method according to claim 1, wherein the AAV expression vector is a cardiac or cardiomyocyte-specific expression vector.
7. The method according to claim 6, wherein in the AAV expression vector is selected from the group consisting of: AAV9, AAV1, AAV6, AAV8, AAV2i8, and AAV9.45.
8. The method according to claim 1, wherein the cardiac-or cardiomyocyte-specific promoter is selected from the group consisting of: a cardiac troponin T promoter (cTnT), a α -myosin heavy chain (α -MHC) promoter and a myosin light chain (MLC2v) promoter.
9. The method according to claim 1, wherein laminopathy is selected from the group consisting of: Acrogeria, Gottron Type; Arrhythmogenic cardiomyopathy; Arrhythmogenic right ventricular cardiomyopathy; Arthropathy syndrome, autosomal recessive; Atrial fibrillation; Atypical progeroid syndrome; Atypical Werner syndrome; Autosomal dominant spinal muscular dystrophy; Axonal neuropathy; muscular dystrophy [; cardiac disease or cardiomyopathy; Axonal neuropathy, muscular dystrophy, cardiac disease, leuconychia; Cardiac arrhythmia; Cardiac conduction defect; Cardiomyopathy with advanced atrioventricular block and arrhythmia; Charcot-Marie-Tooth disease type 2; Congenital fiber type disproportion; Congenital muscular dystrophy; Diabetes Mellitus, Non-Insulin-Dependent (NIDDM); Dilated cardiomyopathy; Distal acroosteolysis, poikiloderma and joint stiffness (DAPJ); Distal motor neuropathy; Dropped Head Syndrome; Emery-Dreifuss muscular dystrophy, autosomal dominant; Familial partial lipodystrophy (Dunnigan Type); Familial partial lipodystrophy, Kobberling; Generalized lipoatrophy syndrome; Hallerman-Streiff syndrome; Heart-hand syndrome, Slovenian Type; Hutchinson-Gilford progeria

syndrome; Lamin-related rigid spine muscular dystrophy; Limb-girdle muscular dystrophy type 1B; Muscular dystrophy; Lone atrial fibrillation; Mandibuloacral dysplasia with type A lipodystrophy; Metabolic Syndrome; Muscular dystrophy and lipodystrophy; Progeroid syndrome, neonatal;-Restrictive dermopathy; Spinal muscular atrophy with cardiac involvement; Type A insulin resistance syndrome; cardiomyopathy associated with Emery-Dreifuss muscular dystrophy (autosomal dominant); cardiomyopathy associated with Emery-Dreifuss muscular dystrophy (autosomal recessive); cardiomyopathy associated with Limb-girdle muscular dystrophy type 1B; cardiomyopathy associated with congenital muscular dystrophy; and a premature aging syndrome; cardiomyopathy associated with Atypical Werner syndrome; and cardiomyopathy associated with Hutchinson-Gilford progeria syndrome.

10. The method of claim 9, wherein the cardiac disease or cardiomyopathy is selected from the group consisting of Arrhythmogenic cardiomyopathy, dilated cardiomyopathy 1 Az, and dilated cardiomyopathy with conduction system defects.

11. The method of claim 9, wherein the muscular dystrophy is selected from the group consisting of Autosomal dominant spinal muscular dystrophy; Congenital muscular dystrophy; Emery-Dreifuss muscular dystrophy, autosomal dominant; Lamin-related rigid spine muscular dystrophy; Limb-girdle muscular dystrophy type 1B; and Spinal muscular atrophy with cardiac involvement.

12. The method of claim 9, wherein the cardiac disease or cardiomyopathy is selected from the group consisting of Cardiac arrhythmia; Atrial fibrillation; Cardiac conduction defect; Lone atrial fibrillation; Sudden cardiac death; cardiomyopathy associated with Emery-Dreifuss muscular dystrophy; Cardiac conduction defect; cardiomyopathy associated with Limb-girdle muscular dystrophy type 1B; cardiomyopathy associated with congenital muscular dystrophy; cardiomyopathy associated with Atypical Werner syndrome; and cardiomyopathy associated with Hutchinson-Gilford progeria syndrome.

13. The method of claim 9, wherein the Progeroid syndrome is selected from the group consisting of Progeroid syndrome, neonatal; Acrogeria, Groton Type; Atypical progeroid syndrome; Atypical Werner syndrome; and Hutchinson-Gilford progeria syndrome.
